

QMaxent — Manual

Maxent species distribution modeling in QGIS

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1. QMaxent Manual

Welcome to the QMaxent user manual. **QMaxent** integrates the elapid Python library into QGIS to deliver the full Maxent species distribution modeling (SDM) workflow — training, spatial cross-validation, jackknife variable importance, projection, and survey planning — through a familiar bilingual interface.

1.1 Where to start

If this is your first visit:

1. Read Introduction for the 2-minute orientation
2. Install the plugin → Installation
3. Set up Python dependencies → Dependencies
4. Download the bundled example data → Example datasets
5. Walk through your first model in 5 minutes → Quick start

If you already use QMaxent and want to dig into a specific feature, the User guide chapters mirror the plugin's five tabs one-for-one. Each can be read on its own.

If you want to see the tool put through its paces on real datasets, the Worked examples cover three case studies of increasing complexity, including a reproduction of the Lee et al. (2025) fairy pitta study published in *Global Ecology and Conservation*.

1.2 Conventions used in this manual

- **Bold** marks UI labels (buttons, tabs, menu items) you click.
- `monospace` marks file paths, code, and parameter values you type.
- Numbered tabs (①, ②, …) refer to the QMaxent Analysis dock tabs.
- Inline citations like *(Phillips et al. 2017)* point to entries in the References chapter.
- Sidebar boxes flag warnings (⚠), tips (💡), and notes worth a second glance — but most of the manual is plain prose; we do not use boxes decoratively.

1.3 Citation

If you use QMaxent in your research, please cite the software using the metadata in the repository's `CITATION.cff`. The same metadata is rendered live on the project home page at osgeokr.github.io/qmaxent and in the References chapter of this manual.

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2. Introduction

QMaxent is a QGIS plugin that brings the full Maxent species distribution modeling (SDM) workflow inside the GIS. From a single docked panel you can choose presence points, register environmental rasters, train a regularised Maxent model, evaluate it with spatial cross-validation, project it across the landscape, and generate field-ready candidate survey sites — without leaving QGIS or writing a line of code.

2.1 What is QMaxent

Under the hood, QMaxent wraps the elapid Python library (Anderson 2023). Elapid implements the maxnet algorithm — a penalised Poisson formulation that is provably equivalent to the original Maxent model on a sufficient sample of background points (Fithian & Hastie 2013) — using the lasso-regularised generalised linear model machinery of glmnet (Friedman, Hastie & Tibshirani 2010). The user-facing model is the same Maxent that the broader SDM literature has used since Phillips, Anderson & Schapire (2006); the difference is the surrounding tooling, not the inferential engine.

2.2 Intended users

QMaxent is for ecologists, conservation planners, and biogeographers who:

- Already use QGIS for spatial analysis and want SDM as an integrated step rather than a separate Java GUI
- Need cross-validated, *spatially honest* performance estimates rather than an over-optimistic in-sample AUC (Roberts et al. 2017)
- Want their results to be **reproducible** — fixed seeds, configuration written to a multi-sheet Excel supplement, model serialised as a portable `.pkl`
- Work bilingually (Korean / English) and need a UI in their language

You do not need to know Python. You do need a working knowledge of vector points and raster layers in QGIS, and enough domain knowledge of your study system to choose appropriate environmental predictors.

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2.3 How it compares to other SDM tools

Tool	Interface	Engine	Status
QMaxent	QGIS plugin	elapid (maxnet, Python)	Active, this manual
Java MaxEnt	Standalone Java GUI	Original Maxent	Stable; broadly used
ENMeval (R)	R package	maxnet / dismo	Strong CV / tuning support (Muscarella et al. 2014)
Wallace	Shiny web app	Multiple (incl. maxnet)	Reproducibility focus (Kass et al. 2018)
dismo (R)	R package	Multiple SDM algorithms	Mature; broad ecosystem

QMaxent occupies the same conceptual niche as the Java MaxEnt GUI for QGIS users, with the added benefit of a modern Python runtime that ships with QGIS, an integrated raster-consistency preflight, and built-in spatial cross-validation. The Pitta nympha worked example reproduces a published Java MaxEnt analysis in QMaxent and discusses where the two pipelines agree or diverge.

2.4 Methodological lineage

QMaxent's defaults follow the recommendations of:

- Phillips, Anderson & Schapire (2006), Phillips & Dudík (2008), and Phillips et al. (2017) for the core Maxent formulation, regularization, and cloglog output transform.
- Elith et al. (2011) and Merow, Smith & Silander (2013) for practitioner guidance on inputs and settings.
- The maxnet R package (Phillips et al. 2017) for the default rule that selects feature classes by sample size. It is a default, not a guarantee of the best model (Merow, Smith & Silander 2013; Radosavljevic & Anderson 2014).
- Roberts et al. (2017) for spatial cross-validation as the default evaluation method.

Every default in the plugin can be overridden in the ② Parameters tab. Full citations are in References.

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3. Installation

QMaxent installs as a regular QGIS plugin. After the plugin itself is in place, a one-time dependency setup downloads the Python libraries Maxent needs into a separate QMaxent folder that does not affect your system Python or QGIS itself. On Linux and macOS this folder is a Python virtual environment; on Windows it is a plain package folder filled by the QGIS Python's own pip (see Dependencies).

3.1 System requirements

- **QGIS** 3.44 LTR or later
- **Operating system**: Windows 10/11, macOS 12+, or any modern Linux distribution
- **Disk space**: about 600 MB free for the QMaxent package folder
- **Internet access** is required only on first run to download the Python dependencies; routine model fitting and projection run fully offline.

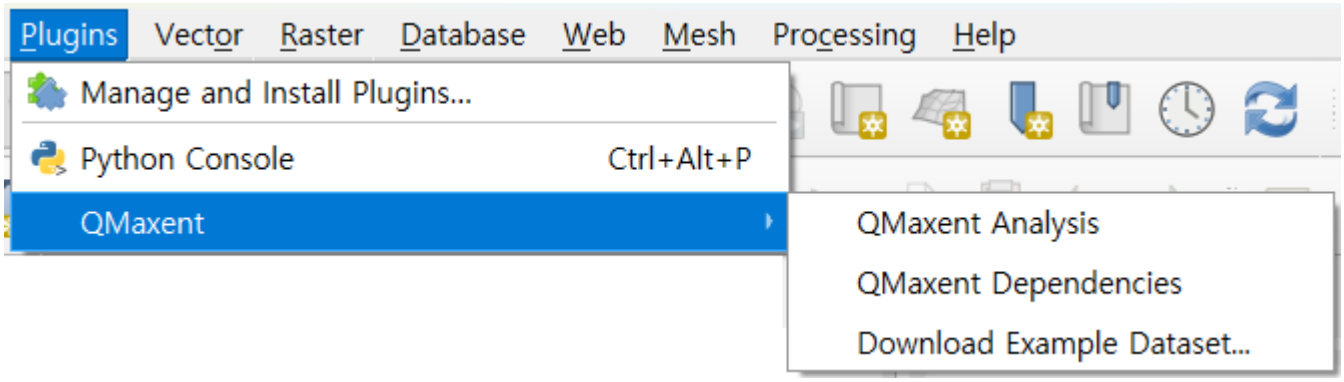
3.2 Install from the QGIS Plugin Repository

QMaxent is published on the official QGIS Plugin Repository.

1. Open QGIS and choose **Plugins → Manage and Install Plugins…**
2. In the **All** tab, type `qmaxent` in the search box. The plugin appears in the result list.
3. Click **Install Plugin**.

Once installed, three entry points appear under **Plugins → QMaxent**:

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Menu item	What it does
QMaxent Analysis	Opens the main analysis dock — the heart of the plugin
QMaxent Dependencies	Opens the dependency-management dialog (next chapter)
Download Example Dataset	Bundled SDM datasets for tutorials (see *Example datasets*)

3.3 Verifying the installation

Choose **Plugins → QMaxent → QMaxent Dependencies**. If the dialog opens, the plugin is correctly installed. The **Environment Status** banner will most likely read *Dependencies not installed* the first time — that is expected and is dealt with in the next chapter.

3.4 Updating to a newer version

When a new release ships, the QGIS Plugin Manager will mark QMaxent as upgradable. Click **Upgrade Plugin** in the **Installed** tab. Dependencies generally do not need to be reinstalled across patch releases (0.1.x → 0.1.y). When the package list or the installation method changes, the **Dependencies** dialog reports that the dependencies need updating; click **Install / Update Dependencies** once more. This is the case when you upgrade to 0.2.0 on Windows, because 0.2.0 changed how packages are installed there.

3.5 Troubleshooting

QMaxent does not appear in the search results

Make sure your QGIS version is 3.44 or later (`Help → About QGIS`). On older versions QMaxent is filtered out as incompatible. Also confirm the **Show experimental plugins** option in **Settings** is set to your preference — QMaxent's stable releases do not require it.

Plugin installs but the menu entry does not appear

Restart QGIS. Some QGIS versions require a restart for top-level menu entries from new plugins to register.

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Windows: installation stops with 'ensurepip ... returned non-zero exit status 1'

QMaxent 0.1.x created a Python virtual environment, and this step fails with QGIS 3.44.15 and later on Windows. QMaxent 0.2.0 no longer needs a virtual environment on Windows: it installs the packages with the QGIS Python's own pip into the QMaxent package folder. Upgrade the plugin to 0.2.0 or later and click **Install / Update Dependencies** again. Details are in Dependencies.

Corporate proxy or restricted network

The plugin install itself uses QGIS's network stack and respects QGIS's proxy settings. The dependency installer (next chapter) downloads from PyPI; if your organisation blocks PyPI, ask your network administrator to allow `pypi.org` and `files.pythonhosted.org`.

Once installed, continue to Dependencies.

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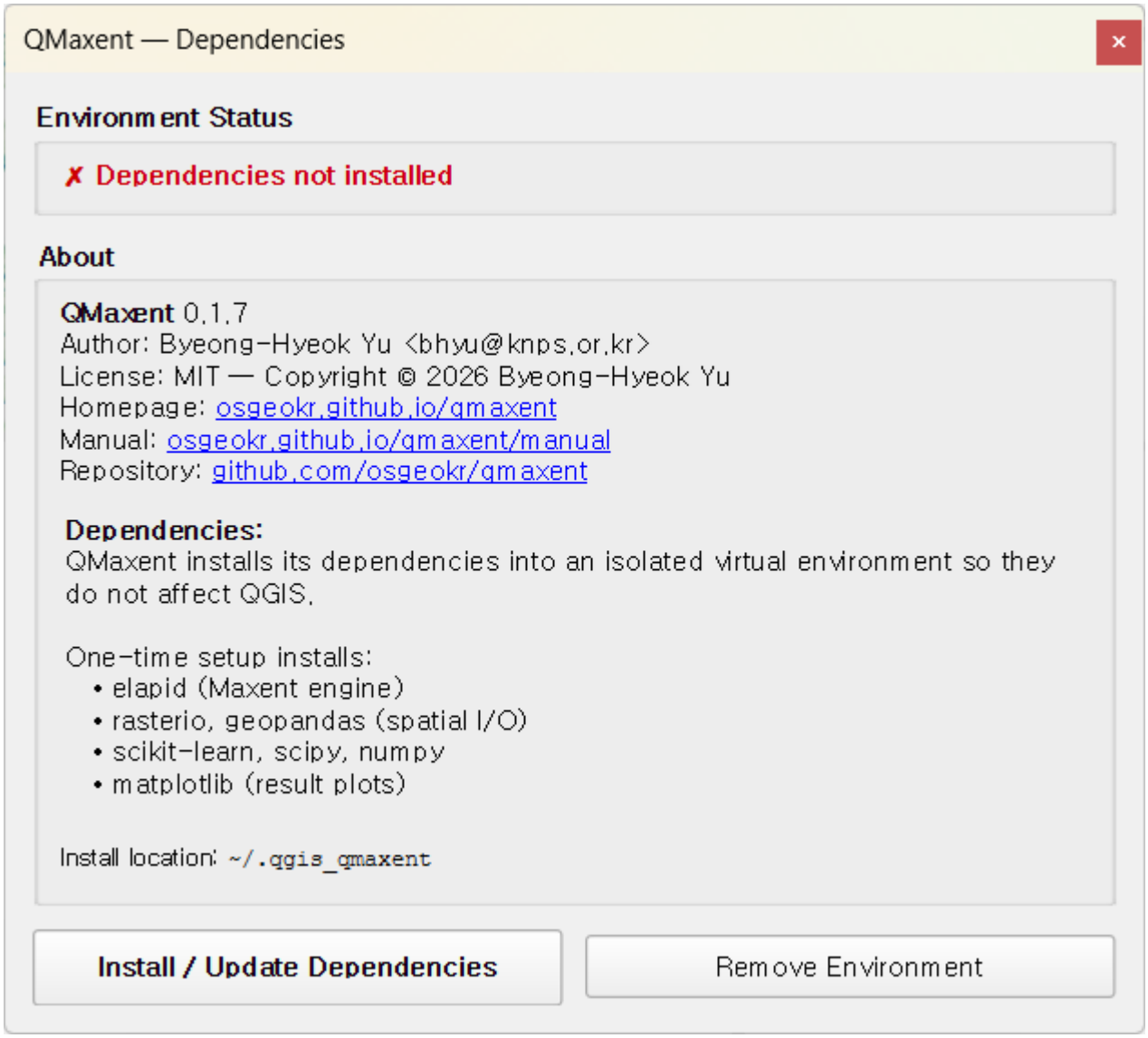
4. Dependencies

QMaxent's modeling engine — the elapid Python library and its companions (rasterio, geopandas, scikit-learn, matplotlib) — does not ship with QGIS. The first time you use the plugin, you install these dependencies into an **isolated, per-plugin package folder** so they do not affect your system Python or QGIS itself. On Linux and macOS this folder is a Python virtual environment; on Windows it is a plain package folder (see Where the packages are installed).

This is a one-time, ~5-minute setup.

4.1 Opening the Dependencies dialog

Choose **Plugins → QMaxent → QMaxent Dependencies**.



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The dialog reports the current state of the QMaxent package folder, the plugin version, and a summary of what will be installed.

4.2 Environment Status panel

The status banner at the top tells you exactly what to do:

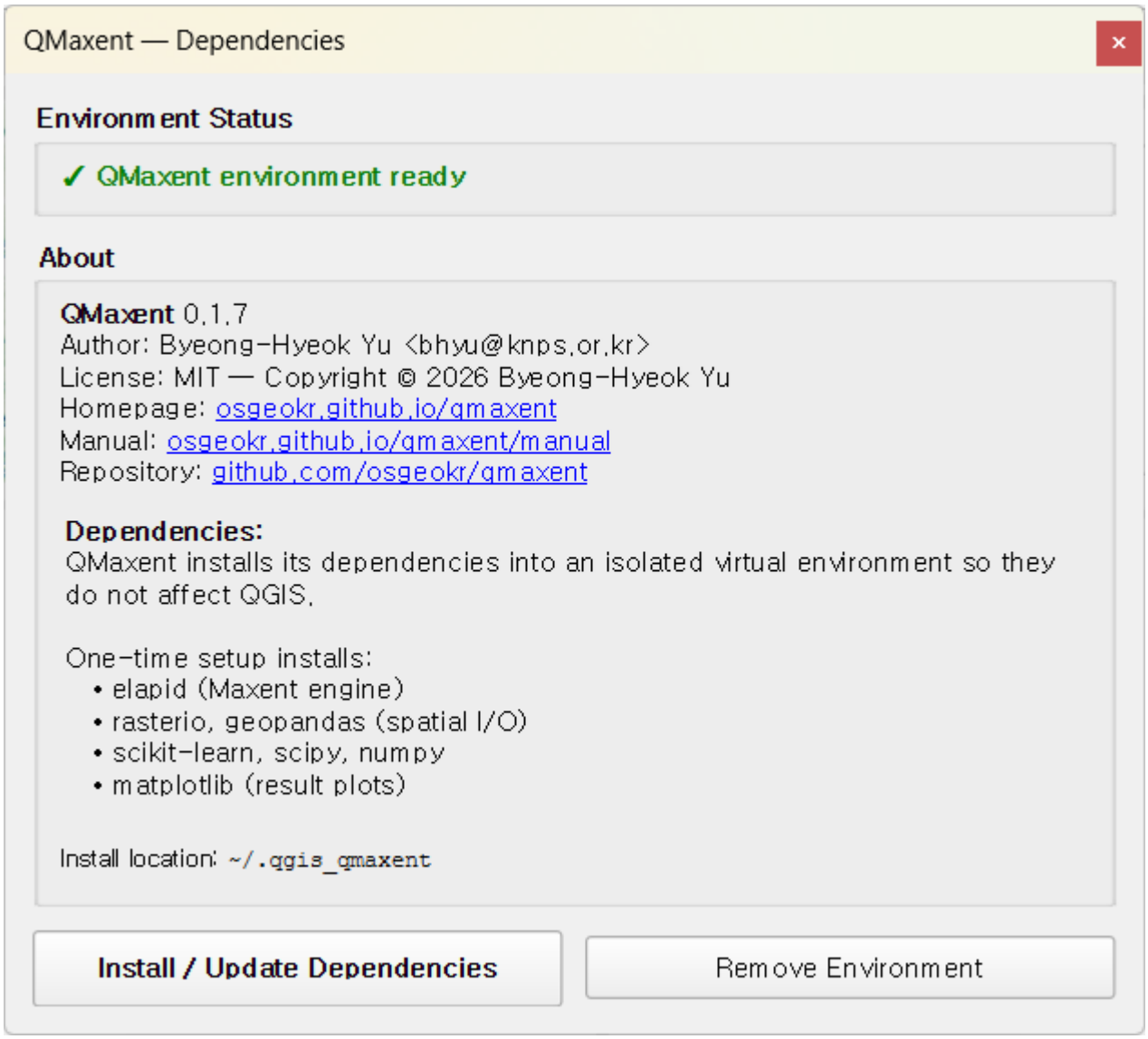
Banner	Meaning
✗ **Dependencies not installed**	First-time setup needed (the case above)
✓ **QMaxent environment ready**	Everything is installed and ready to model
✗ **Dependencies need updating**	The package list or the installation method changed with a plugin upgrade; click **Install / Update Dependencies** again

4.3 Install / Update Dependencies

Click **Install / Update Dependencies**. The plugin downloads the packages from PyPI into the path shown at the bottom of the dialog (default: `~/.qgis_qmaxent` on macOS/Linux, `%USERPROFILE%\.qgis_qmaxent` on Windows). Total download is approximately **300–500 MB** depending on your platform (about 590 MB on disk once installed); allow 3–8 minutes on a typical broadband connection.

When the installer finishes, the status banner turns green:

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You can now close the dialog and open **QMaxent Analysis** — the modeling workflow is unlocked.

4.4 What is installed

QMaxent installs the following packages and their transitive dependencies:

- **elapid** — the Maxent engine (maxnet algorithm, spatial cross-validation, projection); see Anderson (2023).
- **rasterio** — raster I/O.
- **geopandas** — vector I/O for presence points.
- **scikit-learn** — cross-validation, ROC, AUC; see Pedregosa et al. (2011).
- **scipy** + **numpy** — numerical core; see Harris et al. (2020).
- **matplotlib** — response curves, ROC, jackknife plots.

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These are pinned to versions that QMaxent has been tested against; the plugin's release notes record which versions ship with each release.

4.5 Where the packages are installed

All dependencies go into one folder per QGIS Python version, inside `~/.qgis_qmaxent` (`%USERPROFILE%\.qgis_qmaxent` on Windows). The folder name includes the Python version, for example `venv_py3.12` for the Python 3.12 that ships with current QGIS releases.

Platform	How the packages are installed	Package location
Windows	The QGIS Python's own pip installs the packages into the QMaxent package folder (`pip install --target`). No separate Python interpreter is created.	`%USERPROFILE%\.qgis_qmaxent\venv_py3.12\Lib\site-packages`
Linux, macOS	A Python virtual environment (`venv`) is created from the QGIS Python, and its pip installs the packages.	`~/.qgis_qmaxent/venv_py3.12/lib/python3.12/site-packages`

QMaxent only imports these packages inside QGIS, so on Windows it does not need an interpreter of its own. This change was made in 0.2.0: with QGIS 3.44.15 and later on Windows, the virtual-environment approach of 0.1.x stopped with *"ensurepip … returned non-zero exit status 1"* before any package was downloaded.

Two environment variables change this behaviour. Set them before you start QGIS:

- `QMAXENT_CACHE_DIR`: use another folder instead of `~/.qgis_qmaxent`, for example when your home directory is not writable or has little free space.
- `QMAXENT_FORCE_VENV=1`: on Windows, go back to the old virtual-environment method of 0.1.x. Use it only for troubleshooting.

4.6 Remove Environment

Use **Remove Environment** to delete the entire QMaxent package folder (the virtual environment on Linux and macOS) from disk. You might do this to free disk space, force a fresh re-install when something is broken, or switch to a different QMaxent version that pins different package versions.

After removal, the status banner returns to *Dependencies not installed* and **Install / Update Dependencies** becomes available again.

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4.7 Troubleshooting

Installation fails with a network or SSL error

Most often this is a corporate proxy or firewall blocking PyPI. Ask your administrator to allow `pypi.org` and `files.pythonhosted.org`. The installer respects QGIS's proxy settings, so configuring those in **Settings → Options → Network** usually fixes it.

Windows: 'ensurepip ... returned non-zero exit status 1'

You are running QMaxent 0.1.x with QGIS 3.44.15 or later. Upgrade the plugin to 0.2.0 or later, which installs the packages without a virtual environment on Windows, then click **Install / Update Dependencies** again.

Disk space exhausted during install

The installer needs roughly twice the final size as scratch space while extracting wheels. Free at least 1 GB on the drive that holds your home directory before retrying.

"No module named ..." after a successful install

Restart QGIS once. Python's import cache occasionally needs a clean process to pick up the new environment.

Plugin starts but Run Maxent button is disabled

Open the **Dependencies** dialog and confirm the status is green. If it is, restart QGIS; if it is not, click **Install / Update Dependencies** again.

4.8 Security note on the package folder

QMaxent's package folder is populated only from PyPI, by the QGIS Python's pip on Windows and by a standard Python `venv` on Linux and macOS. The plugin does not execute arbitrary code from remote sources at runtime. Saved-model files (`.pkl`) are a separate concern; see Saving and reusing models for the security note on loading pickle files from untrusted sources.

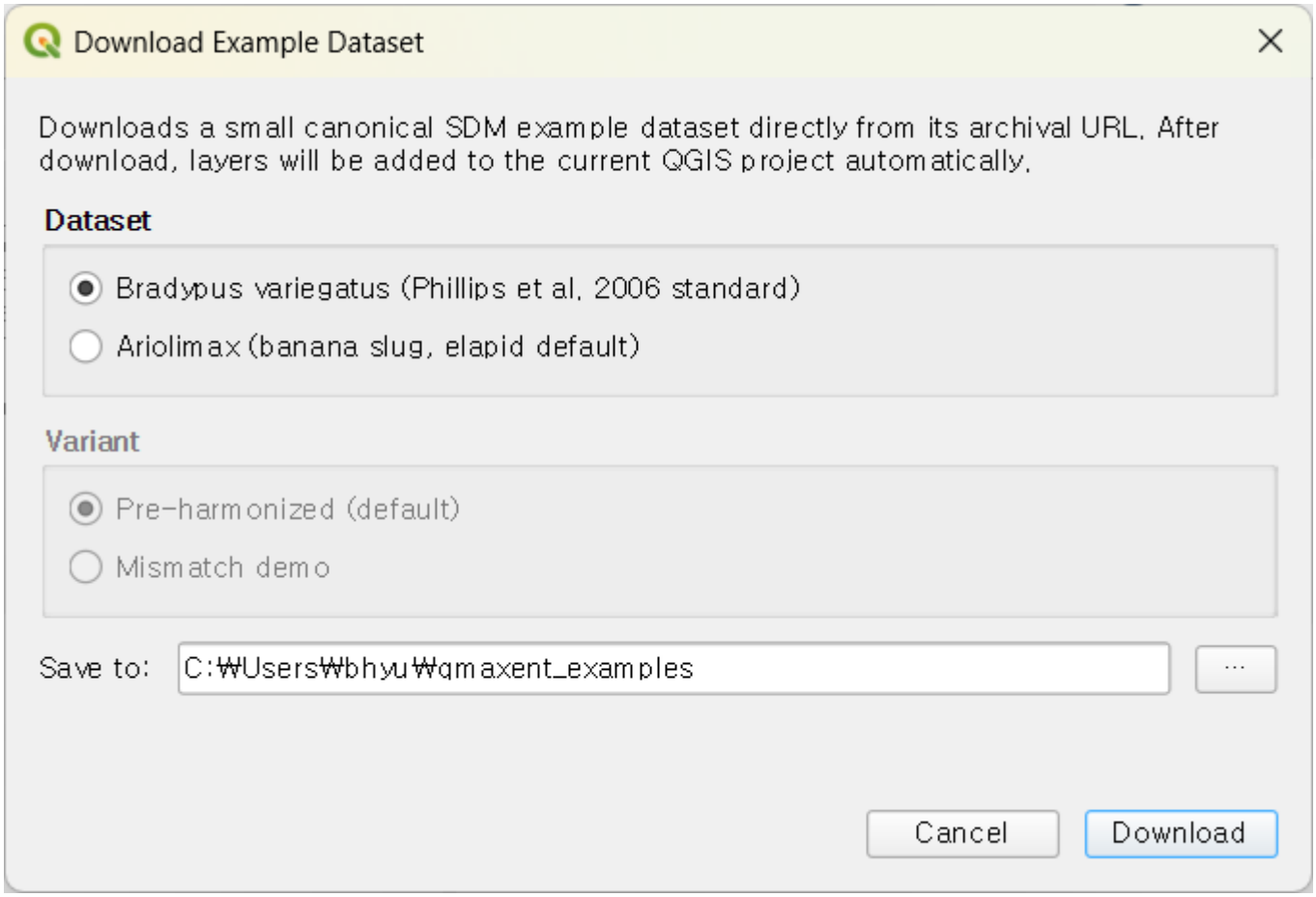
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5. Example datasets

QMaxent ships a one-click downloader for two canonical SDM example datasets, accessed from **Plugins → QMaxent → Download Example Dataset**. They are the same datasets the broader Maxent literature has used for almost two decades, which means you can compare your results directly with published benchmarks.

5.1 Opening the dialog

Choose **Plugins → QMaxent → Download Example Dataset**.



Two datasets are offered. Pick one of the radio buttons, choose where to save the files (the default is a `qmaxent_examples` folder under your home directory), and click **Download**. The plugin fetches the data, unpacks it, and **automatically adds the layers to your current QGIS project** so you can start modeling right away.

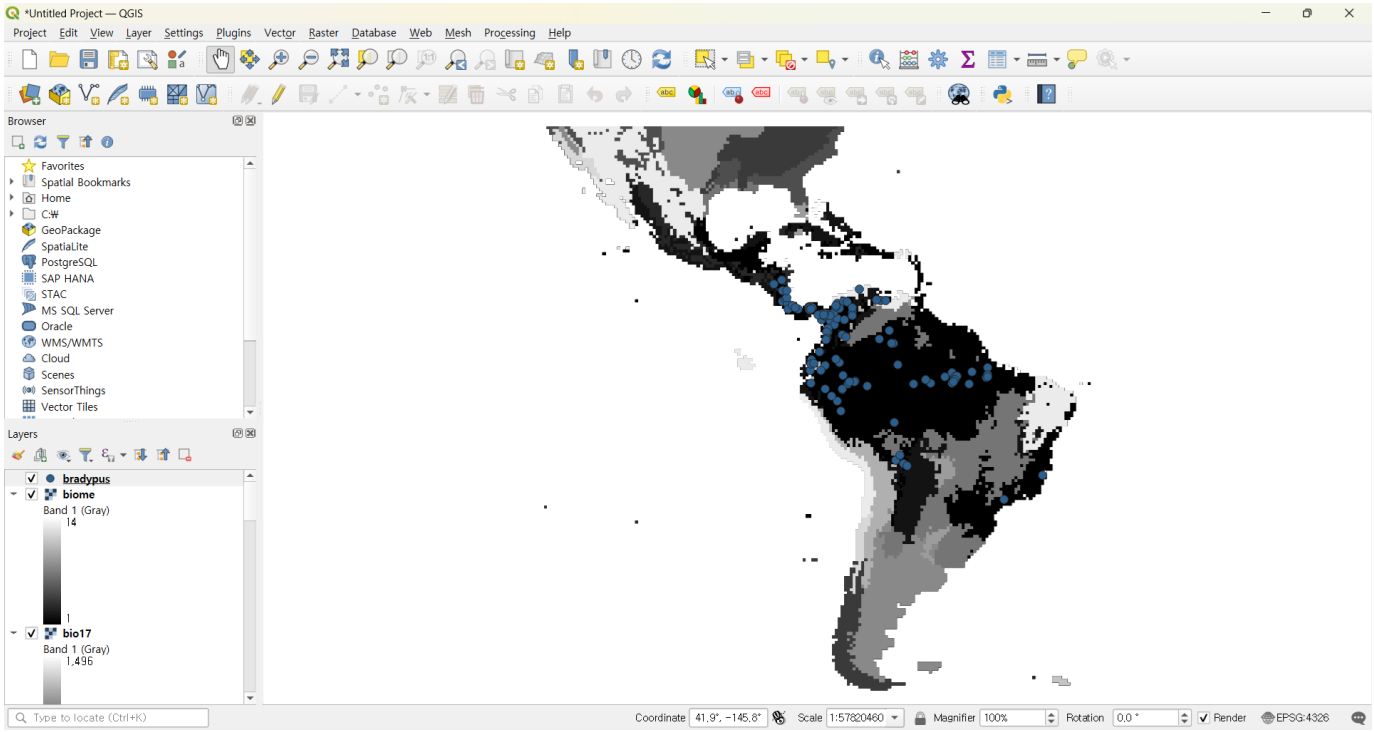
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5.2 Bradypus variegatus dataset

The brown-throated three-toed sloth. This is the canonical Maxent test dataset published with Phillips, Anderson & Schapire (2006) and reused in virtually every subsequent Maxent paper.

Layer	Description
`bradypus.shp`	116 presence points across South and Central America
`bio1, bio5, bio6, bio7, bio8, bio12, bio16, bio17`	WorldClim bioclimatic variables (continuous)
`biome`	Biome classification (categorical)

After download the layers appear immediately in the QGIS canvas:



The dataset is small (well under 100 MB) and the resulting Maxent runs typically finish in under 30 seconds, so it is ideal for first runs and for working through this manual.

5.3 Ariolimax dataset

The Pacific banana slug. This is the default test dataset that ships with elapid.

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About the Ariolimax download vs the Ariolimax worked example

The official Ariolimax download is **already harmonised** — all six rasters share a common CRS, extent, and resolution out of the box. So when you load it, **Check Raster Consistency** will pass immediately and you will not see the mismatch problem the Ariolimax worked example walks through.

The worked example uses a **deliberately desynchronised version** of the same data — the author intentionally reprojected and resampled some rasters to mimic the silent-failure scenario that SDM users frequently encounter in real fieldwork. If you want to follow that example exactly, work from the desynchronised dataset described in the example chapter; the bundled download will not exhibit the mismatch.

5.4 Save location and project structure

Whichever dataset you choose, the downloader creates a sub-folder named after the species:

```
qmaxent_examples/
├── bradypus/
│   ├── bradypus.shp     (and .dbf, .shx, .prj, .cpg)
│   ├── bio1.tif         …
│   └── biome.tif
└── ariolimax/
    ├── ariolimax.shp
    └── …
```

Re-running the dialog with the same Save-to path overwrites previously downloaded files, so it is safe to use as a "reset to defaults" button if you want to start a tutorial cleanly.

5.5 Loading the layers manually

The plugin auto-adds layers, but if you ever need to reload them by hand, just drag the `.shp` and `.tif` files from QGIS's **Browser** panel into the **Layers** panel — exactly as you would for any other dataset.

5.6 Where the data comes from

Both datasets are pulled from their canonical archival URLs at download time; QMaxent does not ship them inside the plugin itself. This keeps the plugin package small and ensures you always get the upstream-canonical version. If you need offline access, run the downloader once on a connected machine and copy the `qmaxent_examples` folder onto offline workstations.

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6. Quick start

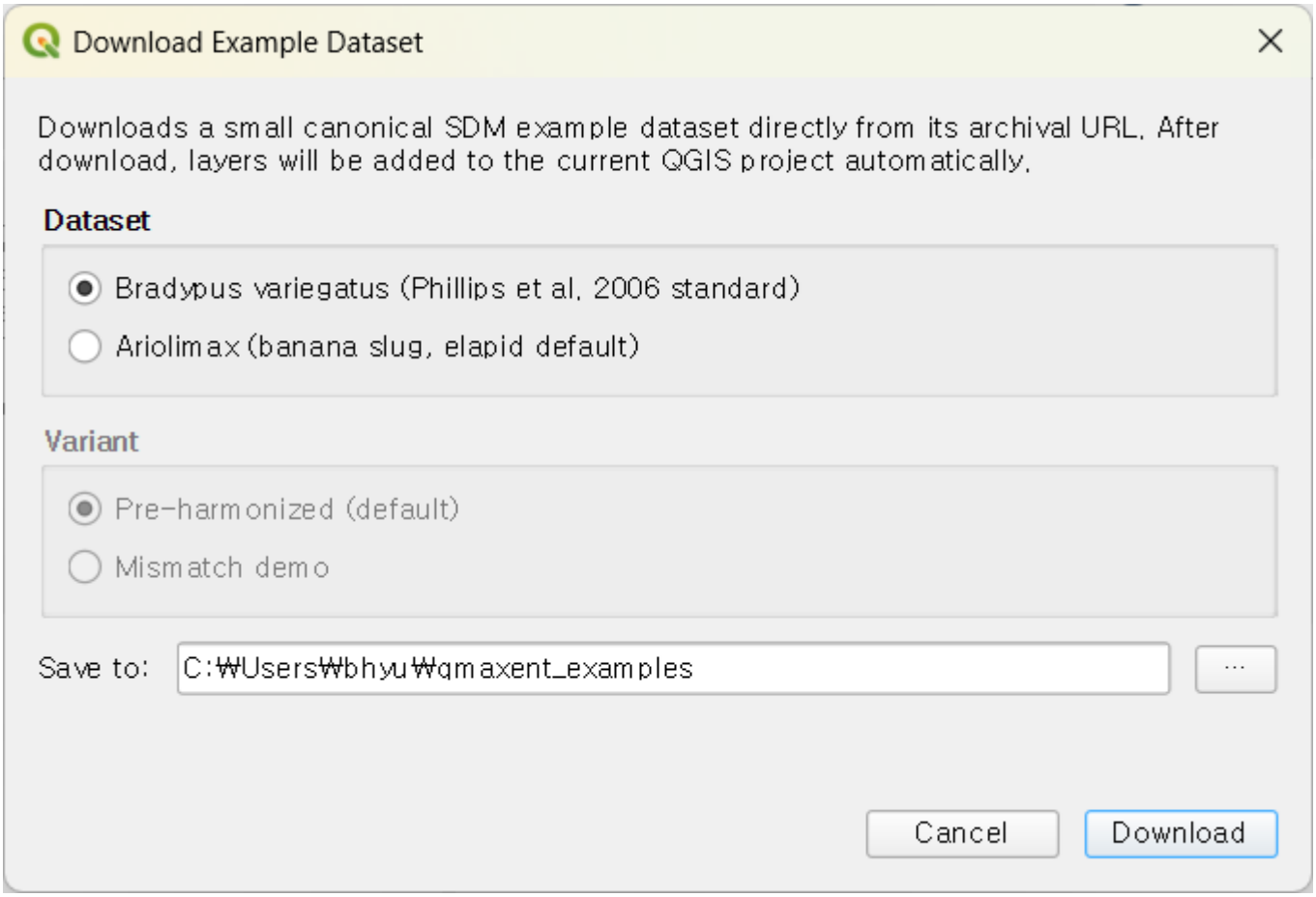
This 5-minute walkthrough takes you from a freshly installed plugin to a finished habitat-suitability map for *Bradypus variegatus*, the brown-throated three-toed sloth — the same species the original Maxent paper used Phillips, Anderson & Schapire 2006. Every screenshot below shows real numbers from a run you can reproduce.

Prerequisites

Make sure you have completed Installation and Dependencies — the **QMaxent environment ready** banner must be green before continuing.

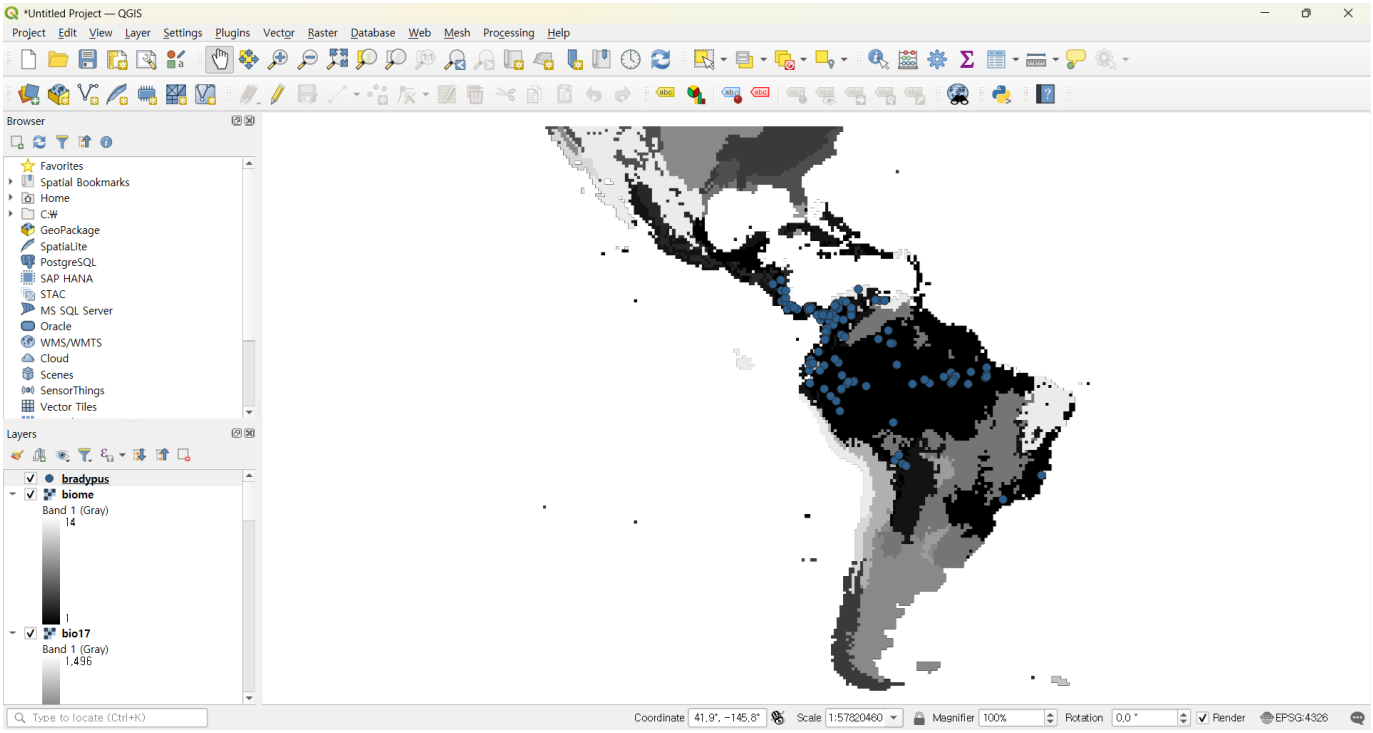
6.1 Step 1 · Download the example dataset

Open **Plugins → QMaxent → Download Example Dataset**, choose **Bradypus variegatus**, and confirm.



QMaxent fetches the bundled dataset (presence shapefile + 9 environmental rasters at 0.5° resolution covering the Americas) and adds every layer to the active QGIS project automatically.

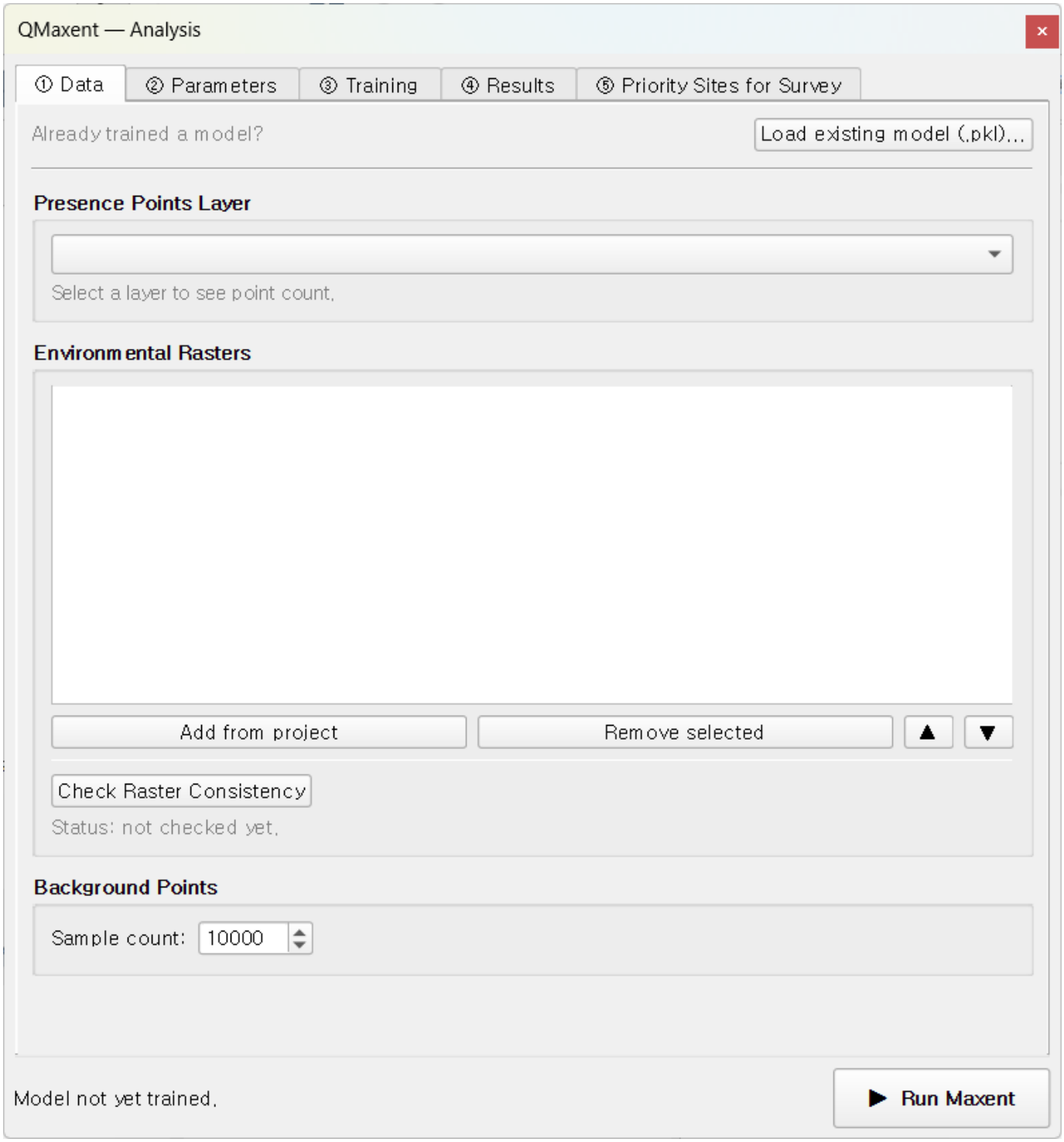
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6.2 Step 2 · Open the Analysis dock

Choose **Plugins → QMaxent → QMaxent Analysis**. The dock opens on the right of the QGIS main window with five numbered tabs you progress through left-to-right.

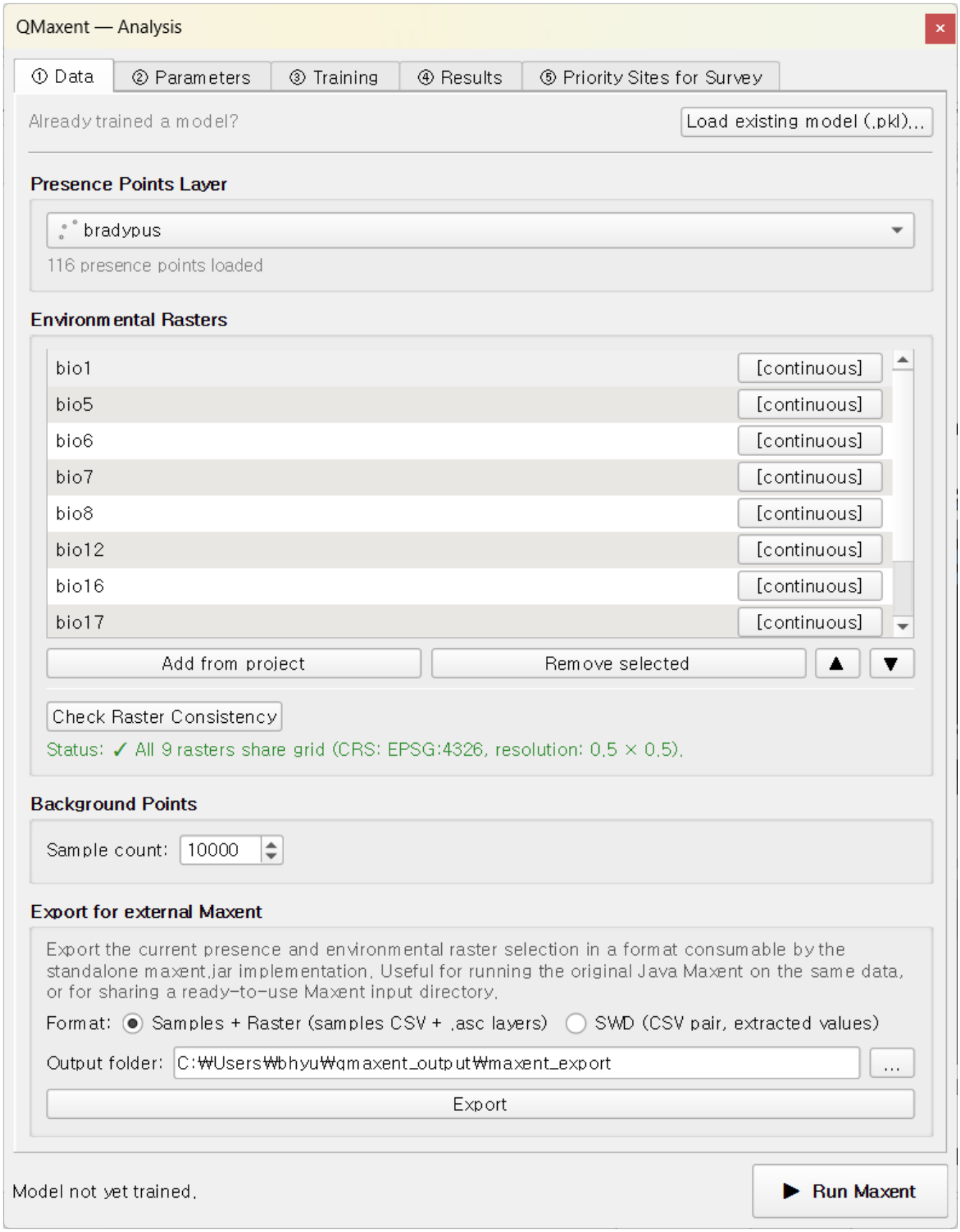
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6.3 Step 3 · Configure the Data tab

On ① **Data**, pick `bradypus` from the **Presence Points Layer** drop-down (QMaxent reports `116 presence points loaded`), click **Add from project** to register every loaded raster at once, then click **Check Raster Consistency** to verify the stack.

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The status line should turn green: `✓ All 9 rasters share grid (CRS: EPSG:4326, resolution: 0.5 × 0.5)`.

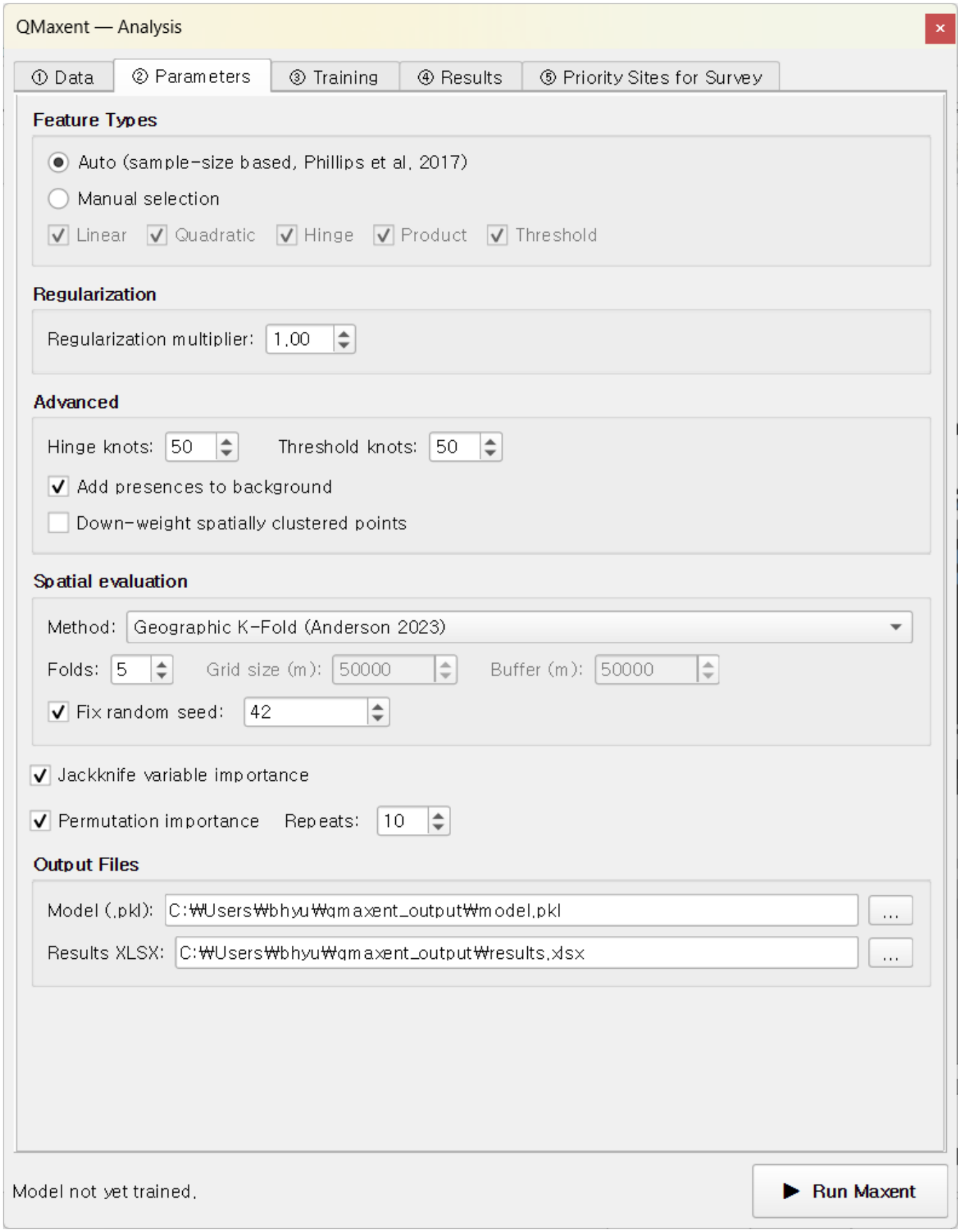
6.4 Step 4 · Accept the Parameters defaults

Switch to ② **Parameters**. For this quick tour, accept the defaults. They are sensible starting values from the SDM literature (Phillips & Dudík 2008, Phillips et al. 2017), not a guarantee of the best model for your data:

- **Feature classes**: **Auto (maxnet default rule by sample size)**, which gives LQPH (linear, quadratic, product, hinge) for 80 or more presences, so LQPH for the 116 Bradypus records
- **Regularization multiplier**: 1.0
- **Spatial CV**: Geographic K-Fold, 5 folds, **Fix random seed** 42 (the seed also fixes the background sample)
- **Jackknife variable importance**: enabled
- **Output files**: `qmaxent_output/model.pkl` and `qmaxent_output/results.xlsx`

The fixed random seed is what lets a reviewer **reproduce your numbers bit-for-bit** — central to the principle of computational reproducibility emphasised in the SDM-best-practice review of Araújo et al. 2019.

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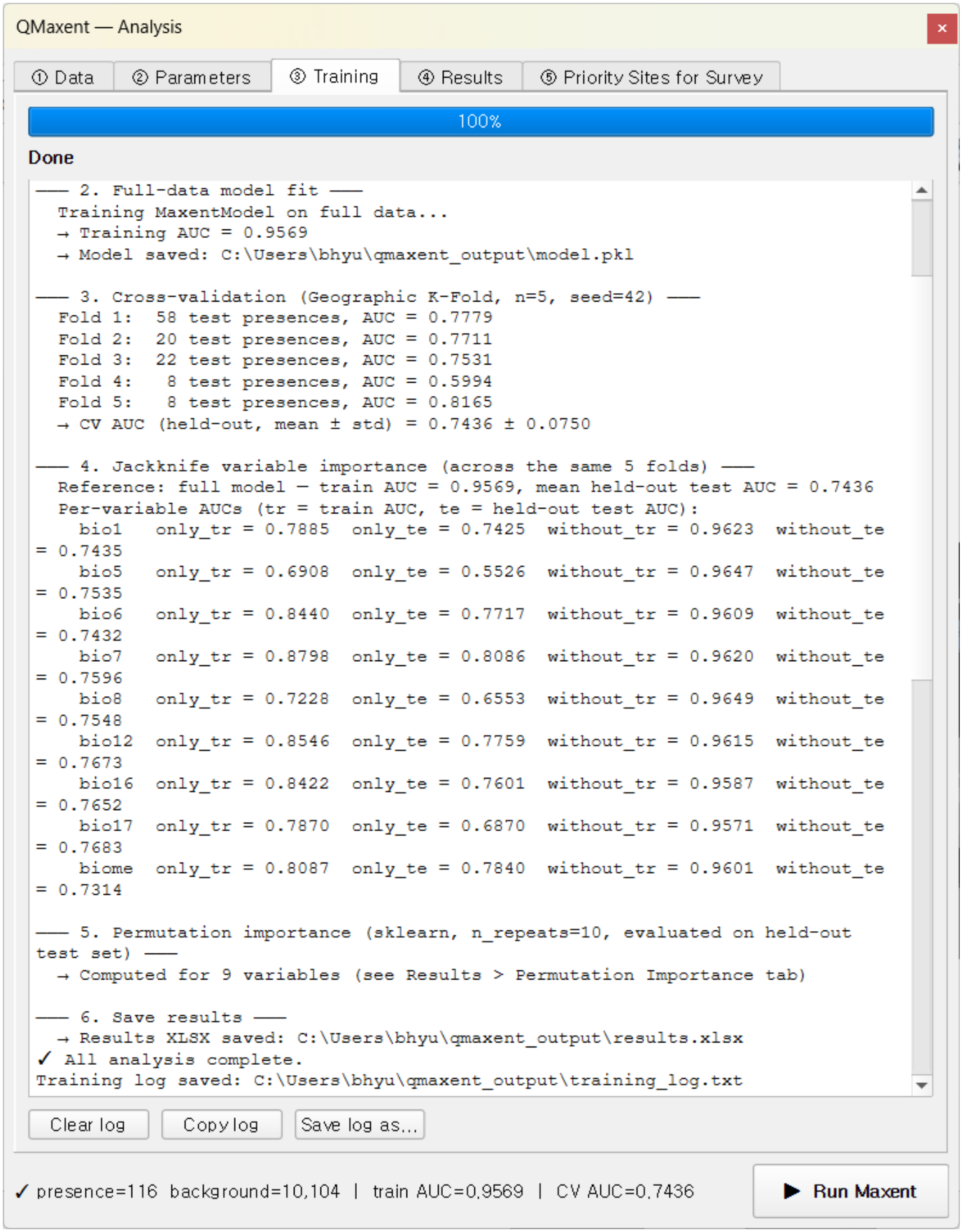


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6.5 Step 5 · Run Maxent

Click ▶ **Run Maxent** at the bottom of the dock. Focus shifts to the ③ **Training** tab; about 30 seconds later you should see:

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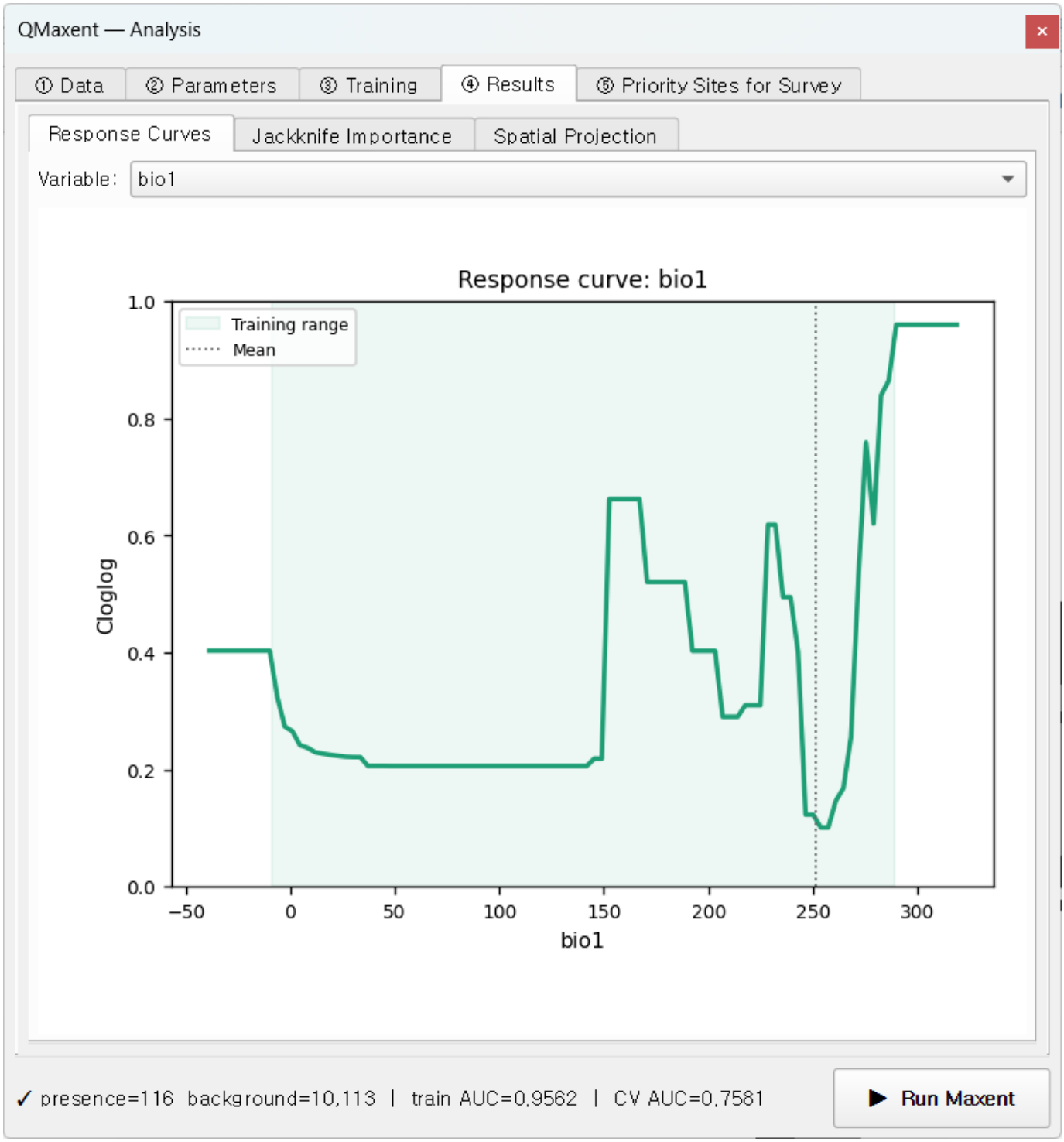
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The log reports a training AUC of about 0.96 and a 5-fold mean CV AUC of about 0.74 (the screenshot was recorded with an earlier version; exact values differ slightly in v0.2.0). The honest gap between training and CV AUC is itself a good sign — it tells you the model has not been silently over-fitted to the training presences (a risk that Roberts et al. 2017 document at length).

6.6 Step 6 · Inspect Response Curves and Jackknife

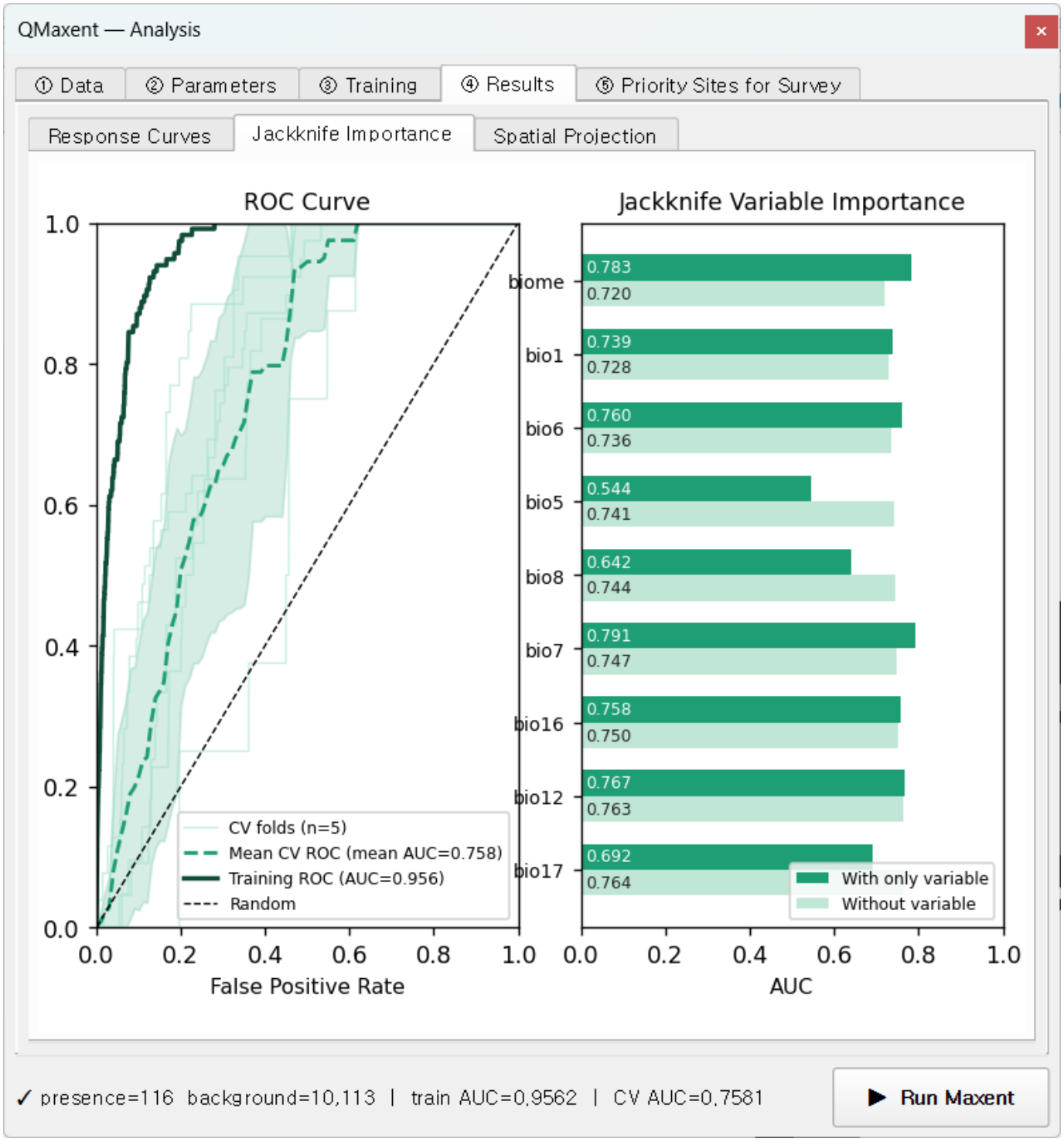
On ④ **Results → Response Curves**, pick `bio1` from the variable selector to see how the model's cloglog suitability responds to mean annual temperature across its training range.

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The **Jackknife Importance** sub-tab combines the ROC curve with per-variable contribution bars — the classic single-figure summary used in Maxent papers since Phillips, Anderson & Schapire 2006.

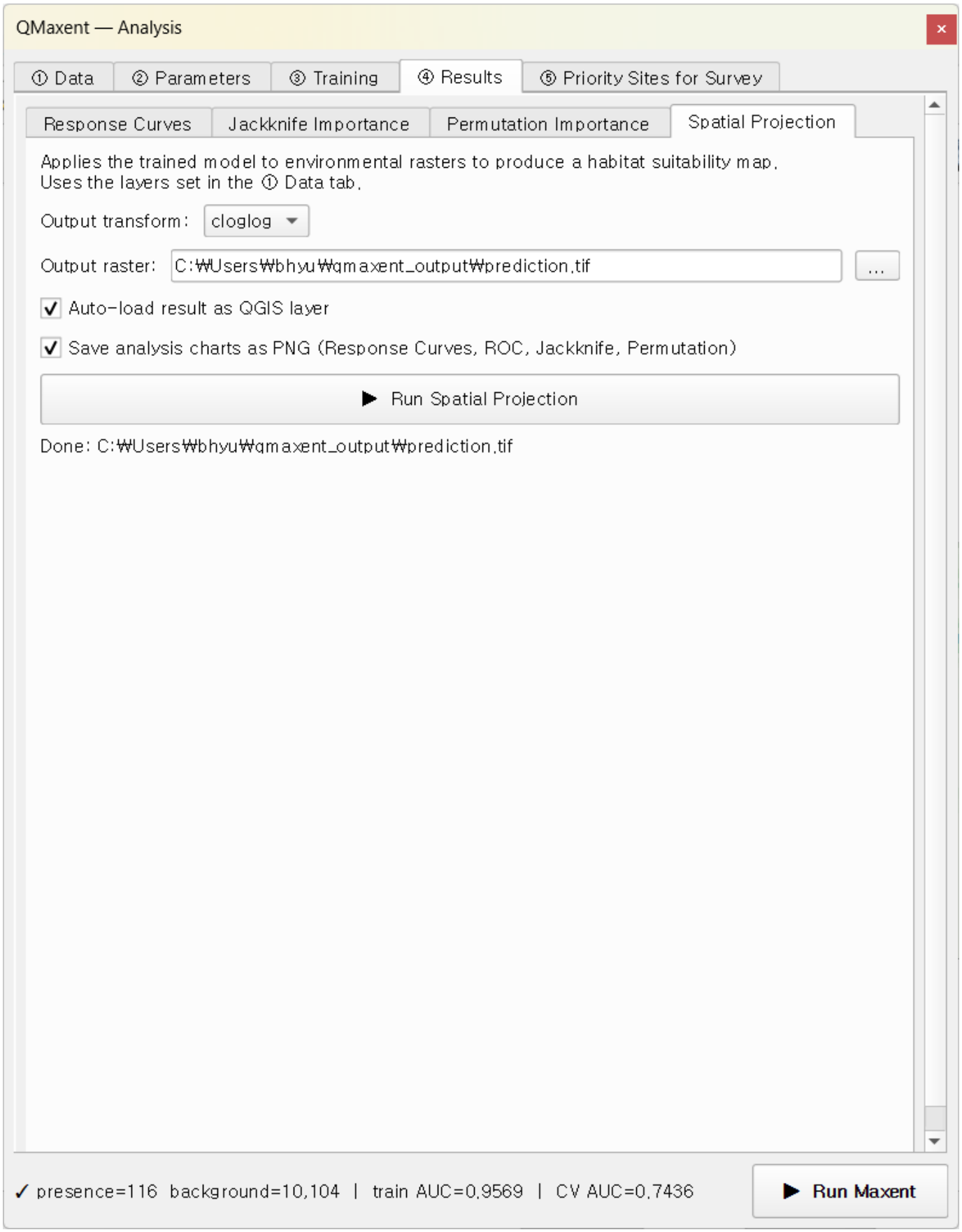
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6.7 Step 7 · Project across the landscape

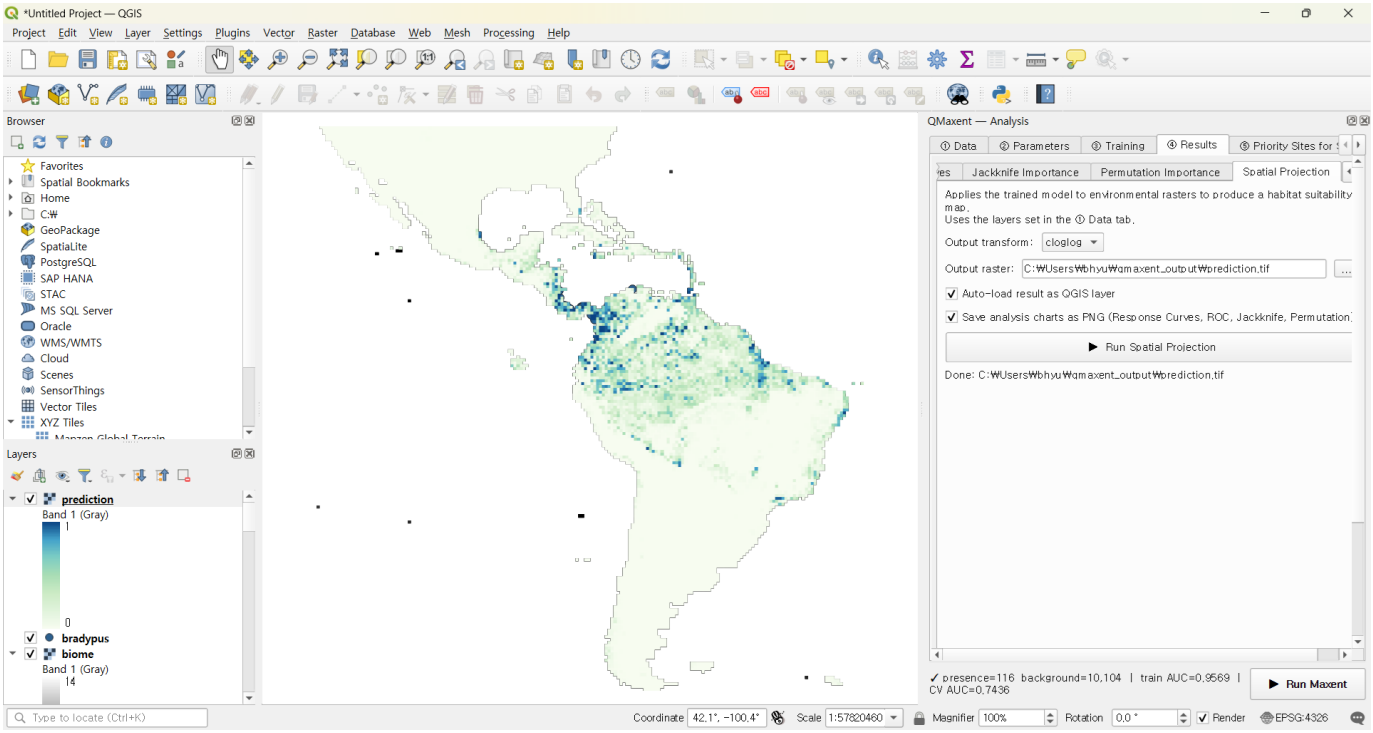
Switch to **Spatial Projection**. Leave **cloglog** as the output transform (Phillips et al. 2017 recommend cloglog as the default probability-of-presence interpretation) and **Auto-load result as QGIS layer** ticked, then click ▶ **Run Spatial Projection**.

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The styled GeoTIFF appears on the canvas:

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That is your finished QMaxent baseline.

6.8 What just happened

In seven clicks you ran a complete species-distribution-model pipeline: sampled background points, fit a regularized Maxent model with auto-selected features, evaluated it with spatial K-Fold CV, computed Jackknife variable importance, and projected the model across the entire dataset extent.

6.9 Next steps

- **Read the User guide chapters** in order to understand what each tab control does and when to override its default.
- **Try the Bradypus worked example** for the same workflow with deeper interpretation of every plot.
- **Move to messy data** with the Ariolimax example, which intentionally breaks raster alignment so you can practise the **Check Raster Consistency** + **Harmonize to Folder…** preflight.
- **Reproduce a published study** with the Pitta nympha example, which mirrors Lee et al. 2025 one parameter at a time.

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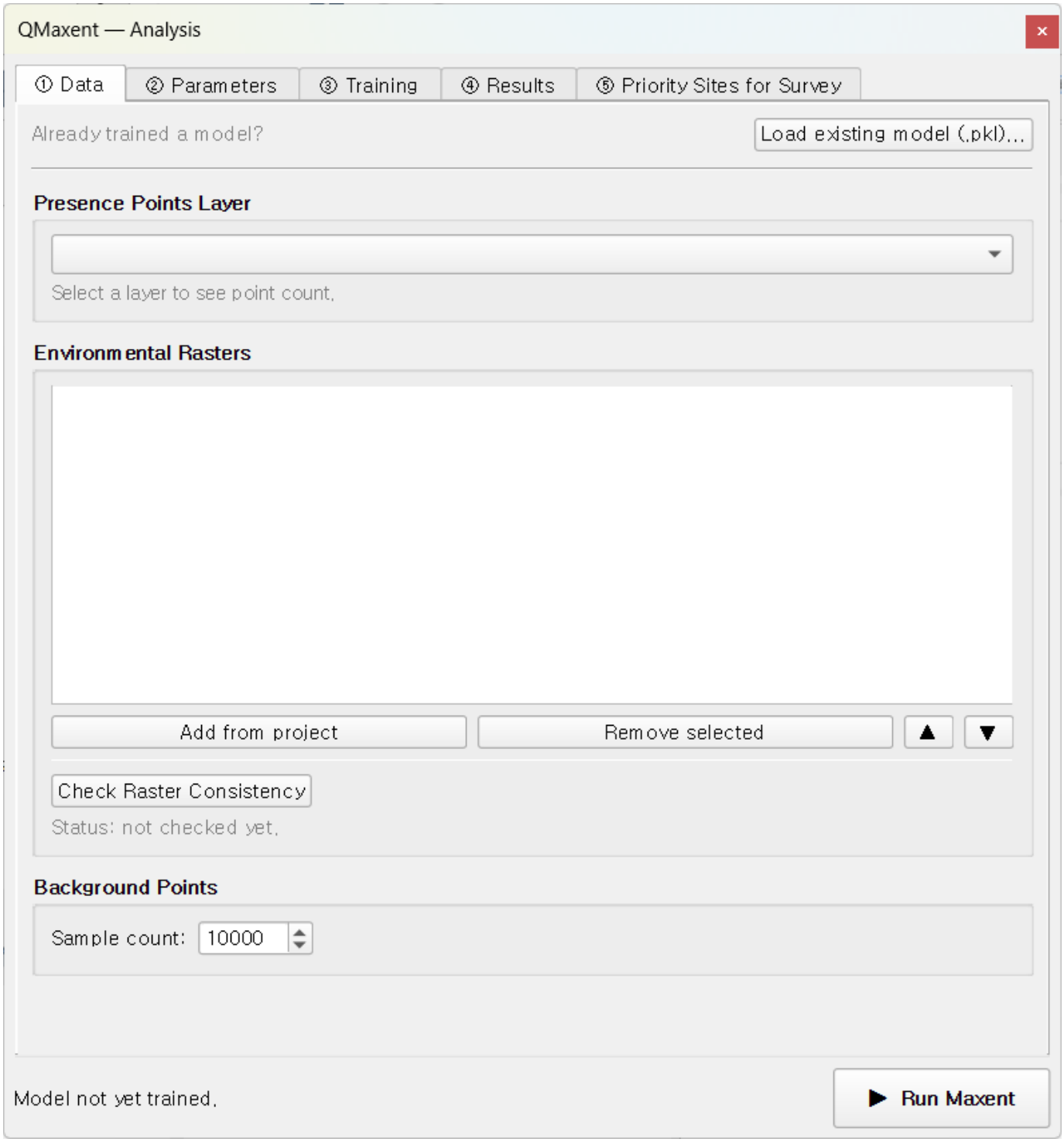
7. The Analysis dock

The QMaxent Analysis dock is the heart of the plugin. It organises the entire SDM workflow into **five numbered tabs** that you progress through in order, left to right. This chapter is a quick tour of the layout; each tab then has its own detailed chapter.

7.1 Opening the dock

Choose **Plugins → QMaxent → QMaxent Analysis**. The dock opens on the right side of the QGIS main window by default and can be detached, floated, or re-docked using the standard QGIS panel handles.

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7.2 The five tabs at a glance

#	Tab	Purpose	Detailed chapter
①	**Data**	Pick or load presence points (point file or CSV); register environmental rasters; pre-flight raster grid and CRS check	① Data tab
②	**Parameters**	Choose feature classes, regularization, spatial CV scheme, output paths	② Parameters tab
③	**Training**	Live progress bar and log while the model fits	③ Training tab
④	**Results**	Response curves, Jackknife importance, spatial projection	④ Results tab
⑤	**Survey Site Selection**	Turn the prediction into field-ready candidate survey sites	⑤ Survey Site Selection

The numbered glyphs are deliberate: even when QGIS is configured in a language other than English, the order is unambiguous.

7.3 Status bar

A persistent **status bar at the bottom of the dock** shows the most recent run summary — presence count, background count, training AUC, and CV AUC. It is updated continuously as you click around, so you can switch projects or reopen QGIS and immediately see whether the dock is showing a fresh state or a previous run's results.

7.4 Persistence across QGIS sessions

QMaxent remembers your last-used presence layer, raster set, parameter values, and output paths inside the QGIS project file (`.qgz`). When you reopen a project, the dock reflects the previous state — no need to re-pick layers each time.

7.5 Buttons across the bottom

The footer of the dock has three persistent buttons:

- ▶ **Run Maxent** — start a training run
- ▶ **Run Spatial Projection** — apply the trained model across all rasters
- ▶ **Select Survey Sites**: sample candidate survey sites from the prediction

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These buttons are **enabled only when their prerequisites are met**: e.g. projection becomes available only after a model has been trained, and survey site selection requires a projection raster on disk.

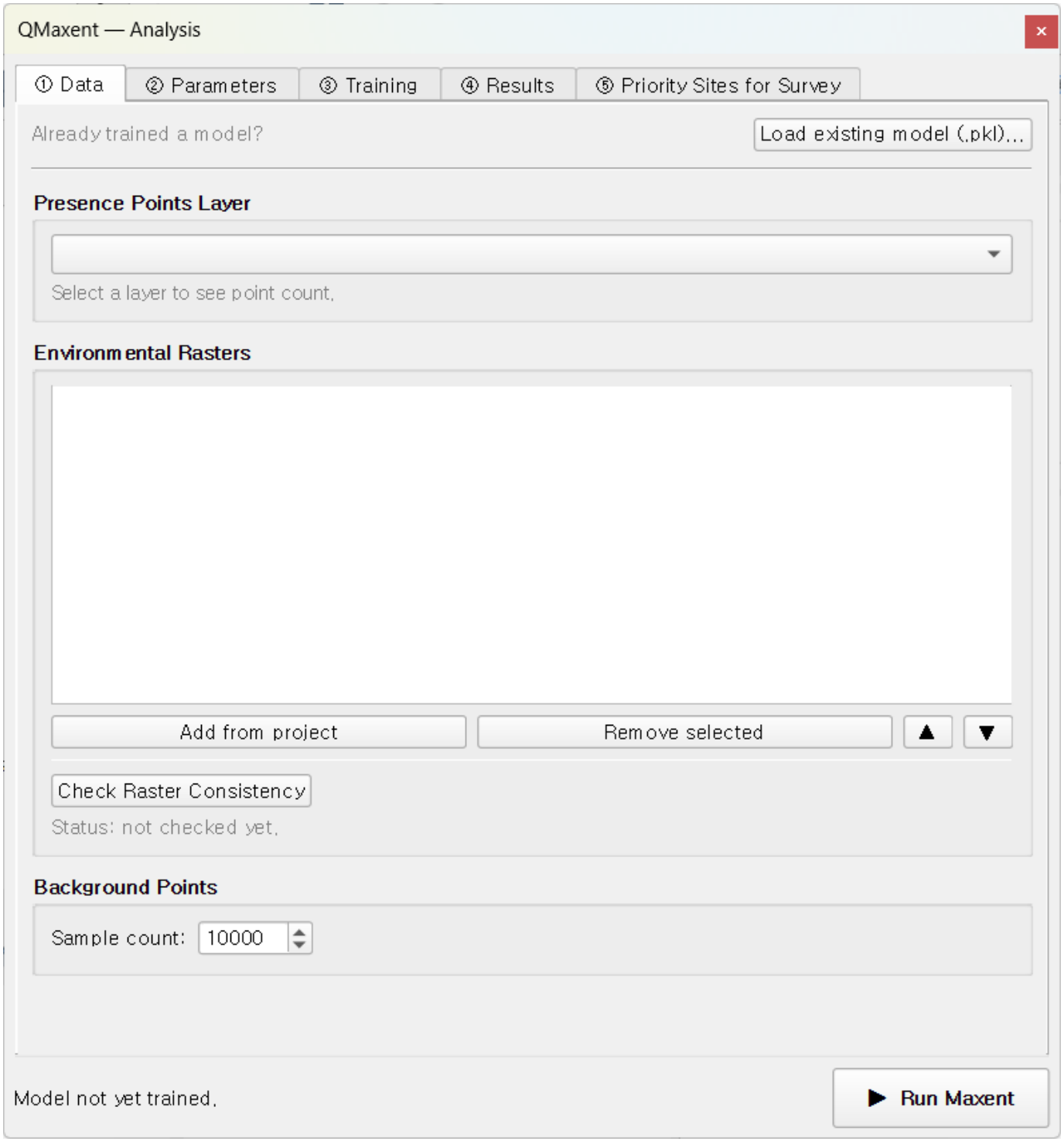
7.6 What this means for your workflow

The dock's design encodes the canonical Maxent workflow described in Elith et al. 2011: assemble inputs → choose hyperparameters → fit → evaluate → project → act. By making each stage a separate tab with its own status, QMaxent forces a discipline that helps catch the silent failures Roberts et al. 2017 and Araújo et al. 2019 warn about — particularly raster mismatch and over-optimistic AUCs from non-spatial CV.

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8. ① Data tab

The Data tab is where you tell QMaxent **which species** you are modeling and **which environmental variables** to use. Three controls deserve attention: the presence-points selector, the environmental-raster list with its **Check Raster Consistency** action, and the background-points sample size.



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8.1 Presence Points Layer

The first drop-down lists every loaded vector point layer in the current QGIS project. Pick the layer that contains your species occurrence records. QMaxent reads the geometry and reports the count beside the selector (e.g. `116 presence points loaded`).

If the records are not in the project yet, two buttons below the drop-down load them without a detour through the QGIS menus. The new layer is added to the project and selected as the presence layer.

- **Open point file…**: any point format QGIS can read (GeoPackage, Shapefile, GeoJSON, KML, GML, FlatGeobuf and others). A file that does not contain points is refused with a message.
- **Load CSV (X/Y columns)…**: a delimited table with one row per record, for example a GBIF download. QMaxent detects the delimiter (comma, semicolon, tab or pipe) and suggests the longitude/latitude or X/Y columns from their names (for example `decimalLongitude` and `decimalLatitude` in GBIF files, or `lon` / `lat`, `x` / `y`). Check the suggested columns and choose the coordinate reference system of the coordinates. The default is EPSG:4326, which is correct for decimal-degree tables.

Tips for preparing this layer:

- **Geometry** must be `Point` or `MultiPoint`. Lines and polygons must be converted to centroids first via **Vector → Geometry Tools → Centroids…**.
- **CRS** can be anything — QMaxent reprojects on the fly to match the raster grid. That said, a projected CRS that matches the rasters avoids a mid-run reprojection step.
- **Duplicates and clustering** are tolerated but distort the model. Boria et al. 2014 recommend **spatial thinning** before modeling; consider running QGIS plugin *NNJoin* or the standalone tool *spThin* before using QMaxent.

8.2 Environmental rasters

Two buttons populate the raster list:

- **Add from project**: adds every raster layer currently loaded in the QGIS project. The fastest way to start.
- **Add raster files…**: opens a multi-select file dialog for any single-band raster GDAL can read (GeoTIFF, ESRI ASCII `.asc`, `.grd`, ERDAS `.img`, VRT, NetCDF and others). The files are added to the QGIS project and to the list in one step.

The list follows the QGIS project: when a raster layer is removed from the project (or you open another project), its row disappears from the list and the status line asks you to run **Check Raster Consistency** again. Use the ▲ and ▼ buttons to change the order of the rows. The order matters for reproducibility, because background points are drawn from the first raster in the list (see Background points).

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QMaxent uses **one band per environmental variable**. A multi-band raster (for example a stack of all bioclimatic variables in one file) is refused when you click ▶ **Run Maxent**, with the names of the affected files. Split it into single-band files first, for example with **Processing > GDAL > Raster conversion > Rearrange bands** (one band per output), then add the single-band files.

Each row in the raster list shows the file name and a `[continuous]` / `[categorical]` tag. **Categorical detection is automatic** when the file carries a PAM (`.aux.xml`) sidecar with the metadata key `LAYER_TYPE=thematic`. Without that tag, click the tag button of a row to switch it between `[continuous]` and `[categorical]`.

> 🔥 **Why categorical handling matters**
>
> Treating a categorical raster (e.g. biome, land-cover class) as continuous is one of the most common silent failures in operational SDM. Maxent will average across class codes as if they were ordinal, producing biologically meaningless responses. Always verify the `[categorical]` tag on integer-coded landscape rasters before training.

8.3 Check Raster Consistency

This button runs the most important pre-flight check in the dock: do all rasters share the **same CRS, extent, and resolution**?

- ✓ **Pass** (green): the stack is grid-aligned and ready to model.
- ⚠ **Mismatch** (amber): one or more rasters disagree.
- **Unusable CRS**: one or more rasters have a coordinate reference system that cannot be reprojected (see below).

When it fails, a second button — **Harmonize to Folder…** — appears. Clicking it lets you nominate an output directory; QMaxent reprojects every raster to the highest-resolution one in the stack (nearest-neighbour for categoricals, bilinear for continuous variables) and re-loads the harmonised set. The Ariolimax worked example walks through this in detail.

Rasters with an unusable CRS

Some rasters cannot be aligned at all because of their CRS. The check reports two cases and names each affected file:

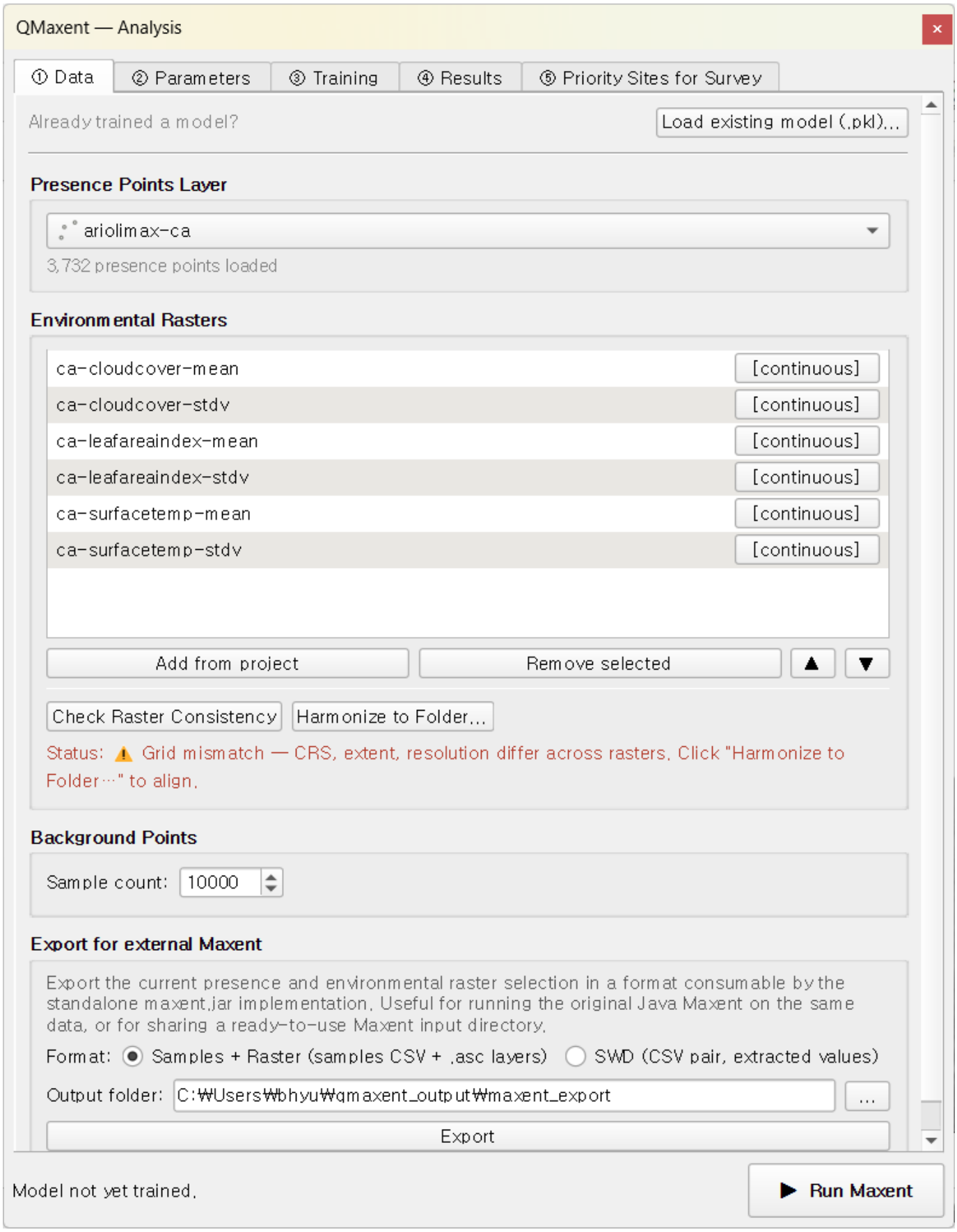
- the raster has **no CRS**, or
- the CRS is a **local (engineering) system without a datum**. This happens when a `.prj` file or GeoTIFF only carries the name of a projection, for example `NAD83(NSRS2007) / Michigan South`, and no datum. PROJ cannot transform such a CRS to any other CRS.

When the name matches an EPSG definition, the message suggests the matching EPSG code. To fix it, in the QGIS **Layers** panel right-click the raster, choose **Layer CRS > Set Layer CRS…**, select the correct CRS

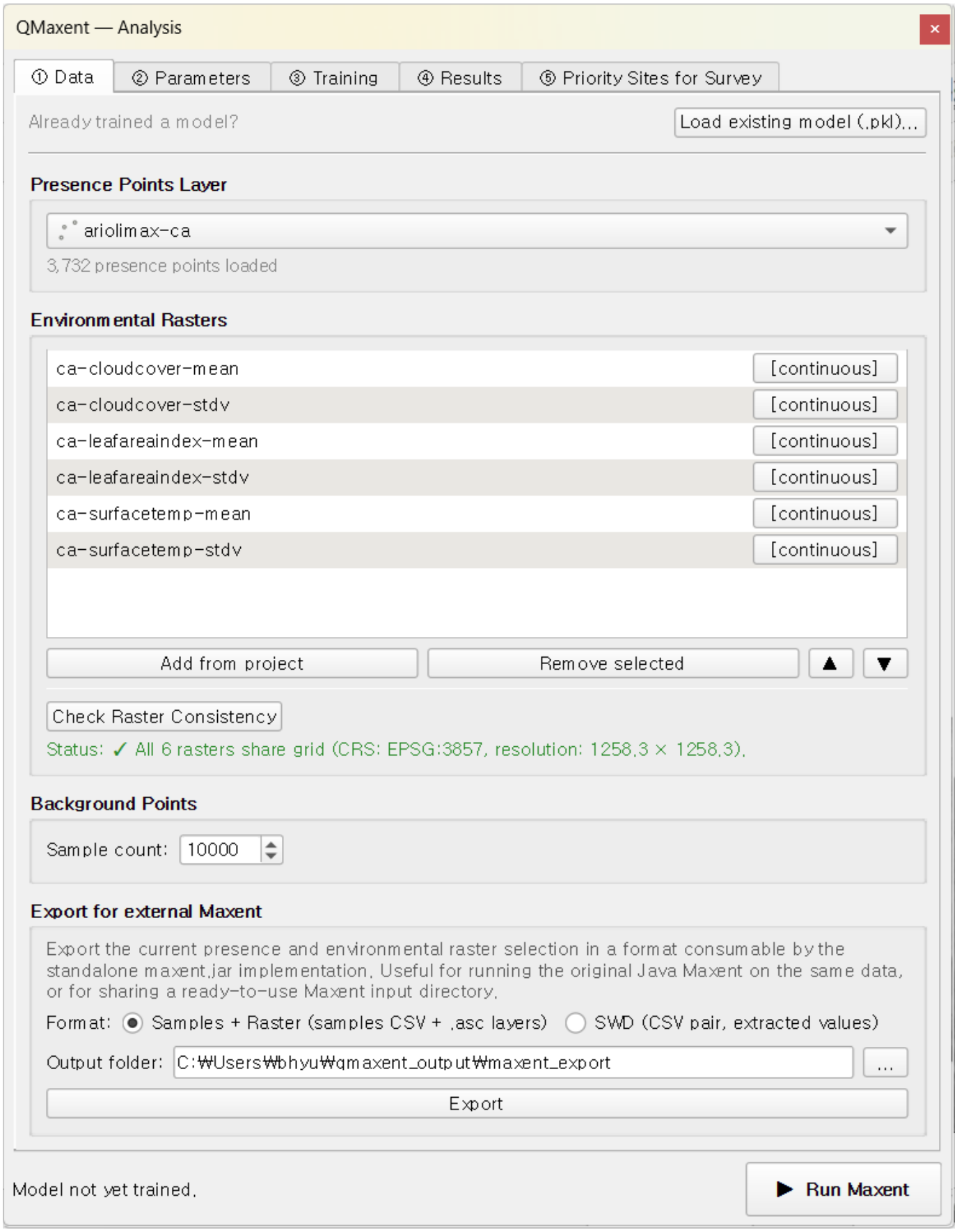
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(for example the suggested EPSG code), and click **Check Raster Consistency** again. From then on, Check and **Harmonize to Folder…** use the CRS you set in QGIS for that raster. Alternatively, write a complete CRS into the file, for example with **Raster > Projections > Assign Projection**. Harmonize no longer stops with a raw PROJ error in this situation.

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8.4 Background points

Maxent is a **presence-background** algorithm (Phillips et al. 2006; Fithian & Hastie 2013) — it contrasts presence locations against a background sample drawn from the rest of the study area. The default background sample is **10,000 points** drawn at random from the valid (non-NoData) cells of the **first raster in the list**, the value Phillips & Dudík 2008 recommend for typical continental- to landscape-scale studies.

The draw is covered by **Fix random seed** on the ② Parameters tab: with the same seed, the same rasters in the same order and the same presences, every run uses the same background sample. (In v0.1.x the background was drawn anew on every run, even with a fixed seed.) Keep the raster order unchanged when you want to reproduce a run exactly.

You can override it in the **Background points** field. Smaller values are appropriate only for very small study areas (<10,000 cells); larger values rarely help and slow the run.

The **Add presences to background** checkbox is **on by default** — this is the convention the original Java MaxEnt applies and is required for the Maxent likelihood to be well-defined (Phillips & Dudík 2008).

8.5 Load existing model (.pkl)…

The bottom of the tab has a **Load existing model (.pkl)...** button for re-using a previously trained model on new rasters. Picking a `.pkl` opens a variable-mapping dialog so you can re-state which raster maps to which saved variable name — a safeguard against silent ordering errors when collaborators use different file names. See Saving and reusing models for the full workflow and the security note on Python pickle files.

8.6 Status line

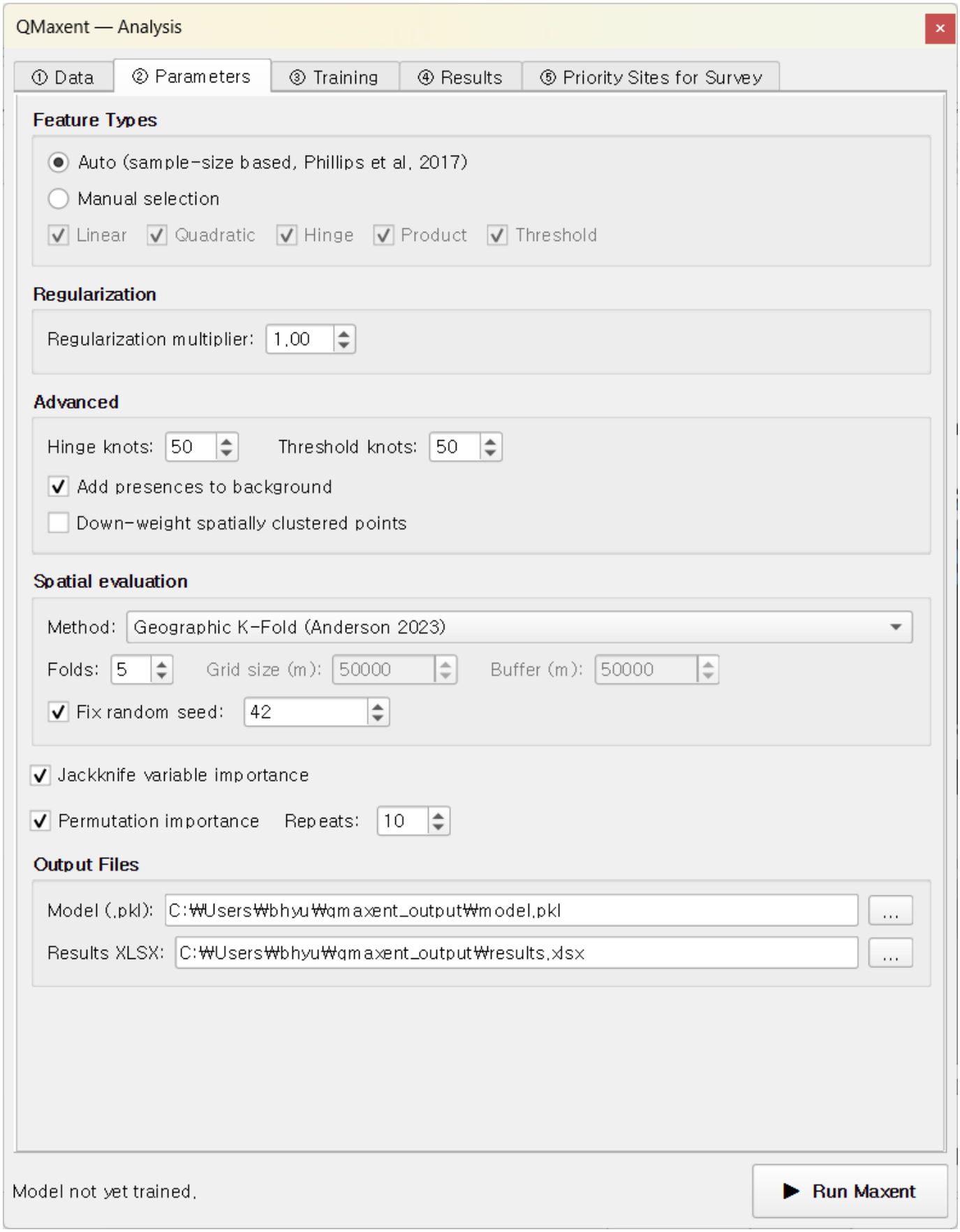
The bottom of the dock shows a one-line summary: `presence=N background=M train AUC=… CV AUC=…`. The AUC fields populate after the ③ **Training** tab finishes a run — they let you tell at a glance whether the dock state matches the most recent training.

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9. ② Parameters tab

The Parameters tab controls the *modeling* and *evaluation* settings: which Maxent feature classes to allow, the regularization multiplier, the spatial cross-validation method, and where to write outputs. Defaults follow established Maxent practice; this chapter explains what each control does and when to override the default.

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9.1 Feature Types

Maxent's "features" are the basis functions it uses to express the response of suitability to each environmental variable (Phillips, Anderson & Schapire 2006; Phillips et al. 2017). The five classes (LQPHT) are:

Feature	Symbol	What it captures
Linear	`L`	Monotonic linear response
Quadratic	`Q`	Optimum-shaped (∩) or U-shaped (∪) response
Product	`P`	Pairwise interactions between two variables
Hinge	`H`	Piecewise-linear "kink" — sharp changes at threshold values
Threshold	`T`	Step-function — a hard cut-off at some value

QMaxent offers two modes:

- **Auto (maxnet default rule by sample size)** (default): uses the default feature classes of maxnet, the R implementation of Maxent by its authors (Phillips et al. 2017; `maxnet.formula(classes = "default")`), based on the number of presence points:

Presences	Feature classes
fewer than 10	L
10 to 14	LQ
15 to 79	LQH
80 or more	LQPH

Threshold features are not added automatically. - **Manual selection**: tick any subset of LQPHT. The boxes start with L, Q, P and H ticked and Threshold unticked. Use this to compare alternatives or to reproduce a published study that fixed a particular feature combination (e.g. the Pitta nympha example fixes feature classes to **LQH** to match Lee et al. 2025).

> **Why Auto no longer adds threshold features**
>
> Up to v0.1.x, Auto followed the maxent.jar rule and added threshold features (LQPHT) for 80 or more presences. With elapid's scikit-learn backend, every feature gets the same regularization strength, while maxent.jar regularizes each feature class separately. Threshold features then overfit. On the Bradypus benchmark (random 10-fold CV), the test AUC was 0.873 with LQPHT and 0.890 with LQPH; maxent.jar with LQPHT reached 0.895. v0.2.0 therefore follows the maxnet default, which does not use threshold features. They remain available under **Manual selection**.

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> **Auto is a default, not the best model**
>
> The rule gives a reasonable starting point, but the most suitable feature classes depend on the data (Merow, Smith & Silander 2013; Radosavljevic & Anderson 2014). To choose them, compare alternatives with **Manual selection** and spatial cross-validation, or use a model-selection tool such as ENMeval (Muscarella et al. 2014). Be aware that repeatedly tuning settings against random (non-spatial) CV gives optimistic AUCs (Roberts et al. 2017).

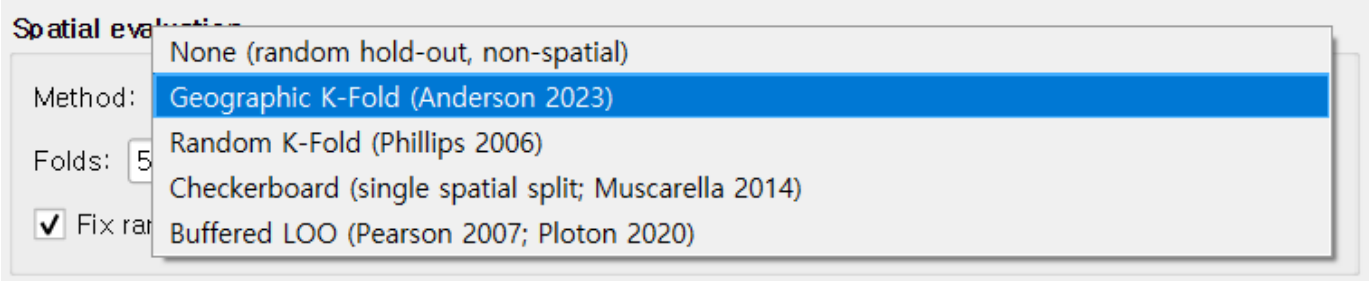
9.2 Regularization multiplier

Maxent fits a *regularized* maximum-entropy distribution — the regularization multiplier (RM) controls how strongly the fit is penalised for complexity. - **RM = 1** (default) — the value Phillips & Dudík 2008 showed yields the best held-out AUC across the full Maxent benchmark. - **RM > 1** (e.g. 2–4) — smoother responses, less over-fit, lower training AUC but often higher CV AUC. Use when occurrences are small in number, spatially clustered, or biased. - **RM < 1** — only justified by a formal hyperparameter search such as ENMeval (Muscarella et al. 2014) or Wallace (Kass et al. 2018).

9.3 Spatial cross-validation

A drop-down picks the spatial CV scheme used to split presences into folds:

Method	Best for	Reference
Geographic K-Fold (default)	General-purpose	Roberts et al. 2017
Random K-Fold	Reproducing studies that used random splits	Phillips et al. 2017
Checkerboard	Strong spatial autocorrelation in presences	Muscarella et al. 2014
Buffered Leave-One-Out	Very small datasets (n < 30)	Hijmans 2012; Valavi et al. 2019



The default **Geographic K-Fold (k = 5, fixed seed = 42)** strikes a balance recommended by Roberts et al. 2017: geographic blocks break the spatial-autocorrelation cheating that random K-Fold permits, yet keep the held-out folds large enough for a stable AUC.

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The fixed seed makes every run **bit-for-bit reproducible** — a property cited as essential by the SDM-best-practice review of Araújo et al. 2019.

9.4 Jackknife variable importance

When the **Jackknife** checkbox is on, QMaxent fits 2 × *N* additional single-predictor and leave-one-out models to estimate each variable's unique contribution. The cost is roughly *2 × N × training time*; for the Bradypus dataset (9 variables, 116 presences) this adds about 30 seconds.

Jackknife output is rendered in two places: 1. The ④ **Results → Jackknife Importance** sub-tab as a paired-bar plot. 2. **Table 4** of the exported XLSX workbook, with all four AUC columns plus computed *Train AUC drop* and *Test AUC drop*.

The interpretation follows the original Phillips, Anderson & Schapire 2006 reading: "with-only" bars show univariate signal; "without" bars show how dependent the full model is on each variable.

9.5 Permutation importance

When **Permutation importance** is ticked, QMaxent shuffles the values of each variable in turn and measures the drop in AUC (scikit-learn `permutation_importance`, **Repeats** shuffles per variable, default 10). The drops are reported as a percentage of the total, the convention of maxent.jar. The **Evaluate on** drop-down sets which data the AUC drop is measured on:

Option	What is evaluated	Use it to
Training data (maxent.jar convention) (default)	The full model on all training presences and background points, as maxent.jar does	Compare variable rankings with maxent.jar studies
Held-out CV test folds (all folds pooled)	Each cross-validation fold model on its own test fold; the results of all folds are pooled	See how much predictive performance at new locations depends on each variable

The held-out option needs **Geographic K-Fold**, **Random K-Fold** or **Checkerboard** cross-validation. With no CV or with Buffered LOO, QMaxent falls back to the training data and says so in the log. (v0.1.x used only the first CV fold, which made rankings unstable.) The training log and the model card record which option was used.

Permutation importance is fast (no refitting), but it can underestimate the importance of variables that are correlated with others. Read it together with the jackknife.

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9.6 Output paths

Two file paths default to a `qmaxent_output/` subdirectory of the current QGIS project:

- **Model file (.pkl)** — the serialised trained model, reloadable from the Data tab. QMaxent also writes `<model>_model_card.json` and `<model>_coefficients.csv` next to it. See Saving and reusing models for these files and the security note on Python pickle files.
- **Results workbook (.xlsx)** — multi-sheet supplementary table. See Exporting results for the sheet-by-sheet layout.

Clear either field to **skip writing that file**. The model object remains in memory for the rest of the session even when not saved to disk.

9.7 Advanced options

A collapsible **Advanced** section exposes:

- **Fix random seed** (default on, value 42): one seed for every random step of a run, that is CV fold partitions, hold-out splits, permutation shuffles, survey-site shuffling and, since v0.2.0, the background-point draw (v0.1.x drew a different background on every run). Background points are drawn from the first raster in the ① Data list, so keep the raster order the same to reproduce a run exactly. The seed value is written to the Overview sheet of `results.xlsx` and to the model card.
- **Down-weight spatially clustered points** — applies the Phillips et al. 2009 sample-bias correction by giving duplicates and very-near-duplicate occurrences less weight in the likelihood. Recommended whenever the presence layer was assembled from ad-hoc records (museum specimens, citizen-science sightings, etc.).
- **Add presences to background** — on by default, matching Java MaxEnt convention. Required for the Maxent likelihood to be well-defined (Phillips & Dudík 2008).

9.8 Next

When everything is configured, click ▶ **Run Maxent** at the dock footer. Focus shifts to the ③ Training tab.

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10. ③ Training tab

When you click ▶ **Run Maxent** at the bottom of the dock, focus shifts to the Training tab. A progress bar shows fold-by-fold progress and a log panel records what the model is doing in real time — useful for diagnosing slow runs, oversized rasters, or convergence issues.



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10.1 What happens when you click Run

QMaxent runs the following pipeline; each step writes a status line to the log:

```
flowchart TD
    A["Sample background points"] --> B["Extract raster covariates<br/>(presence + background)"]
    B --> C["Apply maxnet auto-rule<br/>(or your manual feature choice)"]
    C --> D["Fit full Maxent model"]
    D --> E["Compute training ROC + AUC"]
    E --> F["Run spatial K-Fold CV<br/>(repeat fit per fold)"]
    F --> G["Compute pooled CV AUC"]
    G --> H["(optional) Jackknife loop:<br/>2N additional fits"]
    H --> I["Save .pkl, model card, coefficients<br/>+ results.xlsx"]
```

The fit is performed by elapid's `MaxentModel`, a faithful Python port of the original Java MaxEnt (Phillips et al. 2006; Phillips et al. 2017) that delegates the regularized likelihood maximisation to scikit-learn (Pedregosa et al. 2011). The result is numerically very close to Java MaxEnt for identical inputs and identical hyperparameters — the residual difference comes mostly from how each implementation handles ties in the optimisation step.

10.2 Reading the log

A typical successful run produces a log like:

```
→ 10,000 background points sampled
Extracting raster covariates for presence points…
Extracting raster covariates for background points…
→ Presence: 116, Background: 10,104
→ Feature types: ['linear', 'quadratic', 'product', 'hinge']
Training MaxentModel…
→ Model training complete
→ Model saved: …/model.pkl
Computing ROC curve…
→ Training AUC = 0.9569
Running cross-validation…
  Fold 1: 22 test presences, AUC = 0.7453
  Fold 2: 21 test presences, AUC = 0.7839
  Fold 3: 39 test presences, AUC = 0.8097
  Fold 4: 26 test presences, AUC = 0.8614
  Fold 5:  8 test presences, AUC = 0.5903
→ CV AUC = 0.7436 ± 0.0750  (n=5 fold(s))
```


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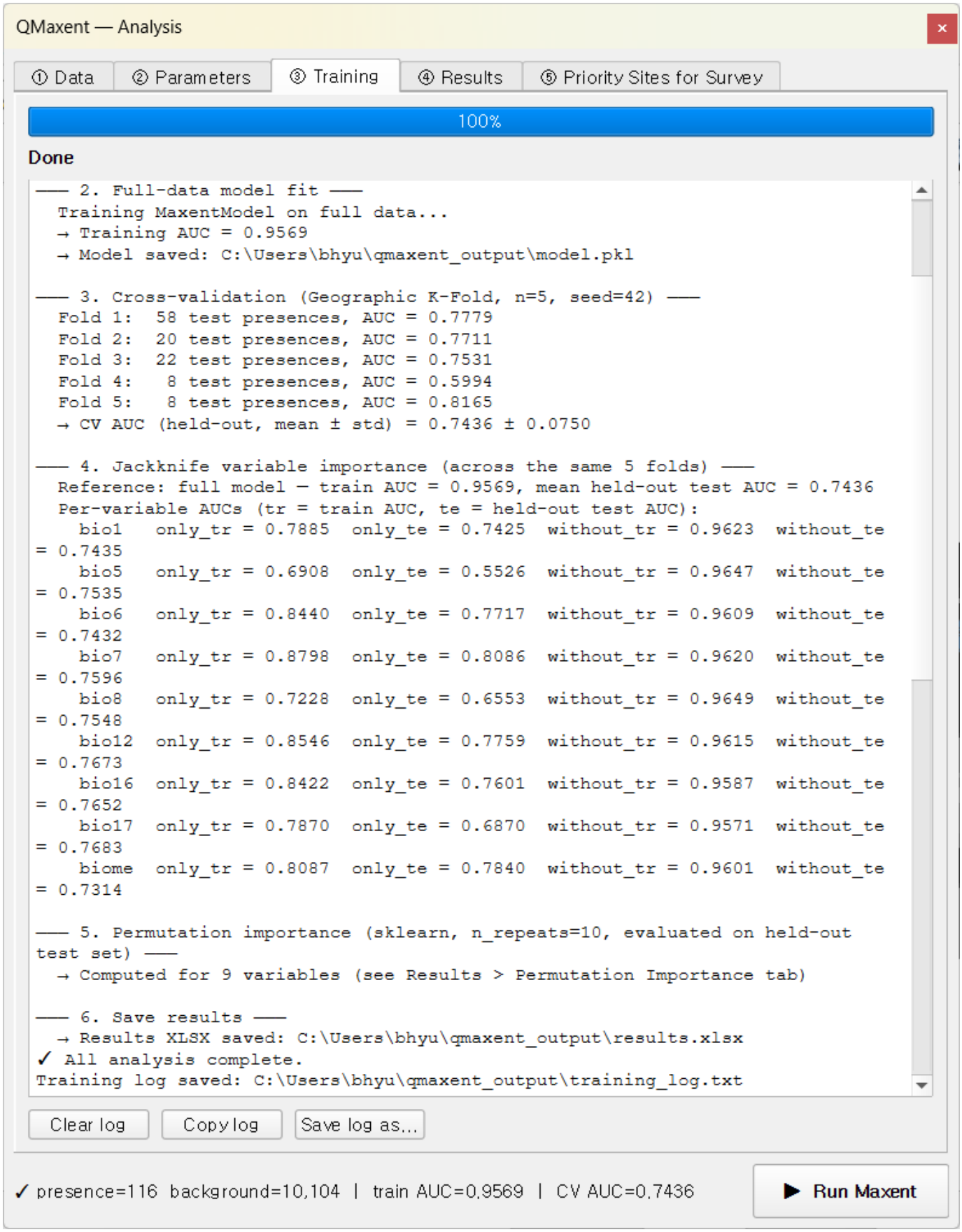
A few patterns to recognise:

- **Presences dropped** (`⚠ N of M presence points fall outside the rasters or on NoData cells and were excluded.`): the log now says how many presence points were lost and why. A large share usually means a CRS problem or rasters that do not cover the study area.
- **Presences sharing a raster cell** (`i N presence point(s) share a raster cell with another presence.`): two or more records fall in the same cell of the first raster. maxent.jar removes such duplicates by default; QMaxent keeps them, so they count more than once in the fit. If this is not intended, thin the records spatially before modeling (Boria et al. 2014).
- **Background = 10,104 (background + presence-added rows)** — points that fell on NoData cells were dropped. Normal and not a cause for concern unless the loss is large (>5 %).
- **Per-fold AUC well below the mean** — for example Fold 5 above (AUC = 0.59 with only 8 presences). Geographic K-Fold deliberately produces uneven folds, and a small fold can land on an atypical region of the study area. The pooled mean smooths this out.
- **Train AUC ≫ CV AUC** — the spatial-CV gap. A *good* sign in the sense that it tells you the model has not been silently over-fitted to the training presences (Roberts et al. 2017). A train AUC of 0.95 with a CV AUC of 0.55 is the warning sign — the CV is telling you the model does not generalise.

10.3 Completion

When the run finishes, you see the green **Done** state:

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The dock's footer status bar populates with `train AUC=… CV AUC=…` and the ④ **Results** tab unlocks.

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10.4 Cancelling a run

Click **Cancel** to abort. QMaxent terminates the current fold cleanly, discards the in-memory model, and leaves disk outputs untouched. You can restart by clicking ▶ **Run Maxent** again.

10.5 Common warnings

Message	Meaning	Action
`⚠ 12 of 116 presence points fall outside the rasters or on NoData cells and were excluded.`	Some points fell outside the rasters or on NoData cells	Inspect the raster stack; may indicate the wrong CRS or a clipped raster
`i 3 presence point(s) share a raster cell with another presence.`	Duplicate records at the raster resolution are kept	Thin the presences if the duplicates are not intended
`⚠ Background points cap reduced from 10,000 to N`	Study area too small for the requested background size	Reduce the background-points field on Data tab
`⚠ Convergence not reached after 500 iterations`	Hard-to-fit model (often very few presences)	Increase the regularization multiplier or simplify the feature set

10.6 Next

Move to the ④ Results tab to inspect response curves, Jackknife importance, and run a spatial projection.

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11. ④ Results tab

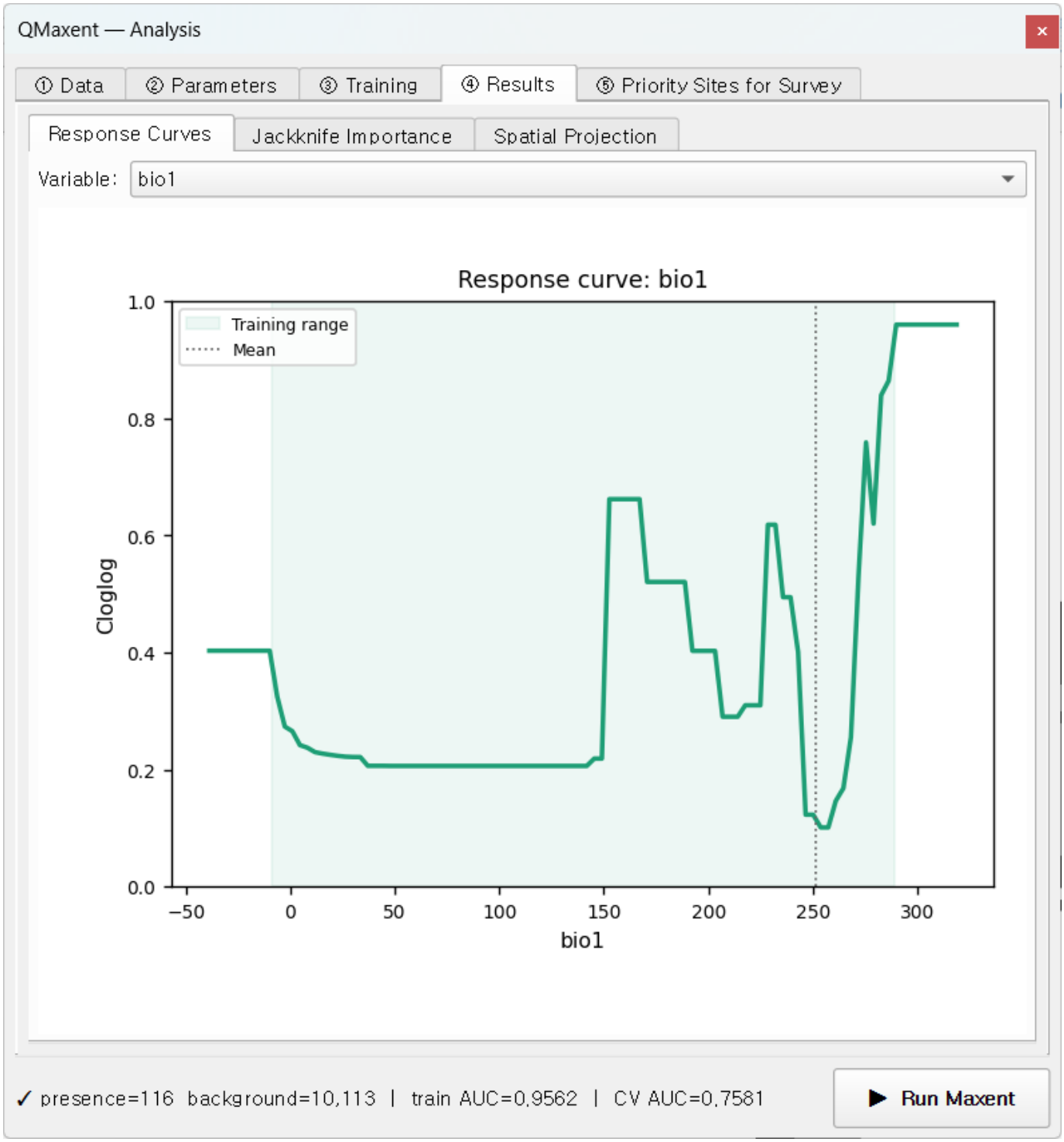
After a successful run, the Results tab unlocks four sub-tabs:

Sub-tab	What it shows
Response Curves	How each variable shapes predicted suitability across its data range
Jackknife Importance	Which variables matter most, with ROC plotted alongside
Permutation Importance	AUC drop when each variable is shuffled, as % of the total; the data it is evaluated on is set by **Evaluate on** in ② Parameters
Spatial Projection	Write the prediction to a GeoTIFF and load it as a styled QGIS layer

11.1 Response Curves

Pick a variable from the drop-down at the top. QMaxent draws the predicted cloglog suitability across the variable's actual training range (shaded band) with the mean as a vertical reference line:

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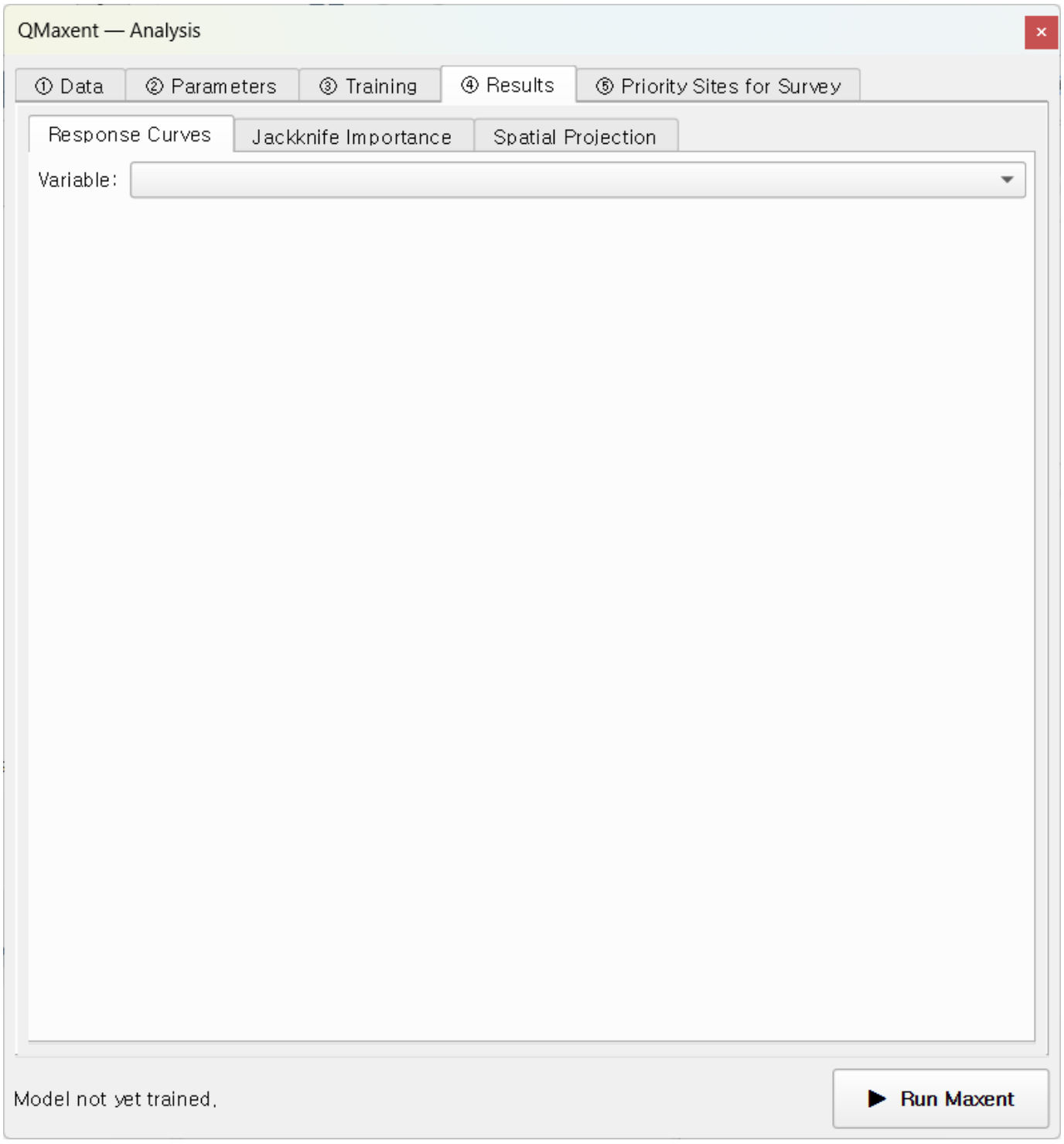
For continuous variables the curve is a marginal-effect plot in the sense of Elith et al. 2011: suitability is computed at the sweep of values for the focal variable while every other variable is held at its mean (or, for categoricals, its modal level).

For categorical variables, QMaxent renders a bar chart with one bar per class, ordered by predicted suitability — visually clearer than the ramped curve a Java-MaxEnt run would produce.

The shaded **Training range** band is important: predictions outside this band are pure extrapolation. Whenever your eventual spatial projection reaches values outside the band, QMaxent shows a one-off

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Multivariate Environmental Similarity Surface-style warning (see Elith, Kearney & Phillips 2010) before writing the output GeoTIFF.



The grid view across all variables is the standard Phillips et al. 2006 summary plot used in publications. QMaxent saves it as `prediction_response_curves.png` (300 dpi) when the **Save analysis charts as PNG** checkbox is on at projection time — see Exporting results.

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11.2 Jackknife Importance

This sub-tab combines the ROC curve and per-variable Jackknife bars in a single panel — the canonical Maxent figure since Phillips, Anderson & Schapire 2006:



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Reading the ROC panel:

- **Training ROC** (solid) — in-sample fit
- **Mean CV ROC** (dashed) — across-fold mean of held-out folds
- **Per-fold ROC** (faint) — variance across folds is the model's spatial sub-sample stability

Reading the Jackknife bars:

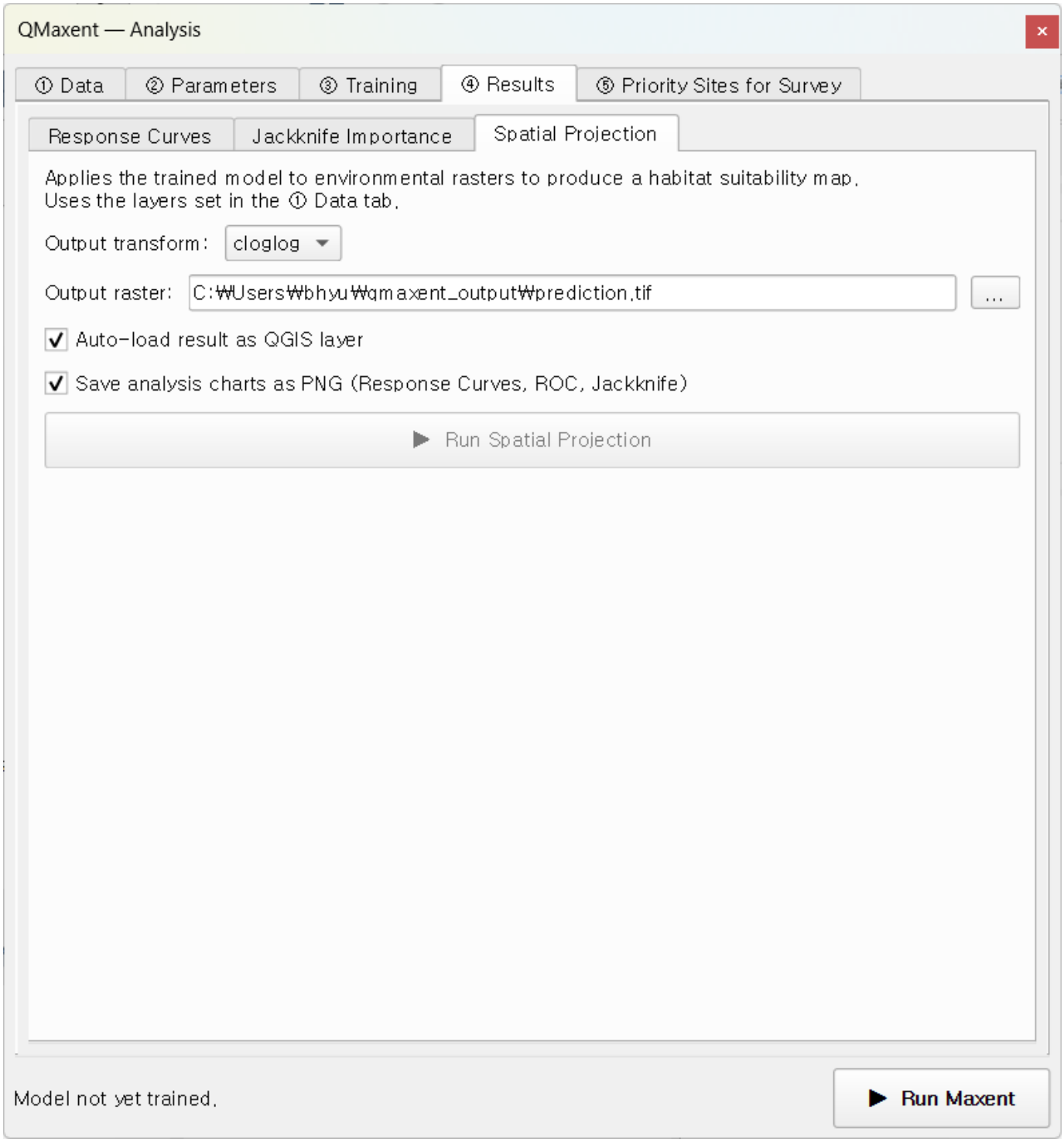
- **With-only** (dark) — AUC of a model fit on this variable alone
- **Without** (light) — AUC of the full model with this variable removed
- A variable with a high *with-only* and a high *without* drop carries a unique signal not redundant with the others (the most informative kind).

The bars are sorted by **Test AUC drop** by default. The same numbers are exported into Table 4 of the XLSX — see Exporting results for the sheet layout.

11.3 Spatial Projection

The third sub-tab projects the trained model across all the input rasters, producing a habitat-suitability GeoTIFF.

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Controls:

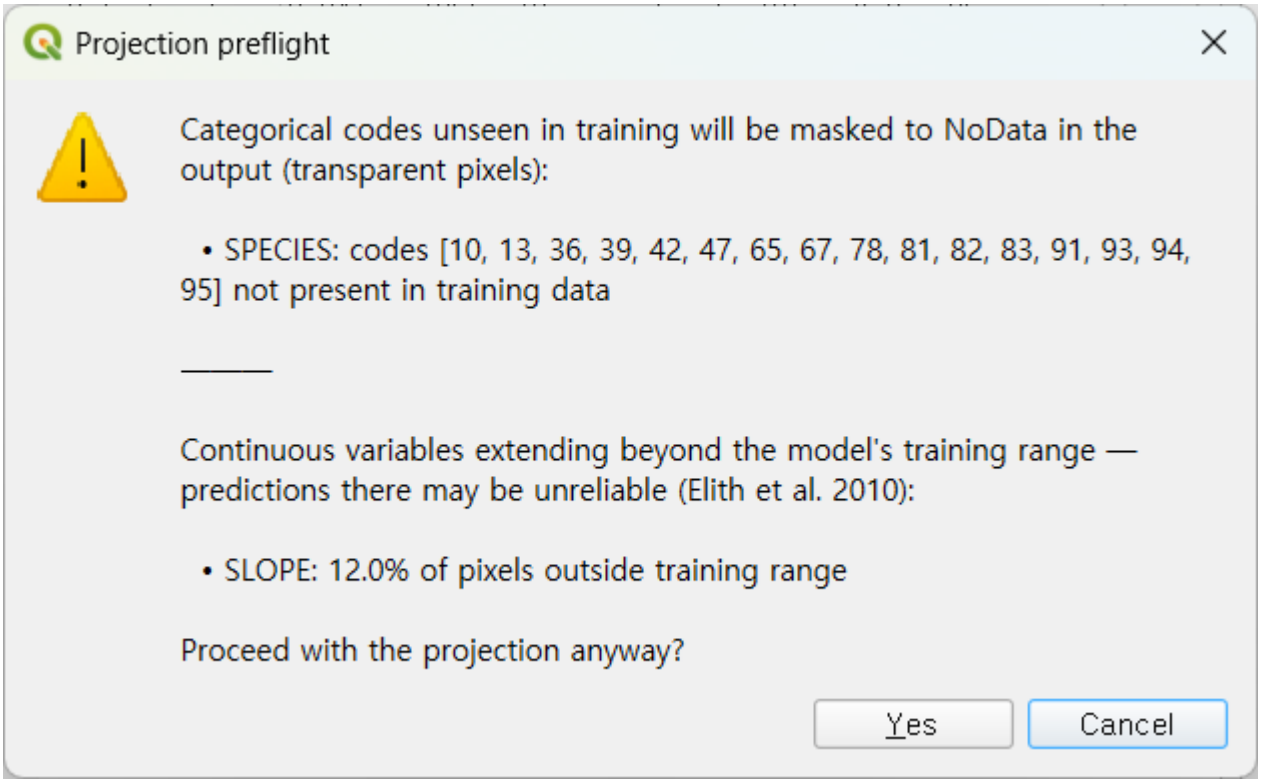
- **Output transform**:
- `cloglog` (default) — interpretable as probability of presence given a typical sample of the species, the form Phillips et al. 2017 recommend as the new default.
- `logistic` — the older (Phillips & Dudík 2008) parameterisation; still used in some published baselines.
- `raw` — the unnormalised exponential output, mostly useful for advanced post-processing.
- **Auto-load result as QGIS layer** (default on) — the GeoTIFF appears on the canvas with a default white-to-green ramp.
- **Save analysis charts as PNG** — when on, three additional 300-dpi PNGs (ROC, Jackknife, response-curve grid) are written next to the GeoTIFF. Sized for direct paste into a single-column manuscript figure.

The unified preflight dialog

Before writing the projection, QMaxent runs **two safety checks** in a single combined dialog:

1. **Categorical-code coverage** — any class code present in the projection rasters but unseen during training is auto-masked to NoData (rather than silently extrapolated to a random class probability). As a further safeguard, since v0.2.0 the projection no longer fails if such a class still reaches the model (for example a class the preflight could not detect): those cells get no categorical contribution, which is how maxent.jar treats a class without training data.
2. **Continuous-variable extrapolation** — if any continuous variable's projection range exceeds its training range (Elith, Kearney & Phillips 2010), the dialog reports the affected variables and the magnitude of the excursion.

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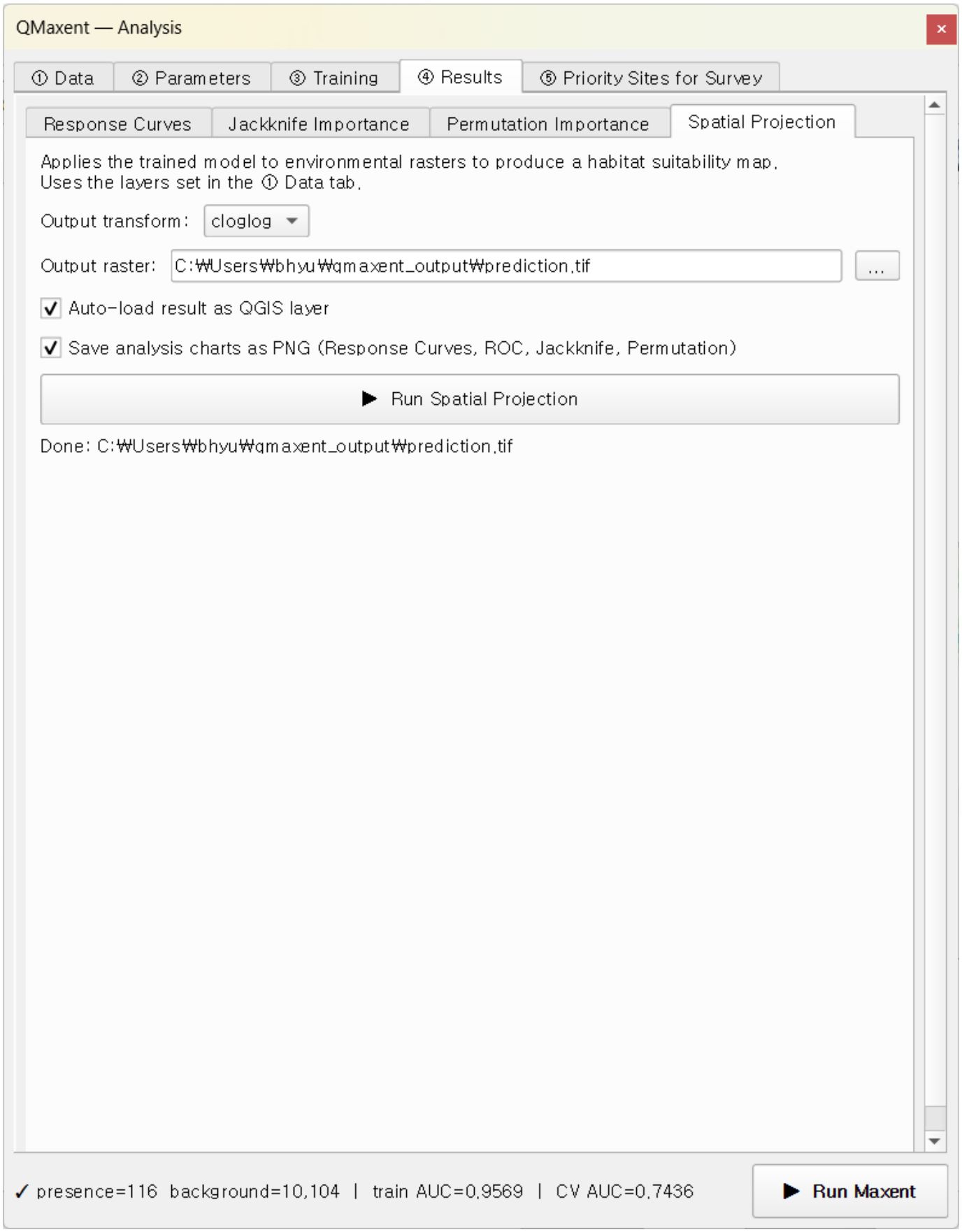


Click **Yes** to proceed; **No** aborts the projection so you can revise the input rasters.

After the projection

A success message appears in the panel:

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The GeoTIFF is loaded into QGIS with a default white-to-green ramp keyed to the cloglog 0–1 range, ready to compose into a publication map.

11.4 Next

Move to the ⑤ Survey Site Selection tab to turn the suitability raster into a field-ready layer of candidate survey sites, or skip ahead to Exporting results to learn what is in the output workbook.

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12. ⑤ Survey Site Selection

The fifth tab, ⑤ **Survey Site Selection** (called *Priority Sites for Survey* before v0.2.0), turns a habitat-suitability raster into a **field-ready list of candidate survey sites**. Field surveys usually have one of two purposes: looking for new populations, or checking the model in the field. QMaxent supports both with separate sampling designs grounded in the survey-design literature.

12.1 Survey purpose

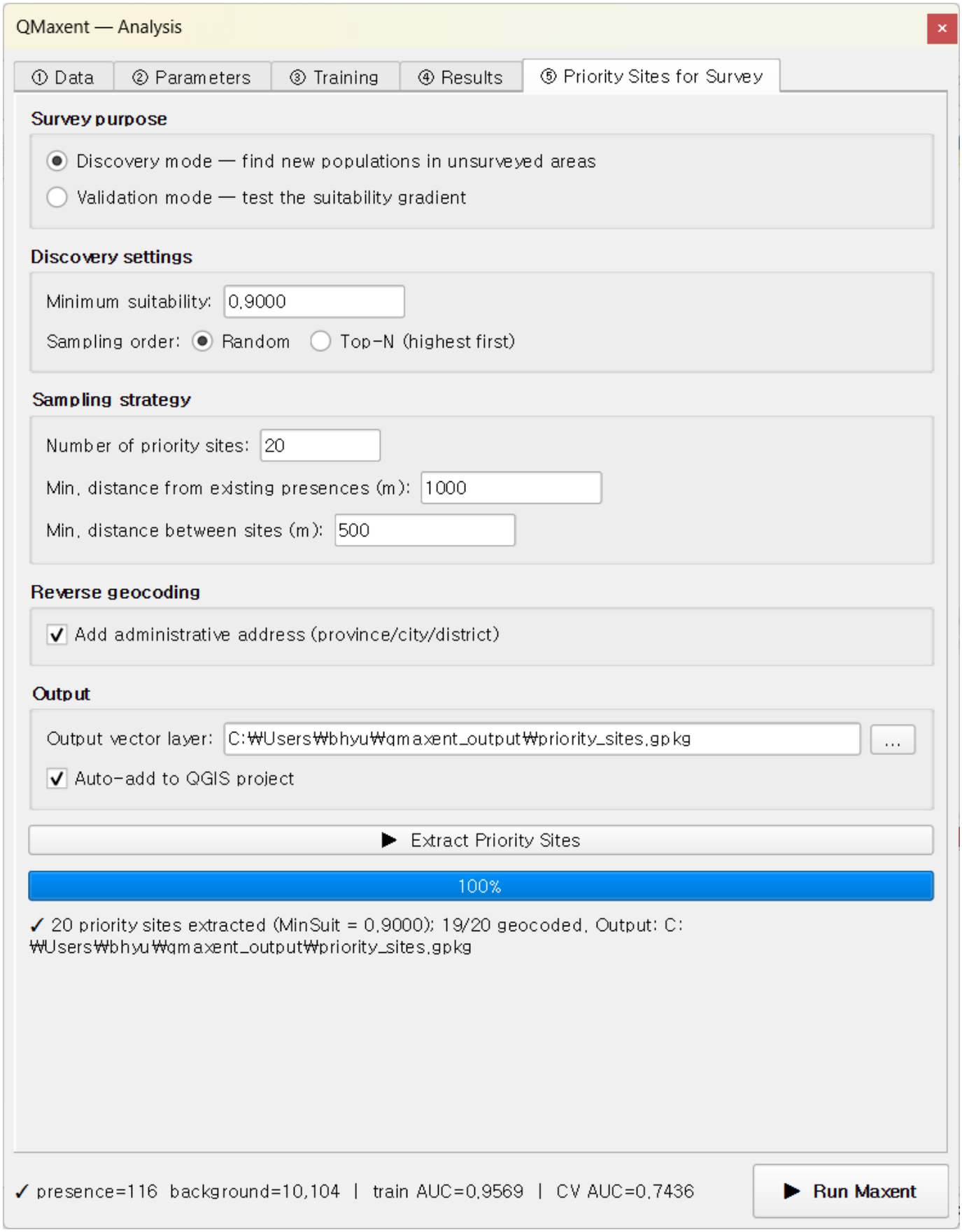
Pick one:

Mode	Goal	Reference
High-suitability search	Look for new populations in unsurveyed high-suitability areas	Williams et al. 2009
Gradient-stratified sampling	Check the model along its suitability gradient: does observed occurrence follow the predicted suitability?	Rhoden, Peterman & Taylor 2017

The two modes ask very different questions of the same map, so they sample it differently:

- **High-suitability search** (formerly *Discovery mode*) samples sites only from a band of high suitability, away from known occurrences. These are the most promising places to find the species where it has not been recorded yet.
- **Gradient-stratified sampling** (formerly *Validation mode*) splits the cells above a threshold into four suitability quartiles and draws the same number of sites from each. Lower-suitability sites are included on purpose, so the field results can show whether the species is found more often where the model predicts higher suitability.

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12.2 Mode settings

Only the settings group of the selected mode is shown.

High-suitability search settings

Control	What it does
Minimum suitability	Cells with suitability at or above this value form the candidate pool. It is filled in automatically as 0.9 × the maximum of the prediction raster when the raster is selected; you can type another value.
Sampling order: Random (default)	Draws sites at random from the candidate pool. This gives an unbiased sample within the high-suitability band.
Sampling order: Top-N (highest first)	Takes the cells with the highest suitability first. Use it when you want to maximise the expected hit rate.

Gradient-stratified sampling settings

Control	What it does
Threshold method	Sets the lower bound of the sampling frame: **10th percentile training presence (T10; Pearson 2007)** (default), **Minimum training presence (MTP; Pearson 2007)**, **Maximum sum of sensitivity + specificity (MaxSSS; Liu 2013)**, or **Custom value…**
Computed threshold value	The threshold computed from the training data for the chosen method (editable only for **Custom value…**)

The threshold **only defines the sampling frame**. The cells above it are split into four quartiles of suitability, and the same number of sites is drawn from each quartile (Rhoden, Peterman & Taylor 2017). No binary suitable/unsuitable map is produced, and the threshold is not a claim that cells below it are unsuitable. Each site carries its quartile in the `quartile` field (0 = lowest band above the threshold, 3 = highest), and the layer is styled by quartile. In High-suitability search the field is always 0.

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12.3 Spacing constraints

The **Sampling strategy** group is shared by both modes. Besides the **Number of survey sites** (default 20), two distance fields define the spatial structure of the sampled set:

- **Min. distance from existing presences (m)** (default 1,000) keeps candidates away from known occurrences. Set it to the species' detection radius (e.g. 1 km for a calling fairy pitta, 200 m for a slug). This prevents the algorithm from re-sampling territory already covered by earlier surveys.
- **Min. distance between sites (m)** (default 500) keeps the candidate set spatially independent. A typical value is half the distance from existing presences. **Spatially balanced** sampling, in the sense of Stevens & Olsen 2004, is approximated by setting this value to the nearest-neighbour distance you want in the final set.

QMaxent enforces both constraints exactly: candidates are dropped, not moved, when a constraint is violated, so the surviving set is always within the specification. If the candidate pool cannot supply the requested number of sites under these constraints, QMaxent returns the sites it found and reports the actual count.

12.4 Reverse geocoding

When **Add administrative address (OpenStreetMap)** is ticked, QMaxent calls the Nominatim service of OpenStreetMap for each site. No API key is needed. Reverse geocoding works worldwide. It adds four generic OpenStreetMap address levels and the full address:

Field	Address level	Examples
`country`	Country	Dominican Republic, South Korea
`province`	First-level administrative division	a Dominican *provincia*, a Korean province, a US state
`city_county`	Municipality	a Dominican *municipio*, a Korean city or county
`district`	Locality	a suburb, village, neighbourhood or hamlet
`display_name`	Full address text returned by Nominatim	

A level that OpenStreetMap does not hold for a location is left blank.

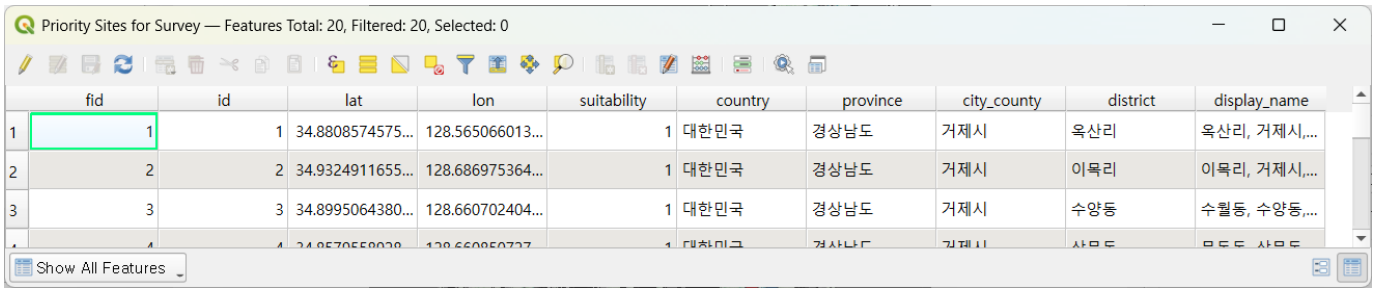
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Address language sets the language of the names:

- **Local names** (default): each country's own names, for example Spanish names in the Dominican Republic and Korean names in Korea.
- **QGIS interface language**: the language of your QGIS interface, with English as a fallback.
- **English**: English names where OpenStreetMap has them.

The resulting GeoPackage is directly usable for permit applications and field-team coordination, which matters in jurisdictions where survey access requires prior notice to the local administration.

Requests are limited to one per second to comply with Nominatim's fair-use policy, so 20 sites take about 20 seconds. For 30 or more sites QMaxent first shows the expected time; for large sets, untick the option or select the sites in smaller batches. Results carry the attribution © OpenStreetMap contributors.



12.5 Outputs

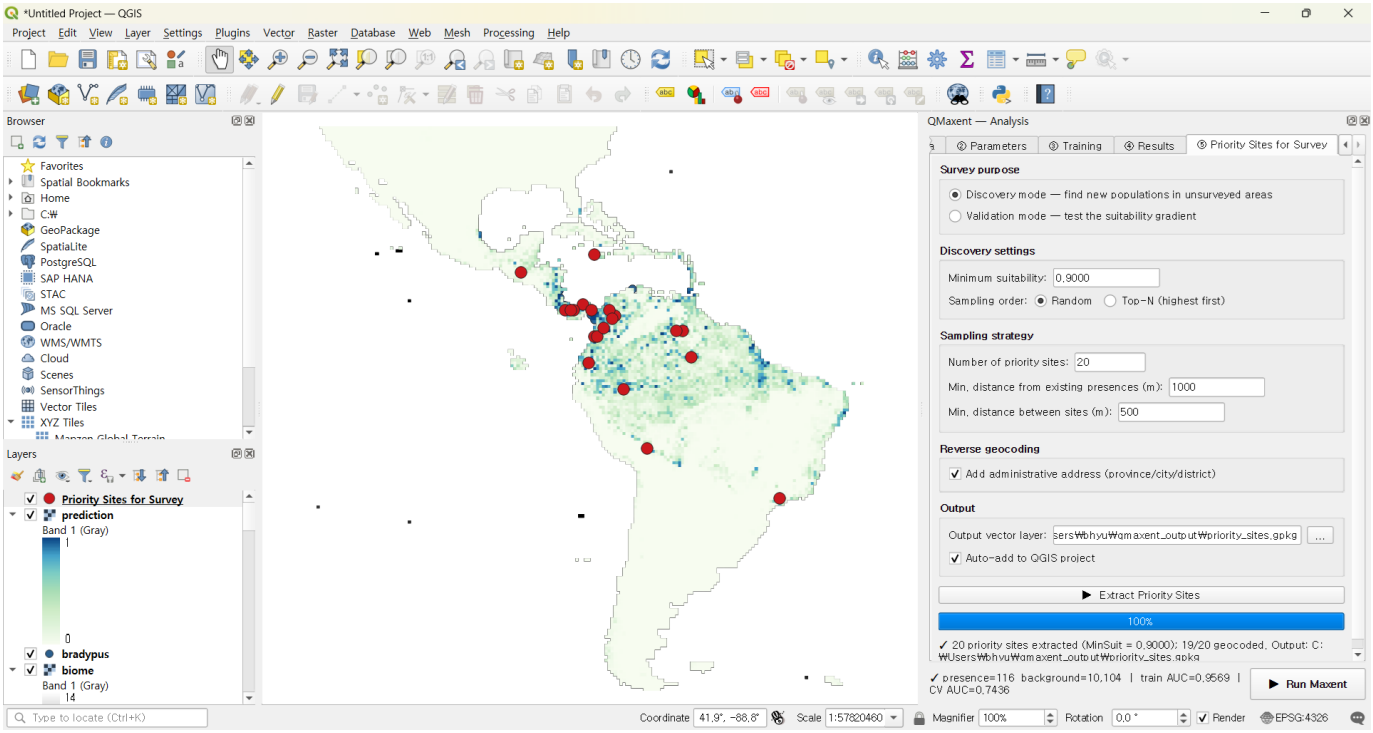
After clicking ▶ **Select Survey Sites**, QMaxent writes:

- `survey_sites.gpkg` (default name, in the QMaxent output folder): a GeoPackage point layer with the fields `id`, `lat`, `lon`, `suitability`, `quartile` and the address fields above. With **Auto-add to QGIS project** ticked, it is added to the map as the layer **Survey sites**.

If the output GeoPackage already exists, it is replaced, and any layer of the current project that reads it is removed first. If another program holds the file open, QMaxent writes to the next free name instead, such as `survey_sites_1.gpkg`, and reports the name it used.

The training thresholds (MTP, T10, MaxSSS) that Gradient-stratified sampling can use are also recorded in the model card (`<model>_model_card.json`, see Saving and reusing models).

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The sites are immediately ready for export to mobile-GIS apps such as **QField**, **Mergin Maps**, or **Locus Map** for offline field use.

12.6 Choosing parameter values

Some practical starting points:

- **Number of survey sites**: 20–30 per species per season is a typical field-survey budget for a single team (Robinson et al. 2018).
- **Min. distance from existing presences**: the species' typical home-range diameter is a good starting point. Too small and you re-sample known sites; too large and you push candidates into marginal habitat.
- **Minimum suitability** (High-suitability search): the automatic value (0.9 × raster maximum) gives a focused search; lower it to widen the candidate pool when too few sites satisfy the spacing rules.
- **Threshold method** (Gradient-stratified sampling): T10 or MTP give a wide frame that includes lower-suitability cells; MaxSSS gives a narrower one.

12.7 Worked applications

- The Bradypus example walks through both modes at continental scale.
- The Pitta nympha example shows a High-suitability search at municipality scale, with reverse-geocoded administrative addresses ready for field permits.

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12.8 Next

Move to Saving and reusing models to learn how to share the trained model with collaborators or re-project it onto a new raster stack later.

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13. Saving and reusing models

Persisting a trained model and loading it back later — with confidence that variables map to rasters correctly — is what enables collaboration, re-projection onto updated environmental data, and long-term reproducibility of published QMaxent results.

13.1 Saving a trained model

Every successful Maxent run writes a `model.pkl` file to the path you specified on the ② **Parameters** tab (default `<project>/qmaxent_output/model.pkl`). This is a Python pickle of the trained `elapid.MaxentModel` instance — it carries:

- The fitted regularized-likelihood coefficients
- The complete list of variable names (in order)
- For each variable: type (`continuous` / `categorical`), training data range (continuous) or class set (categorical)
- The full set of hyperparameters (feature classes, RM, CV scheme, random seeds)
- A QMaxent build-version stamp

Treat the `.pkl` as the **canonical record of the trained model**. Distribute it alongside the `results.xlsx` workbook when sharing with collaborators or publishing supplementary material — that is the pair of files Araújo et al. 2019 cite as the minimum reproducibility unit for a published SDM.

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13.2 Model card and coefficient table

Since v0.2.0, QMaxent writes two plain-text files next to the `.pkl`, named after it (for `model.pkl`: `model_model_card.json` and `model_coefficients.csv`):

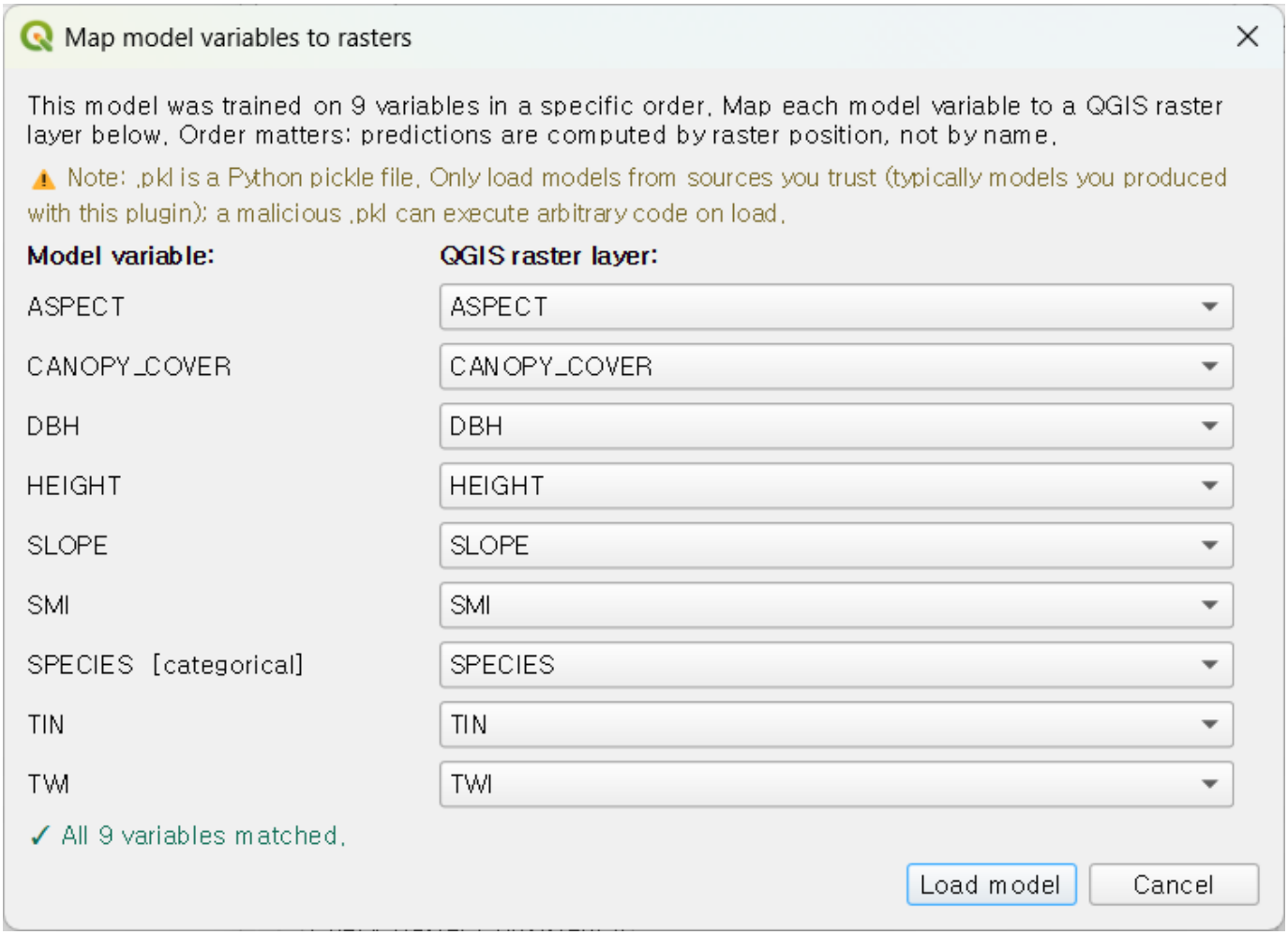
File	Contents
`<model>_model_card.json`	Software versions (QMaxent, elapid); settings (feature classes and whether Auto or Manual selection chose them, regularization multiplier, output transform, hinge and threshold knots, CV method and folds, random seed, permutation-importance evaluation data); data summary (presence and background counts, each variable with its type, training range and raster file); performance (training AUC, per-fold and mean CV AUC); thresholds (MTP, T10, MaxSSS); variable importance (permutation and jackknife); model constants
`<model>_coefficients.csv`	One row per Maxent feature: index, feature type (linear, quadratic, product, threshold, hinge, categorical), variable, a short description, the knot value for hinge and threshold features, the fitted coefficient, and whether it is non-zero

Use the model card to report the run in a methods section or a supplement, and the coefficient table to see which features the regularization kept. Unlike the `.pkl`, both files are **safe to open and share**: they contain only text and numbers, no executable code.

13.3 Reloading a saved model

On the ① **Data** tab, click **Load existing model (.pkl)…** and select the saved file. QMaxent reads the variable list out of the pickle and opens a **variable-mapping dialog**:

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The dialog has one row per saved variable. For each row, QMaxent picks a default raster from the current QGIS project by matching variable names to layer names. When the names match exactly, the dialog is just a confirmation step. When they differ, the dialog **forces an explicit mapping** — the only way to load a `.pkl` whose variable names do not match the current naming convention is to *consciously* re-state which raster maps to which saved variable.

This safeguard prevents the silent ordering errors that have been documented as a recurring source of "but it worked yesterday" bugs in collaborative SDM projects.

13.4 The variable-mapping dialog in detail

The dialog renders three columns:

- **Saved variable name** — read from the pickle, not editable.
- **Type** — `continuous` or `categorical`, also from the pickle. A variable saved as categorical can only be mapped to a raster the current project has marked as categorical.
- **Mapped raster** — a drop-down listing the rasters in the current project. Pick one for each row.

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A footer status line reads `0 of N mapped` and updates live as you populate the drop-downs. The **OK** button enables only when all rows are mapped.

After confirmation, QMaxent rewires the dock as if you had just trained the model fresh — the ③ **Training** tab is skipped and ④ **Results** is enabled directly. You can immediately run a spatial projection or select survey sites against the new raster stack.

13.5 Re-projecting onto an updated raster stack

The most common reason to reload a model is to **apply it to a more recent or higher-resolution version of the same environmental layers** without re-fitting. Because re-fitting on different rasters would produce a different model, this is the only correct way to ask "what does the original model predict about the new data?".

The workflow:

1. Train the model once on the original rasters; save `model.pkl`.
2. Load the new raster stack into QGIS.
3. **Load existing model (.pkl)…**, map saved variables to the new rasters in the dialog, click **OK**.
4. Switch to ④ **Results → Spatial Projection** and run.

The preflight dialog will flag any extrapolation introduced by the new data — particularly important when the new rasters extend the geographic extent or include time periods outside the original training range.

13.6 Security note on Python pickle files

`.pkl` files are **Python pickles**, which can contain executable code. Loading a malicious `.pkl` can run arbitrary code on your machine. Only open `.pkl` files that you produced yourself or that come from trusted collaborators. The model card (`.json`) and the coefficient table (`.csv`) are safe to share and open.

QMaxent has no way to verify the provenance of a third-party `.pkl` beyond what Python's `pickle` module itself does (which is nothing). For published QMaxent results, the safe pattern is:

- Distribute the `model.pkl` alongside the `results.xlsx` workbook and the model card so reviewers have the deterministic seed, hyperparameters, and inputs needed to **re-train an equivalent model from scratch** if they prefer not to load the pickle. The model card and coefficient table let them inspect the model without opening the pickle at all.
- Sign the `.pkl` with a SHA-256 hash in your supplementary materials.

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13.7 Next

The companion document is Exporting results, which describes the multi-sheet XLSX workbook that ships alongside every `model.pkl`.

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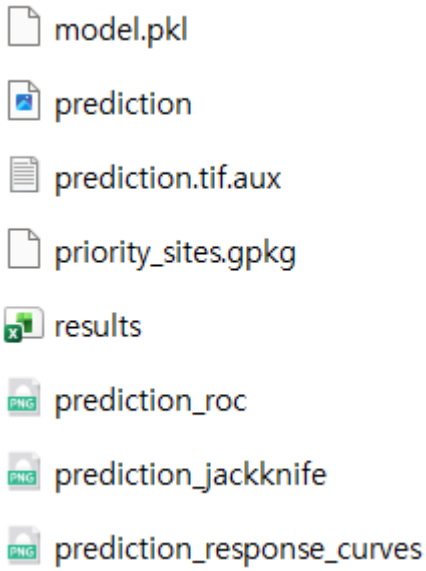
14. Exporting results

Every QMaxent training run writes a **multi-sheet XLSX workbook** alongside the `.pkl` model. The workbook is intentionally formatted to read as a **Supplementary Table block** for an academic manuscript: Times New Roman, table-numbered headings, row footers explaining how each metric was computed. You can paste sheets directly into a paper's appendix.

14.1 Output folder layout

After a training run that also produced a spatial projection and the `Save analysis charts as PNG` checkbox was on, the output folder looks like this:

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File	Produced by	Use
`model.pkl`	Training	Reload later via *Load existing model (.pkl)...* (Python pickle: open only trusted files)
`model_model_card.json`	Training	Settings, data, performance, thresholds, importance and software versions; safe to share (see Saving and reusing models)
`model_coefficients.csv`	Training	Every Maxent feature with feature type, variable, knot and coefficient; safe to share
`training_log.txt`	Training	Copy of the training log
`prediction.tif`	Spatial projection	Habitat-suitability raster
`prediction.tif.aux`	Spatial projection	QGIS auxiliary metadata
`results.xlsx`	Training	Multi-sheet supplementary table
`survey_sites.gpkg`	Survey Site Selection tab	Field-ready candidate survey sites
`prediction_roc.png`	Spatial projection	ROC curve (300 dpi)
`prediction_jackknife.png`	Spatial projection	Jackknife bars (300 dpi)
`prediction_response_curves.png`	Spatial projection	All response curves (300 dpi)

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14.2 Sheet-by-sheet description

The XLSX has four to five sheets depending on which features you ran. The screenshots below are from a Pitta nympha run — the *format* is identical for every dataset; only the values change.

Table 1 — Experimental setup

A one-page record of the full run configuration. Reviewers can verify seed, variable count, regularization, CV scheme, and training/CV AUCs at a glance. This is exactly the information Araújo et al. 2019 flag as the minimum methods-section reporting standard for a published SDM.

	A	B	C
1	**Table 1. Experimental setup, training data, and run-level metrics.**		
2	**Category**	**Item**	**Value**
3	**Run**	Generated	2026-05-09 20:41:07
4		QMaxent version	0.1.1
5		Random seed	0
6	**Training data**	Presence points	47
7		Background points	9448
8		Categorical variables	AGE, SPECIES
9	**Model**	Output transform	cloglog
10		Beta multiplier	4
11		Feature types	linear, quadratic, hinge
12		Distance weights	no
13	**Cross-validation**	Method	Random K-Fold
14		Folds requested	4
15		Folds completed	4
16	**Performance**	Training AUC (full data)	0.8681
17		CV AUC (mean ± std)	0.7825 ± 0.0542
18		Jackknife full-model test AUC	0.7826
19	*Generated by QMaxent. CV AUC ± std is across CV folds; the jackknife full-model test AUC is averaged across the same folds and serves as the reference for the per-variable Jackknife sheet.*		

This sheet is the single best artefact for **methods-section reproducibility**. Cite it as your "Table S1" and a reader can re-execute the analysis bit-identically.

Table 2 — Predictor variables

Lists every environmental variable, its type (`continuous` / `categorical`), and its training-data range. For continuous variables the range is `[min, max]`; for categorical, the discrete set of levels encountered during training.

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	A	B	C	D
1	**Table 2. Predictor variables, types, and training-data range.**			
2	**#**	**Variable**	**Type**	**Range / Levels**
3	1	AGE	categorical	2, 3, 4, 5, 6, 7
4	2	ASPECT	continuous	[0.0019, 359.9804]
5	3	CANOPY_COVER	continuous	[1.0, 3.0]
6	4	DBH	continuous	[1.0, 3.0]
7	5	HEIGHT	continuous	[6.0, 30.0]
8	6	SLOPE	continuous	[0.0098, 60.798]
9	7	SMI	continuous	[-2.3807, 1.649]
10	8	SPECIES	categorical	11, 14, 15, 17, 18, 30, 31, 33, 34, 35, 39, 44, 45, 46, 49, 60, 77
11	9	TIN	continuous	[10.0, 566.8892]
12	10	TWI	continuous	[4.0037, 18.7304]
13	*Range gives the [min, max] observed at training time for continuous variables; for categorical variables, the levels encountered are listed. Use this range to identify extrapolation regions when projecting the model.*			

Use this sheet to document **extrapolation scope**: any prediction applied to environmental values outside these ranges is by definition extrapolation (Elith, Kearney & Phillips 2010) and should be flagged in your results discussion.

Table 3 — Cross-validation results

Per-fold held-out AUC and the across-fold mean ± standard deviation — the headline performance estimate.

	A	B
1	**Table 3. Cross-validation results (Random K-Fold).**	
2	**Fold**	**AUC**
3	Fold 1	0.8334
4	Fold 2	0.8342
5	Fold 3	0.7063
6	Fold 4	0.7562
7	**Mean ± std**	**0.7825 ± 0.0542**
	AUC computed on each fold's held-out presences (Random K-Fold). Folds with degenerate test sets (e.g. all-positive after spatial partitioning) are reported as NaN and excluded from the mean.	

Folds with degenerate test sets (e.g. all-positive after spatial partitioning) are reported as `NaN` and excluded from the mean. This matches the way the original Maxent literature handles the rare cases where a fold cannot produce a valid AUC.

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Table 4 — Jackknife variable importance

For each variable, four AUCs (only-train, only-test, without-train, without-test), plus a "Train AUC drop" and "Test AUC drop" column to make the contribution magnitudes scannable. Variables are sorted by *descending Test AUC drop* so the most uniquely-informative variables are at the top.

Table 4. Jackknife variable importance — AUC for models with only / without each variable.

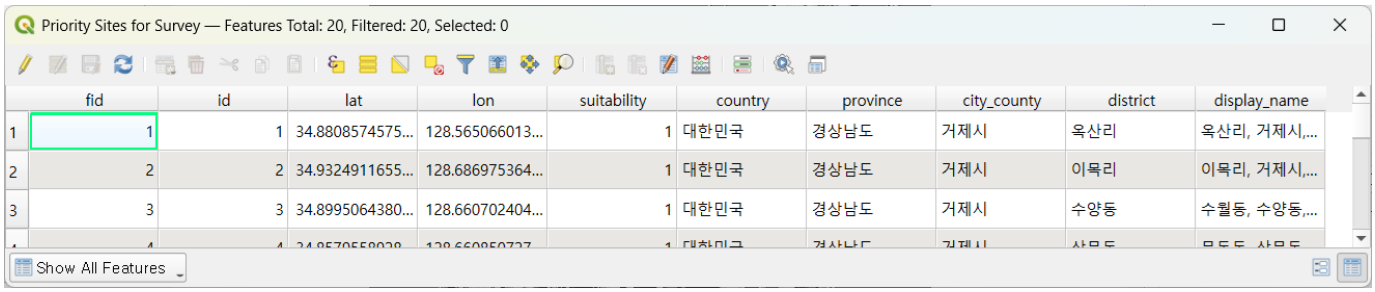
Variable	Only — train AUC	Only — test AUC	Without — train AUC	Without — test AUC	Train AUC drop	Test AUC drop
ASPECT	0.7215	0.6806	0.8488	0.7476	0.0192	0.0350
TWI	0.7745	0.7721	0.8581	0.7543	0.0100	0.0283
SPECIES	0.6555	0.6075	0.8593	0.7647	0.0087	0.0179
DBH	0.5525	0.5172	0.8732	0.7747	-0.0052	0.0079
SLOPE	0.7044	0.6971	0.8776	0.7753	-0.0095	0.0072
HEIGHT	0.6188	0.5553	0.8712	0.7817	-0.0032	0.0009
TIN	0.5766	0.5533	0.8743	0.7821	-0.0063	0.0005
CANOPY_COVER	0.4962	0.4962	0.8793	0.7829	-0.0112	-0.0003
SMI	0.6385	0.5200	0.8765	0.7851	-0.0085	-0.0025
AGE	0.5901	0.4281	0.8747	0.7885	-0.0066	-0.0060

Only — model fit using a single variable. Without — model fit using all other variables. Drop = full-model AUC − without-variable AUC; larger drops indicate variables whose unique contribution is harder to recover from the others. Each AUC is the mean across CV folds. Variables are sorted by descending test-AUC drop.

The footer note explains the drop columns: *Drop = full-model AUC − without-variable AUC; larger drops indicate variables whose unique contribution is harder to recover from the others.* This is the metric Phillips, Anderson & Schapire 2006 recommend as the most robust indicator of *unique* contribution.

Survey sites

The candidate sites from the ⑤ **Survey Site Selection** tab are written to their own GeoPackage (`survey_sites.gpkg` by default), not to the workbook. Its attribute table has one row per site with `id`, `lat`, `lon`, `suitability`, `quartile` and, when reverse geocoding was enabled, the OpenStreetMap address fields `country`, `province`, `city_county`, `district` and `display_name`. Open the attribute table in QGIS, or export it to a spreadsheet with **Export > Save Features As…**, if you need it as a supplementary table. See ⑤ Survey Site Selection for the field definitions.



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14.3 Customizing the export

The default file path on the ② **Parameters** tab is `<home>/qmaxent_output/results.xlsx`. Change it to anywhere writable; relative paths are resolved against the current QGIS project folder.

To **stop the XLSX from being written** (rare — usually faster to ignore the file than to disable it), simply clear the path field. QMaxent will skip the export and emit a notice in the training log.

The PNG analysis charts are controlled by the **Save analysis charts as PNG** checkbox on the ④ **Results → Spatial Projection** sub-tab. With it ticked, three additional PNGs are written next to the GeoTIFF at the moment you run projection. They are sized for **direct paste into a single-column manuscript figure** (300 dpi, ~1.6 MB each).

14.4 Citing the workbook

The XLSX was designed to be cited the way published supplementary tables are cited:

> Yu, B.-H. (2026). QMaxent results workbook for [Supplementary table]. https://github.com/osgeokr/qmaxent

If your model is reproducible (fixed seed + same input rasters) you can also distribute the `model.pkl` alongside the XLSX so reviewers can re-load the model and inspect predictions independently.

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15. Worked examples

Three end-to-end case studies, each highlighting a different aspect of QMaxent and a different real-world workflow.

15.1 Overview of the examples

Example	Species	Highlights	What you will learn
Bradypus variegatus	Three-toed sloth	Bundled dataset; 8 bioclimatic variables + biome, 116 presences	The complete QMaxent feature tour — same workflow as the original Maxent paper
Ariolimax	Pacific banana slug	Mismatched rasters	The **Check Raster Consistency** + **Harmonize to Folder...** preflight
Pitta nympha	Fairy pitta	Reproduction of Lee et al. 2025	How to reproduce a published study in QMaxent and compare results with maxent.jar on identical inputs

The three examples are designed to be **completed in order**: Bradypus introduces the workflow, Ariolimax adds raster harmonisation, Pitta nympha adds publication-quality reproduction with academic comparison.

15.2 How to follow along

Every example assumes you have completed Installation and that the **QMaxent environment ready** banner is green.

Two of the three datasets are bundled with the plugin — load them via **Plugins → QMaxent → Download Example Dataset → \<species name>**:

- **Bradypus variegatus**: 9 variables (8 bioclimatic variables and a categorical biome raster), 116 presences, ships with the plugin. Bit-identical to the Phillips et al. 2006 dataset as distributed with the R package dismo.
- **Ariolimax** — 6 variables, 3,732 presences, the elapid library's default Pacific-coast slug dataset. **The official download is pre-harmonised** — to follow the Ariolimax worked example exactly, you will need to use the deliberately desynchronised version described in the example's header.
- **Pitta nympha**: 10 variables (2 categorical), 47 nests in Geoje-si, South Korea. Sourced from the supplementary archive of Lee et al. 2025; not bundled with the plugin (licence considerations).

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15.3 Reproducibility

All three examples are run with **fixed random seeds**. Since v0.2.0 the seed also fixes the background sample, so with the same inputs, the same raster order and the same settings, a run reproduces the same numbers. Some screenshots and numbers in the Bradypus and Ariolimax chapters were recorded with v0.1.x (LQPHT under Auto, unseeded background) and differ slightly from what v0.2.0 gives; those places are marked.

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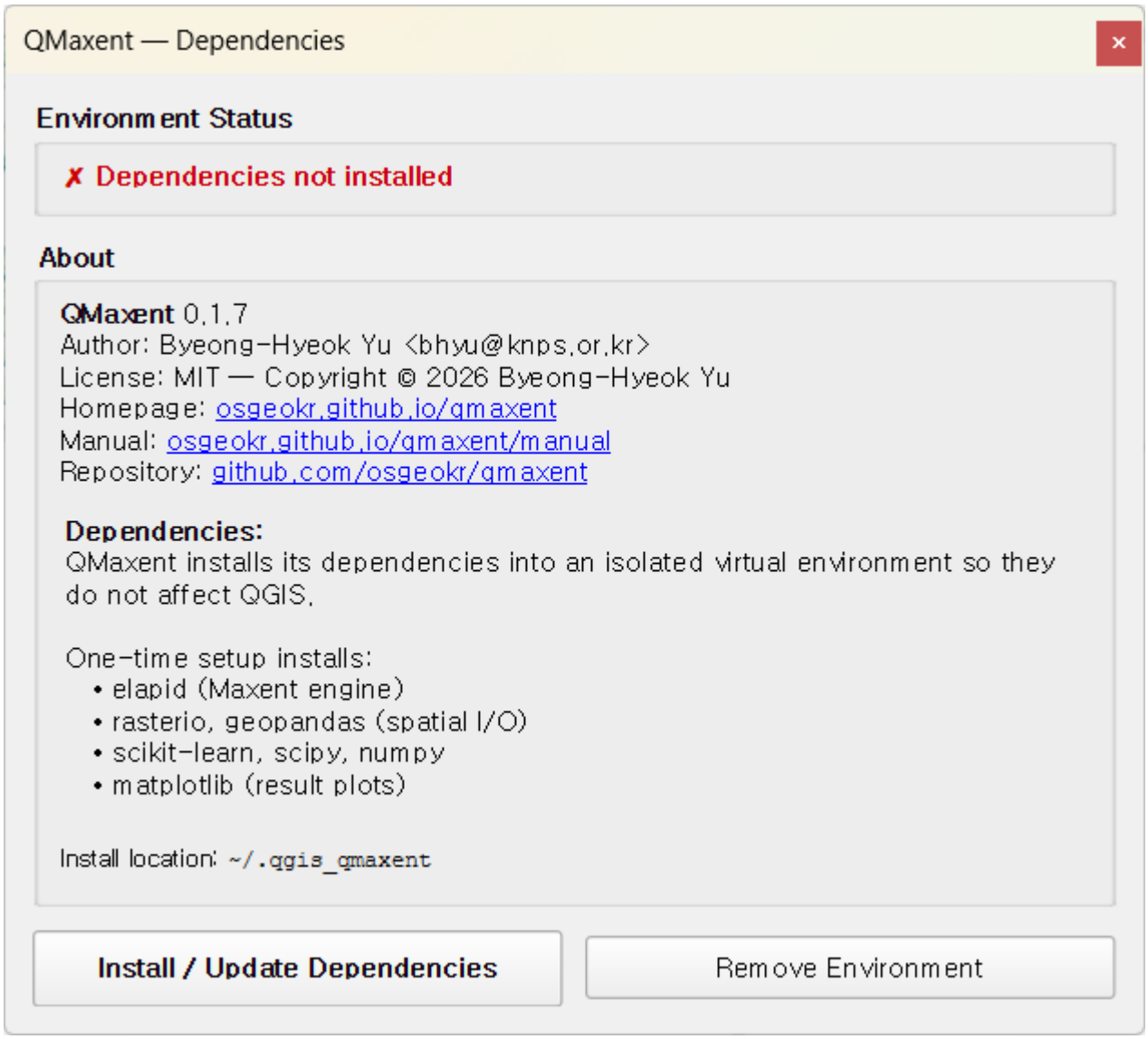
16. Bradypus variegatus

The brown-throated three-toed sloth — *Bradypus variegatus* — is the canonical Maxent test dataset, originally published with Phillips, Anderson & Schapire 2006 and reused in virtually every subsequent Maxent paper. We use it here as a **guided tour of every QMaxent feature**: dependency setup, data loading, parameter selection, spatial cross-validation, jackknife and permutation importance, projection, and survey planning. By the end of this chapter you will have produced — and be able to defend academically — a complete Bradypus habitat-suitability model.

16.1 0. Before you start: dependencies

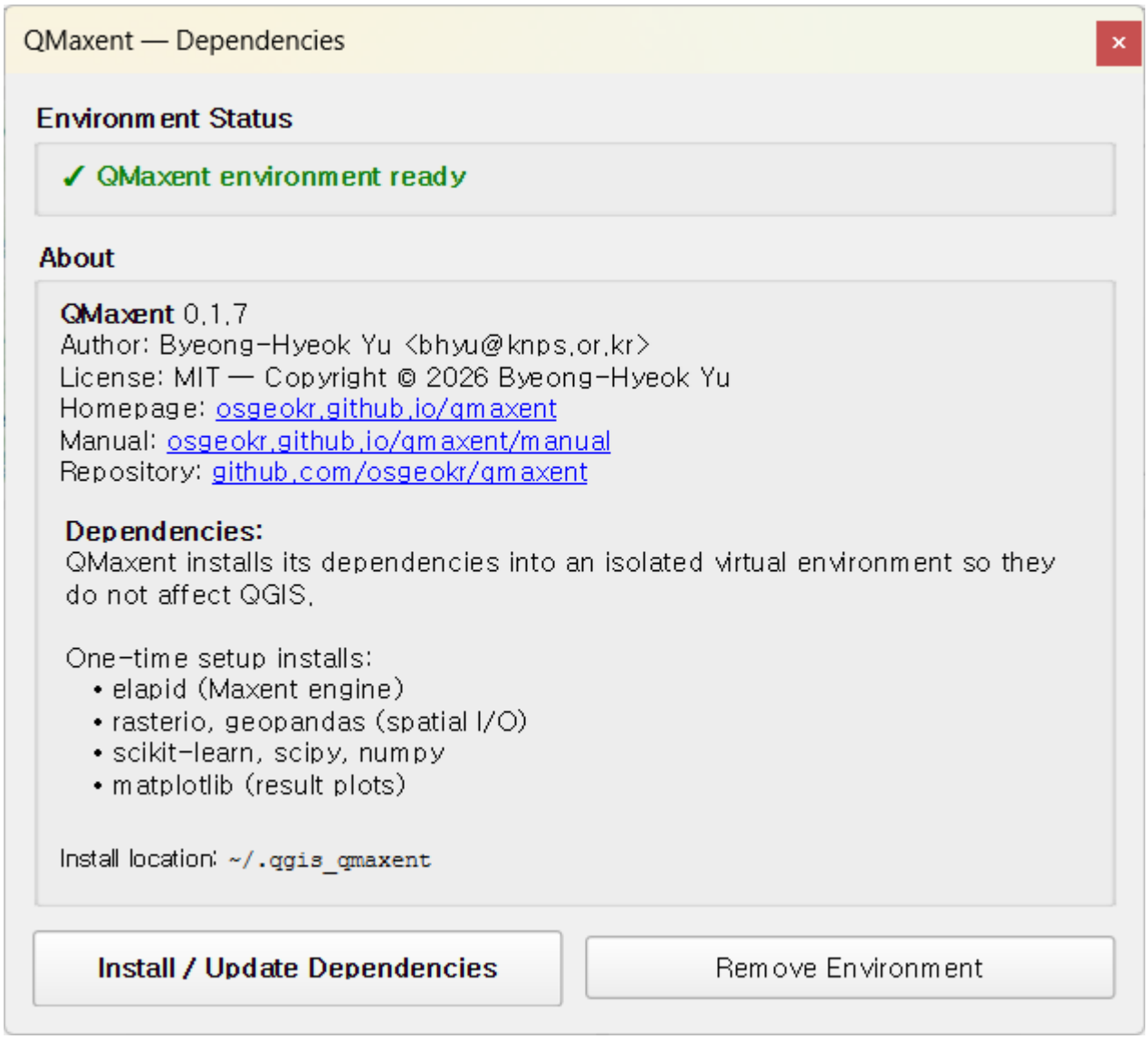
QMaxent installs its third-party Python libraries (elapid, rasterio, geopandas, scikit-learn, scipy, numpy, matplotlib) into an isolated package folder so they cannot conflict with QGIS's own Python (a virtual environment on Linux and macOS; on Windows, a package folder filled by the QGIS Python's own pip, see Dependencies). The first time you open the plugin, you are invited to perform a one-time install:

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Click **Install / Update Dependencies**. The progress bar walks through the four pip phases (collecting, downloading, building, installing) and finishes with the green badge below — at that point every QMaxent feature is ready to run.

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If you ever need to recreate the package folder (e.g. after a major QGIS upgrade), the **Remove Environment** button on the same dialog is the clean way to start over.

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16.2 1. Dataset

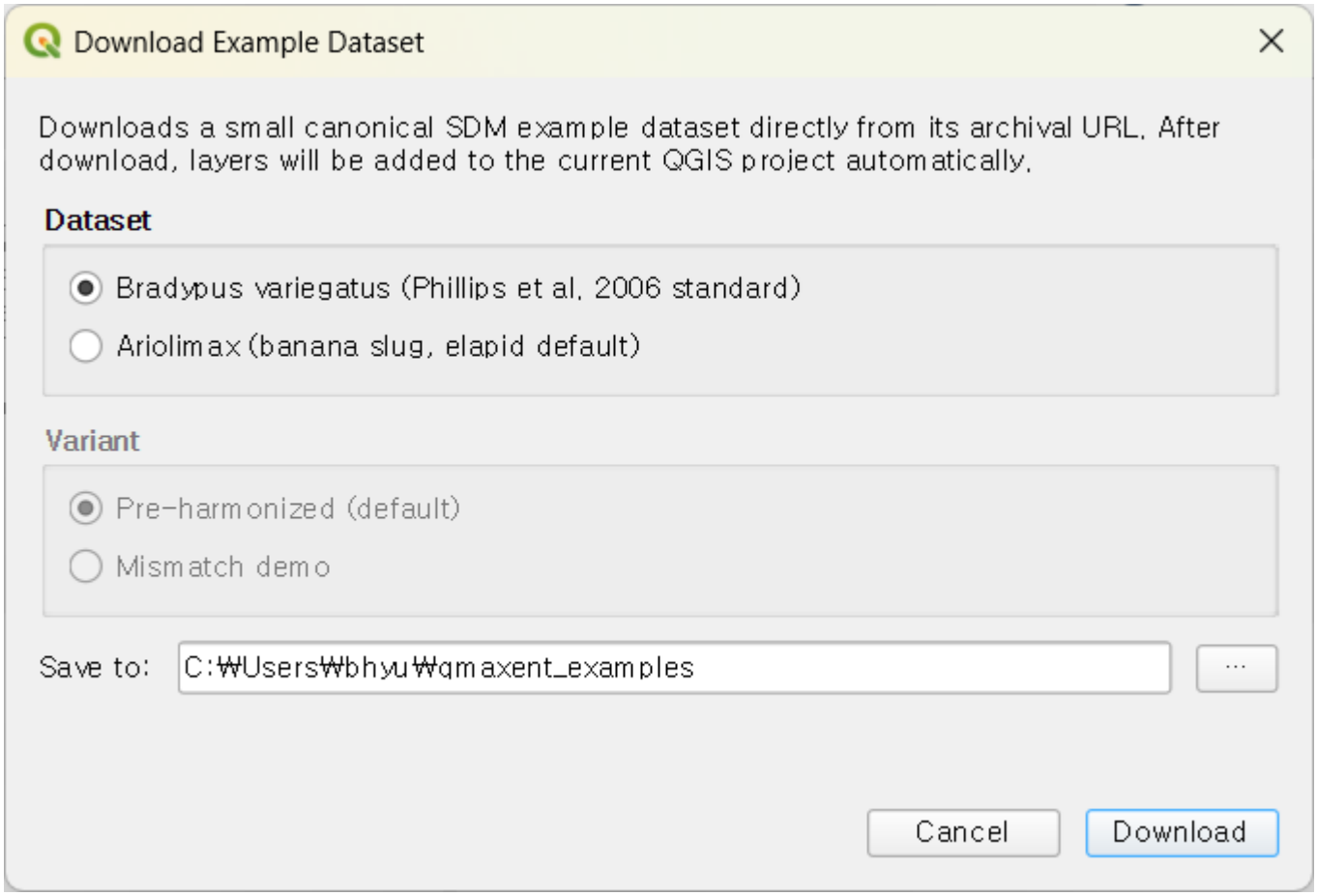
The Phillips et al. (2006) dataset, as distributed with the R package dismo, contains 116 presences, 8 bioclimatic variables and one categorical biome raster (9 rasters in total):

Layer	Type	Description
`bradypus`	Vector point	116 occurrence records across South and Central America
`bio1, bio5, bio6, bio7, bio8`	Continuous raster	Temperature variables (WorldClim)
`bio12, bio16, bio17`	Continuous raster	Precipitation variables (WorldClim)
`biome`	Categorical raster	Biome type (Olson et al. 2001)

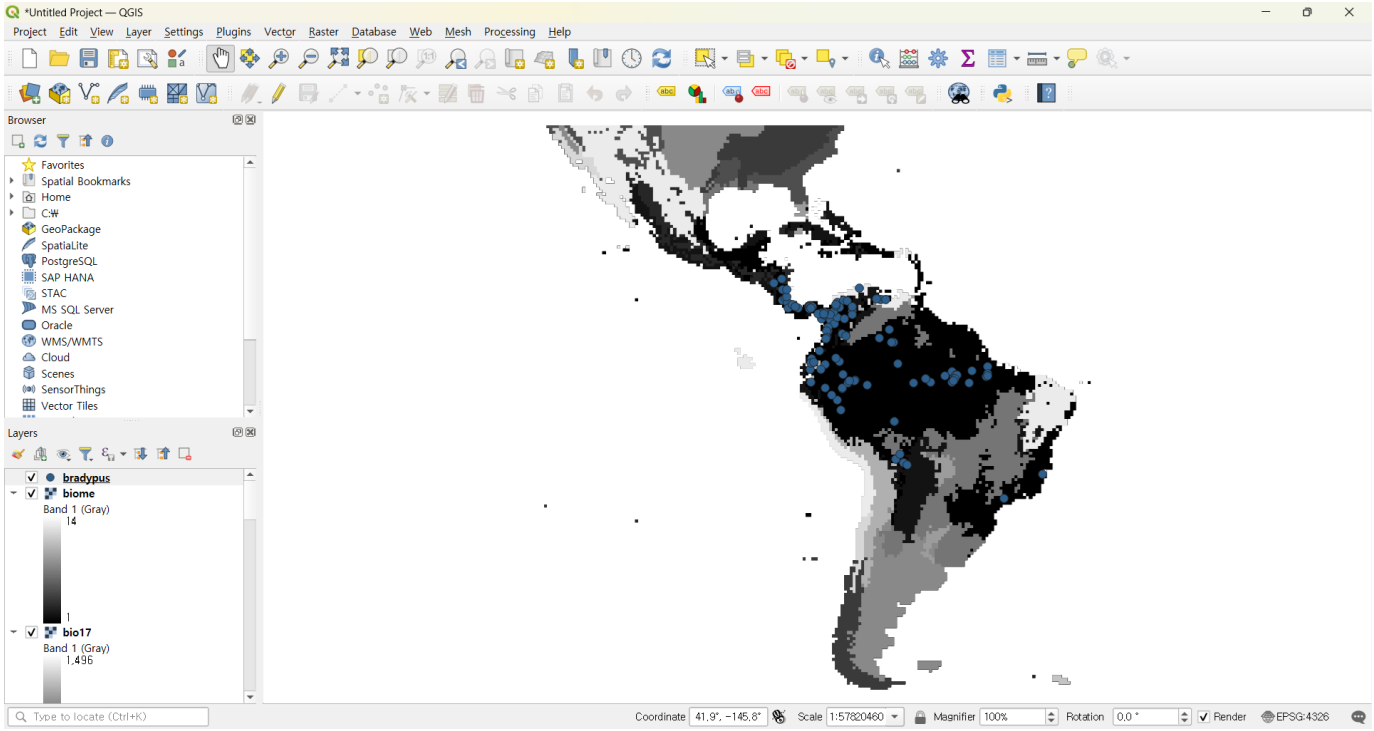
All rasters share the same grid: EPSG:4326, 0.5° × 0.5° cells, full Americas coverage. Total dataset size < 100 MB.

Download via **Plugins → QMaxent → Download Example Dataset**, pick **Bradypus variegatus (Phillips et al, 2006 standard)**, choose a destination folder, and click **Download**:

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Layers are added to the QGIS project automatically:



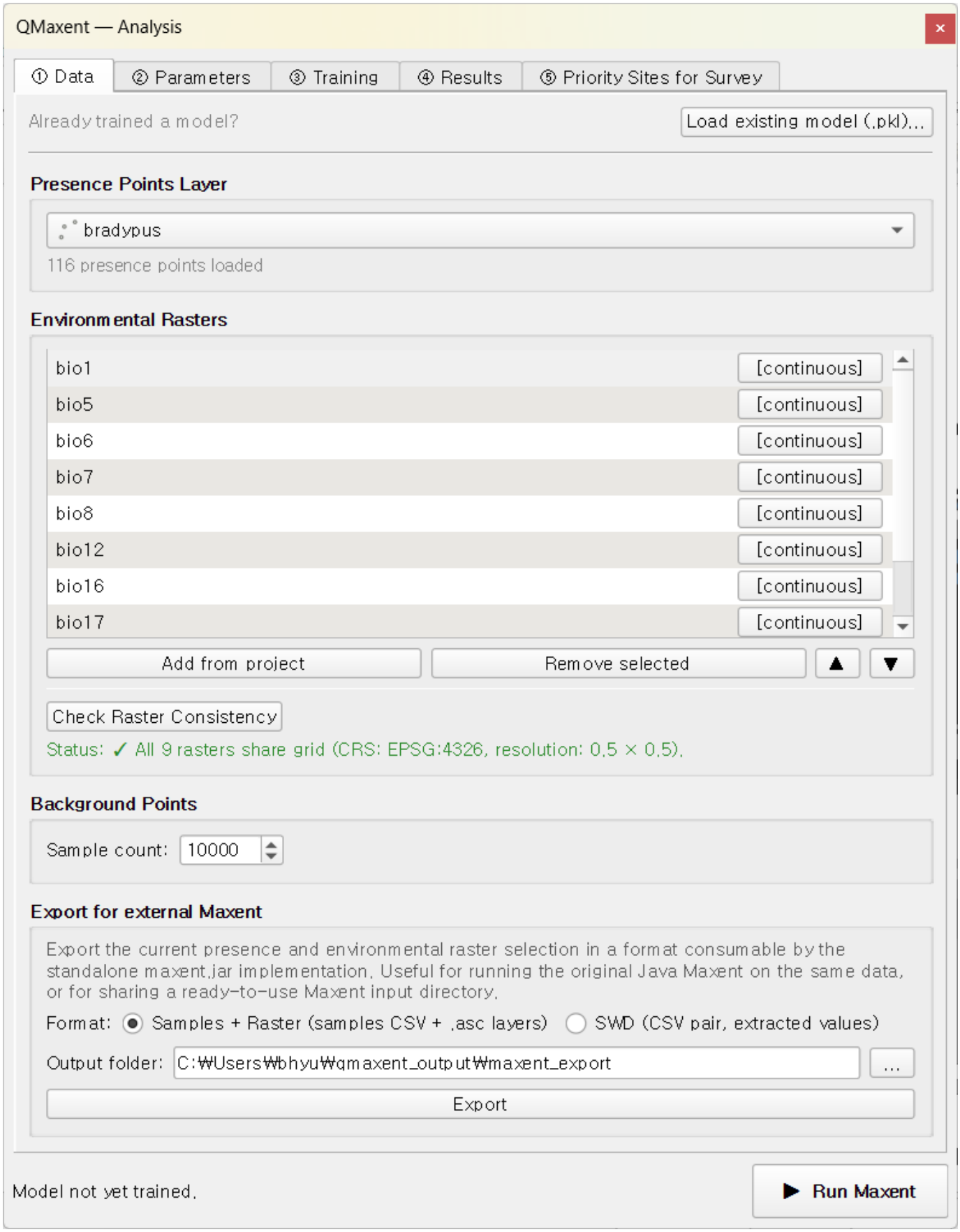
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16.3 2. Loading data into the Analysis dock

Open **Plugins → QMaxent → QMaxent Analysis**. On ① **Data**, pick `bradypus` from the **Presence Points Layer** drop-down — QMaxent immediately reports `116 presence points loaded`.

Click **Add from project** to register every loaded raster at once. The `biome` row gets a `[categorical]` tag (its sidecar metadata says so). Click **Check Raster Consistency** to verify the grid; the status line should read `✓ All 9 rasters share grid (CRS: EPSG:4326, resolution: 0.5 × 0.5)`, exactly what you expect from the bundled dataset:

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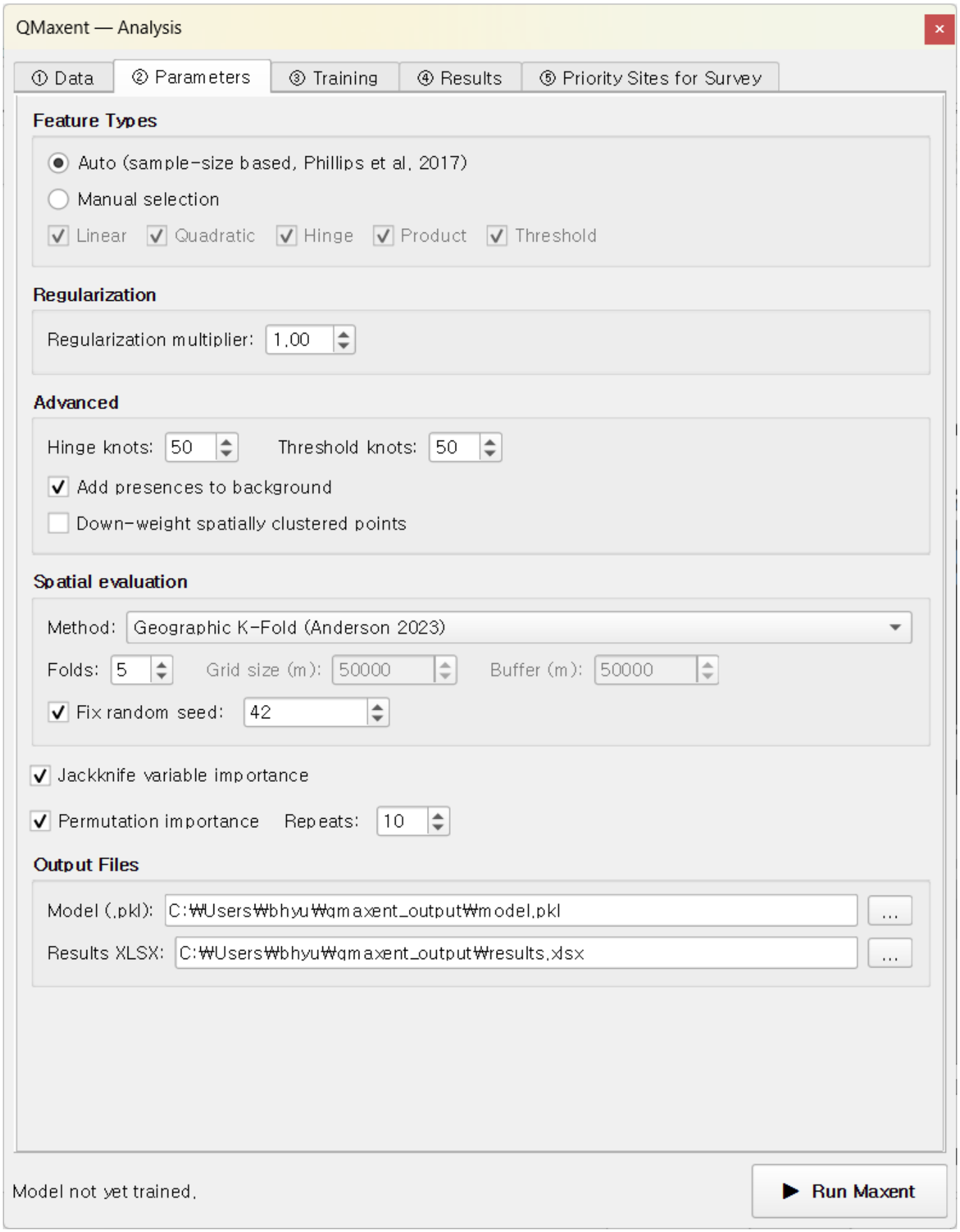
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Background Points is left at its default of 10,000, the value Phillips & Dudík 2008 recommend for continental extents. The **Export for external Maxent** panel at the bottom is optional — leave it alone for this tutorial; see Exporting results for when it matters.

16.4 3. Model setup

Switch to ② **Parameters**. For this tour we accept every default — each default is the literature-recommended value, and accepting them lets us inspect what those choices produce:

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- **Feature Types**: **Auto (maxnet default rule by sample size)**. The maxnet default (Phillips et al. 2017) gives LQPH (linear, quadratic, product, hinge) for 80 or more presences, so LQPH for 116 presences. Threshold features are not added automatically (see ② Parameters tab).
- **Regularization multiplier**: 1.0 (Phillips & Dudík 2008 recommendation).
- **Spatial evaluation**: Geographic K-Fold (Anderson 2023), 5 folds, grid 50,000 m, buffer 50,000 m, **Fix random seed** = 42 (Roberts et al. 2017 default). Since v0.2.0 the seed also fixes the background sample.
- **Jackknife variable importance**: enabled.
- **Permutation importance**: enabled, 10 repeats, **Evaluate on: Training data (maxent.jar convention)**.
- **Output files**: `qmaxent_output/model.pkl` and `qmaxent_output/results.xlsx`.

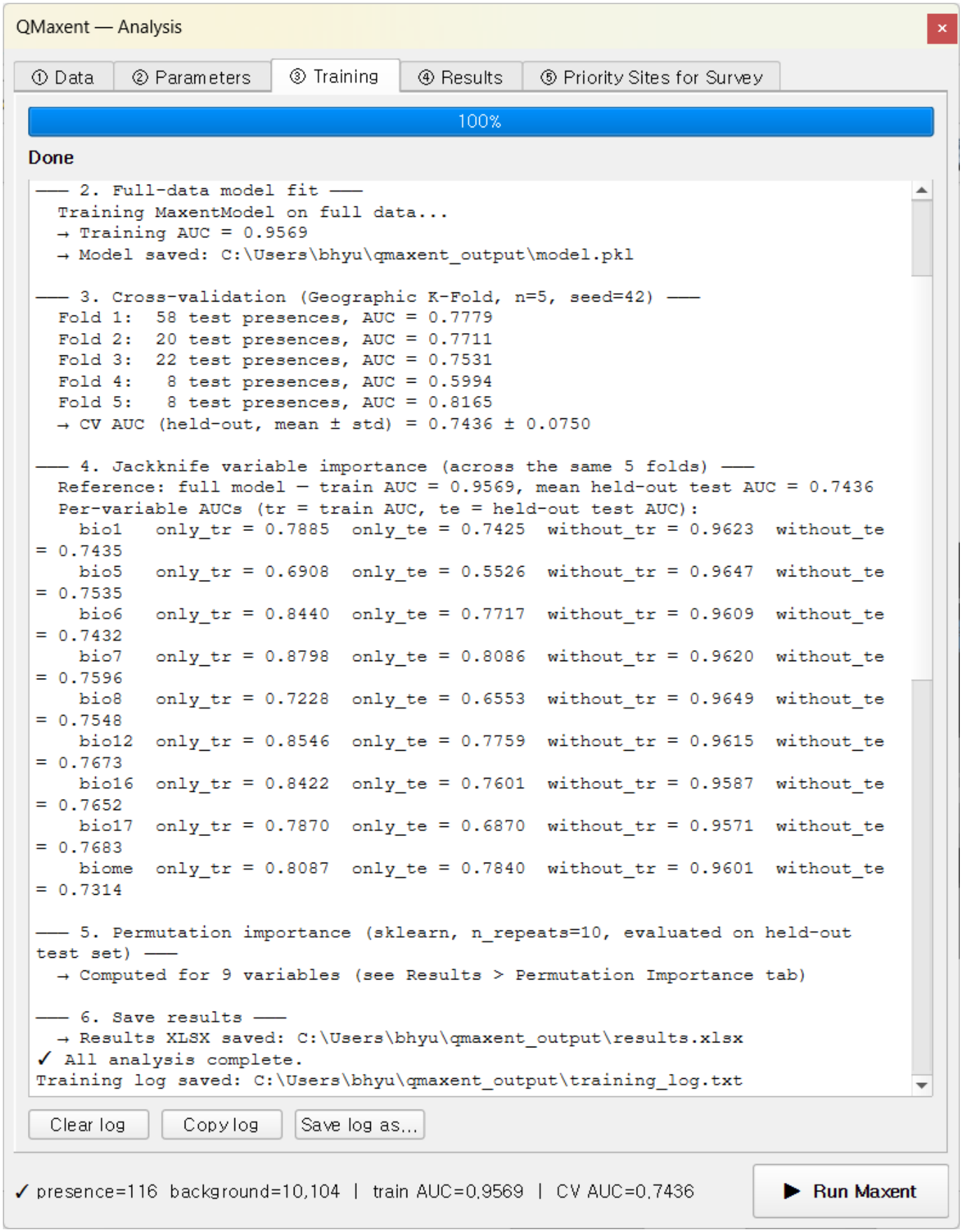
The fixed random seed means **anyone re-running this tutorial gets bit-identical results** — central to computational reproducibility (Araújo et al. 2019).

16.5 4. Running training and cross-validation

Click ▶ **Run Maxent**. The ③ **Training** tab takes over and finishes in about 30 seconds:

> **Numbers from an earlier version**
>
> The AUC values and fold sizes in this section were recorded with QMaxent v0.1.x, which used LQPHT under Auto and drew a new background sample on every run. With v0.2.0 (LQPH, seeded background) your values will differ slightly; the interpretation is unchanged.

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The status bar at the bottom summarises the run: `presence=116 background=10,104 | train AUC=0.9569 | CV AUC=0.7436`. Reading the log section by section:

- **Full-data model** — `Training AUC = 0.9569`, saved to `qmaxent_output/model.pkl`.
- **Cross-validation** — Geographic K-Fold n=5, seed=42:

Fold	Test presences	AUC
1	58	0.7779
2	20	0.7711
3	22	0.7531
4	8	0.5994
5	8	0.8165

Pooled mean ± std = **0.7436 ± 0.0750**.

- **Jackknife variable importance** — per-variable AUC across the same 5 folds in three modes (full-model reference, *only this variable*, and *without this variable*). The numbers feed the Jackknife plot below.
- **Permutation importance**: sklearn `permutation_importance` with `n_repeats=10`, evaluated on the training data (presences and background), as maxent.jar does. Results feed the Permutation Importance sub-tab.
- **Save results**: `results.xlsx`, `training_log.txt`, the model card (`model_model_card.json`) and the coefficient table (`model_coefficients.csv`) are persisted next to the `.pkl`.

Train AUC = 0.957 while **CV AUC = 0.744 ± 0.075**. That gap is the cost of holding out spatially-distinct validation sets — and a far more honest measure of real-world predictive performance than the inflated training AUC alone, exactly the point Roberts et al. 2017 make.

Folds 4 and 5 have the smallest validation set (8 presences each) and the largest spread in AUC. In spatial CV, folds are intentionally uneven in area, and one fold can land on a small, atypical region. The pooled CV AUC averages over this variance — the **± 0.075 standard deviation** is precisely the quantity you would cite alongside the mean in a publication.

The full log is exportable via **Save log as…** at the bottom of the tab; **Copy log** drops it into the clipboard, handy for pasting into an issue report.

16.6 5. Inspecting variable behaviour

Response curves

On ④ **Results → Response Curves**, pick `bio12` (annual precipitation):

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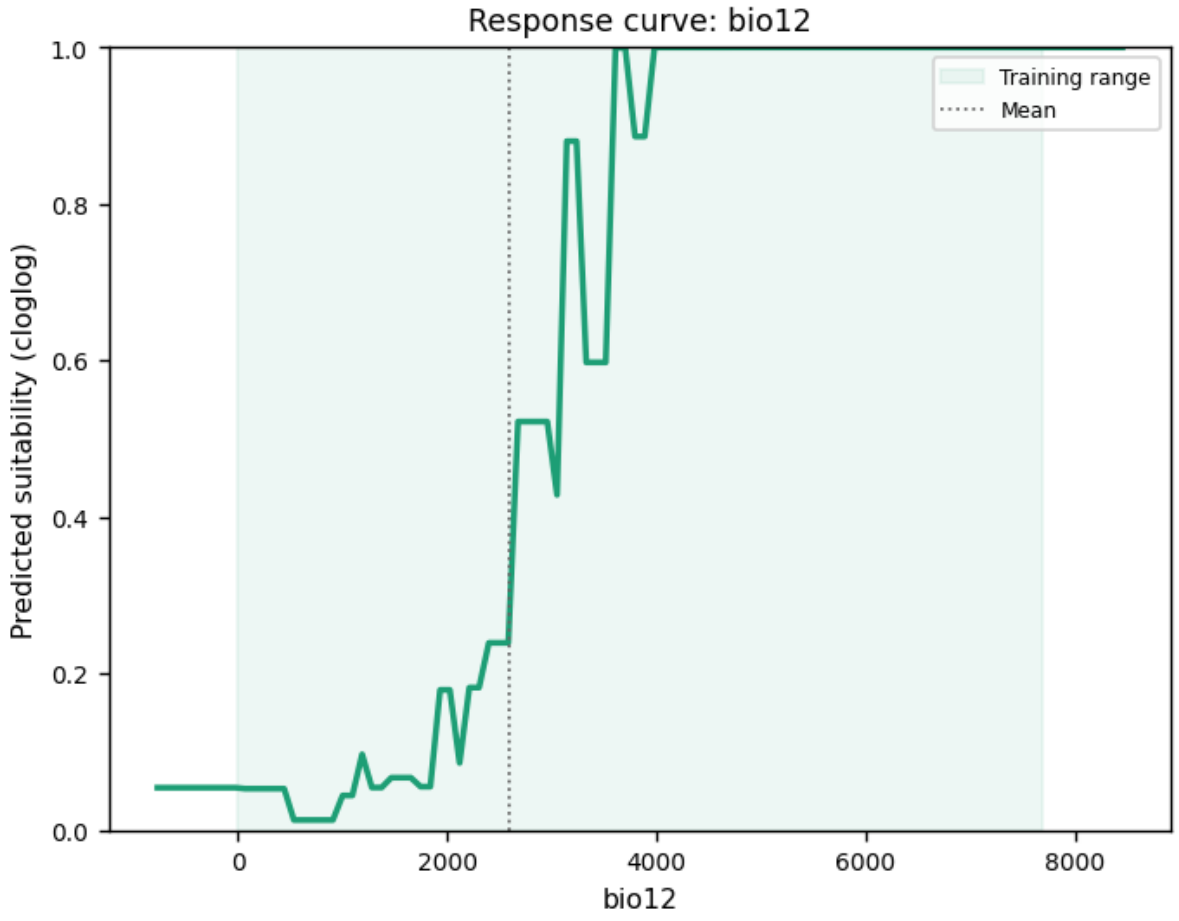
QMaxent — Analysis

① Data | ② Parameters | ③ Training | ④ Results | ⑤ Priority Sites for Survey

Response Curves | Jackknife Importance | Permutation Importance | Spatial Projection

Marginal effect: how the predicted suitability changes across the range of one variable while the other variables are held at their training-set means. The Y-axis reads as relative habitat suitability (cloglog: 0 to 1, saturates near 1 in the most suitable region). The shaded green band marks the variable's training-data range — curves outside it are extrapolation and should be interpreted with caution; the dotted line marks the training-data mean. Useful for inspecting whether the model has learned ecologically plausible shapes (e.g. unimodal temperature optima) and for spotting variables whose curves are nearly flat — candidates for removal in a parsimonious model.

Variable: bio12



✓ presence=116 background=10,104 | train AUC=0.9569 | CV AUC=0.7436

▶ Run Maxent

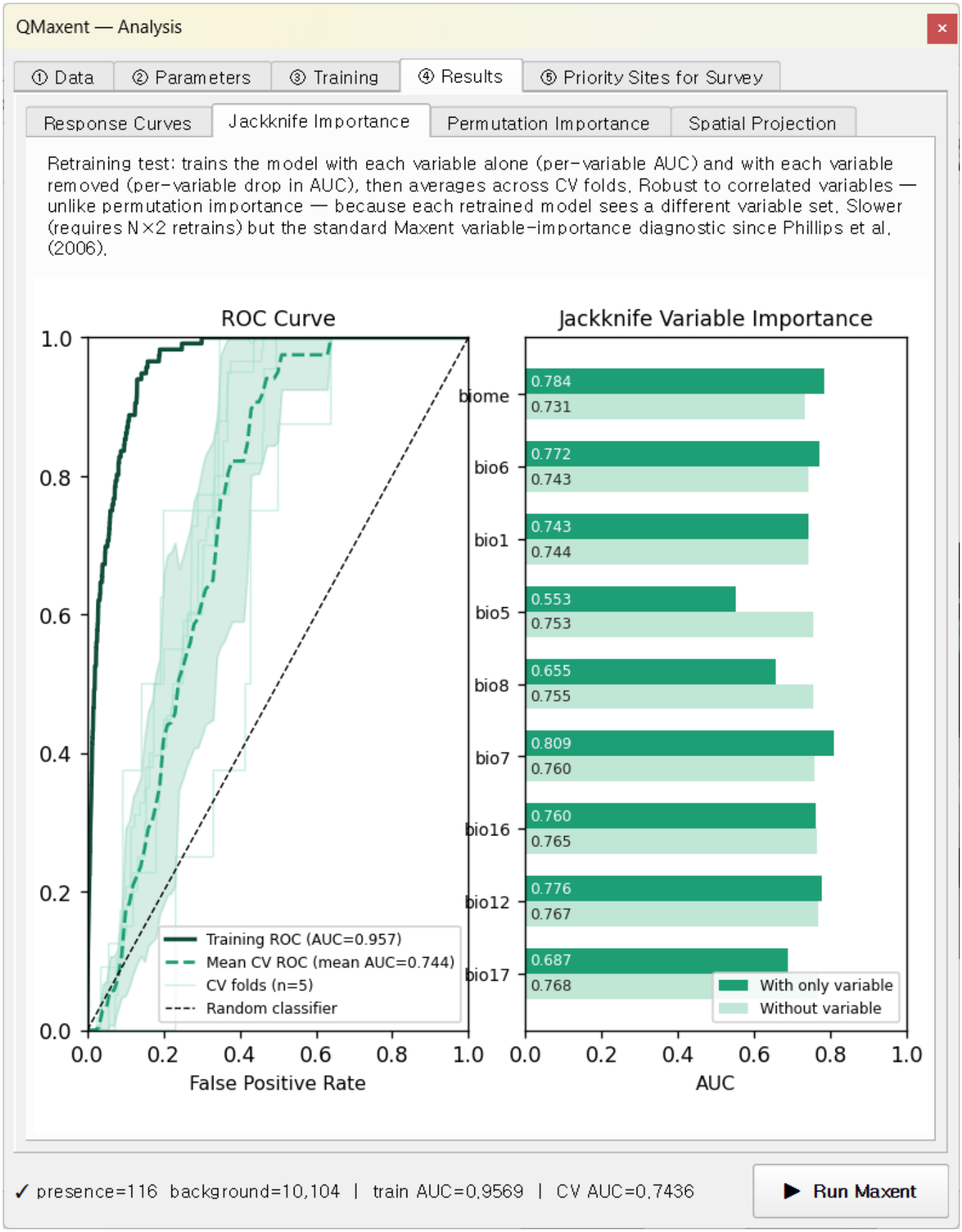
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The model assigns highest suitability to the 1,500–3,500 mm/year band — the climatology of South American rainforest — with a sharp drop-off below ~800 mm. The shape combines hinge and quadratic features. Try other variables in the drop-down: smooth U- or peak-shaped curves indicate quadratic terms; angular kinks come from hinge features. (Step-shaped jumps would come from threshold features, which Auto no longer uses.)

Jackknife importance

The **Jackknife Importance** sub-tab compares each variable's stand- alone signal against its incremental contribution. Dark bars (*Only this variable*) and light bars (*Without this variable*) tell you each variable's unique value:

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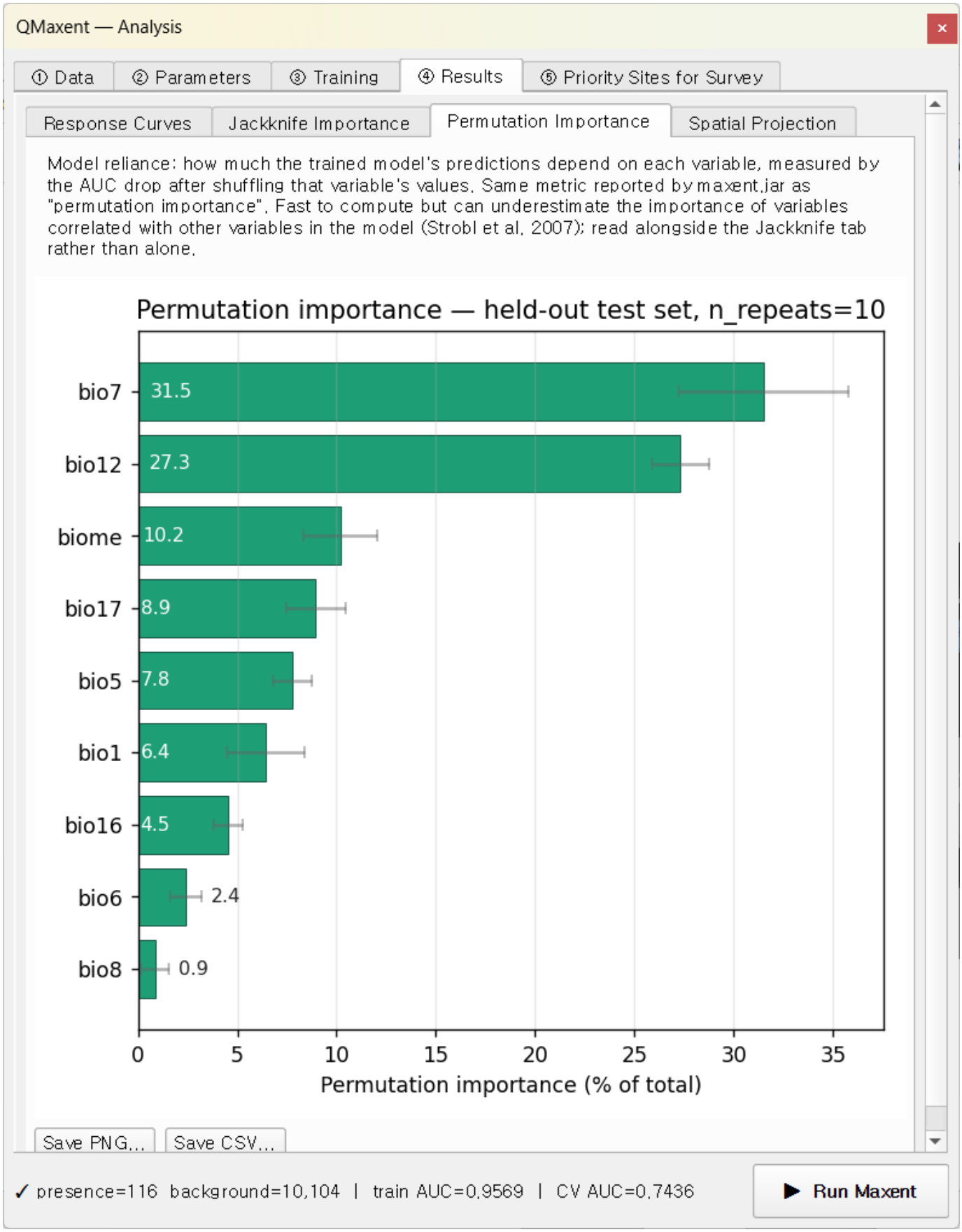
For Bradypus:

- **`bio7`** (annual temperature range) and **`bio12`** (annual precipitation) have the strongest stand-alone signals.
- **`biome`** and **`bio6`** (min temp of coldest month) come next.
- The "without" bars are tightly bunched in the high-0.9s — Maxent recovers from removing any single variable because the climate variables are correlated. This is the textbook Phillips, Anderson & Schapire 2006 pattern.

Permutation importance

The **Permutation Importance** sub-tab reports the same idea from a different lens: scikit-learn's `permutation_importance` shuffles each variable's values, measures the AUC drop, repeats 10 times, and normalises to 100% of the total. With the default **Evaluate on: Training data (maxent.jar convention)**, the AUC drop is measured on the training presences and background, as in maxent.jar. Choose **Held-out CV test folds (all folds pooled)** to measure it on the CV test folds instead:

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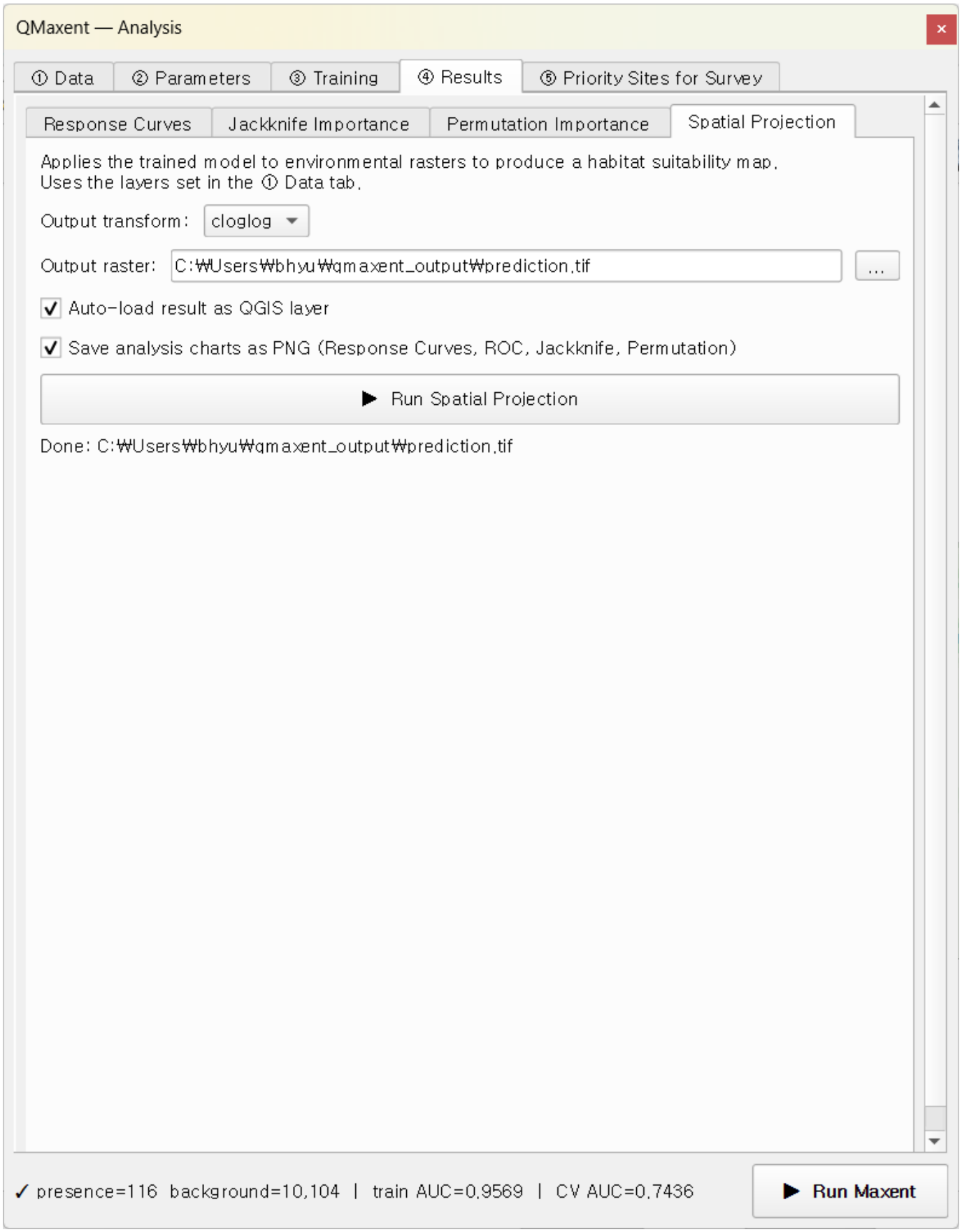
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`bio7` and `bio12` again dominate; the permutation view distributes the total importance across all variables and, evaluated on the training data, is directly comparable to maxent.jar's per-variable percentage table.

16.7 6. Spatial projection

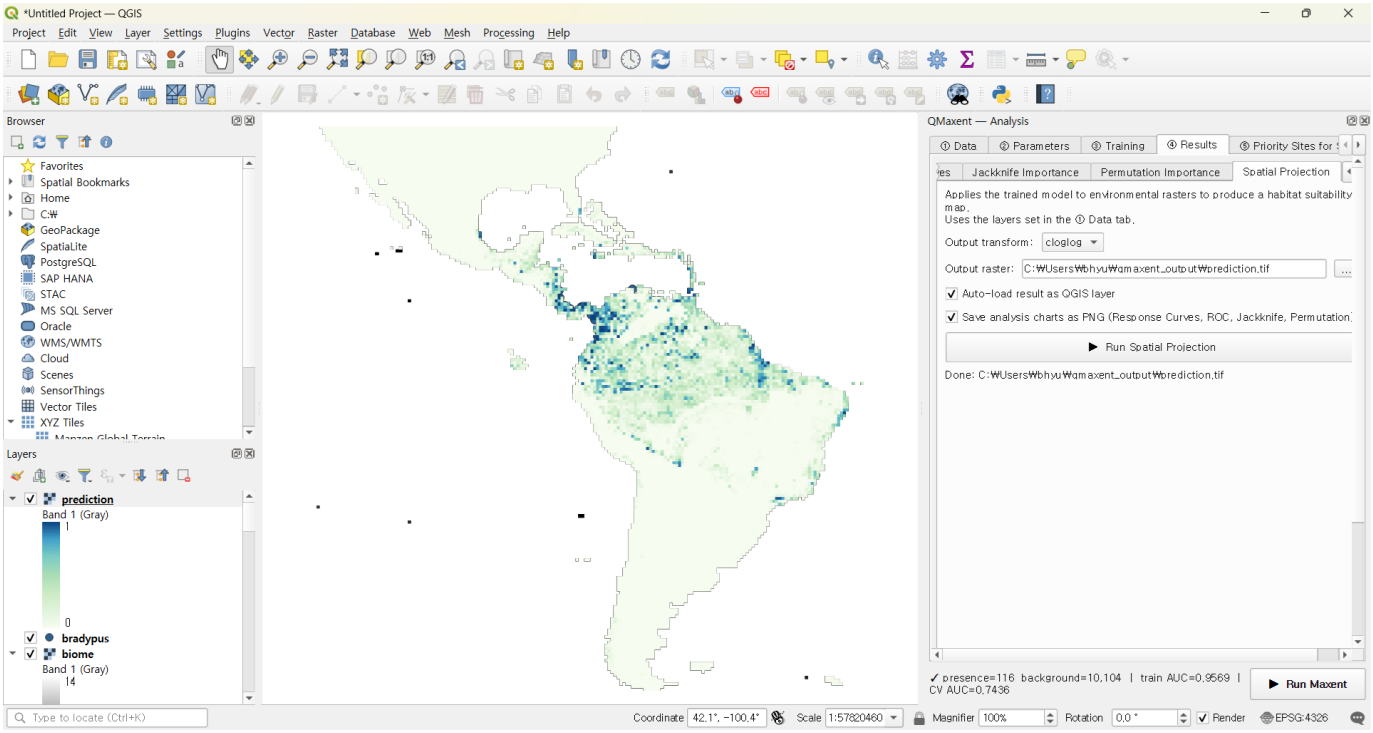
Switch to **Spatial Projection** in the same Results tab. Leave **cloglog** as the output transform (the Phillips et al. 2017 recommended default) and **Auto-load result as QGIS layer** ticked, then click ▶ **Run Spatial Projection**:

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The map appears in QGIS with auto-styled white-to-green ramping:

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The high-suitability core covers Brazil's southeastern Atlantic Forest and the Amazon basin — both well-known sloth strongholds — plus secondary patches across Central America. The model correctly identifies the unsuitability of the Andes (cold, high altitude) and the very dry Brazilian Northeast (Caatinga).

16.8 7. Saving outputs

These files are written automatically:

- `qmaxent_output/model.pkl` — the serialised trained model. Reload it later from the Data tab's **Load existing model (.pkl)...** button or share it with trusted collaborators. Security note in Saving and reusing models.
- `qmaxent_output/model_model_card.json` and `qmaxent_output/model_coefficients.csv`: a plain-text record of the run (software versions, settings, data, performance, thresholds, variable importance) and every Maxent feature with its coefficient. Unlike the `.pkl`, both are safe to share.
- `qmaxent_output/results.xlsx` — the multi-sheet supplementary table containing experimental setup, variable list, CV results, jackknife, permutation, response-curve breakpoints, and threshold tables. See Exporting results for the sheet-by-sheet layout.

If you ticked **Save analysis charts as PNG** before projection, four additional 300-dpi PNGs of the response curves, ROC, jackknife, and permutation panels are written next to the GeoTIFF — sized for direct paste into a single-column manuscript figure.

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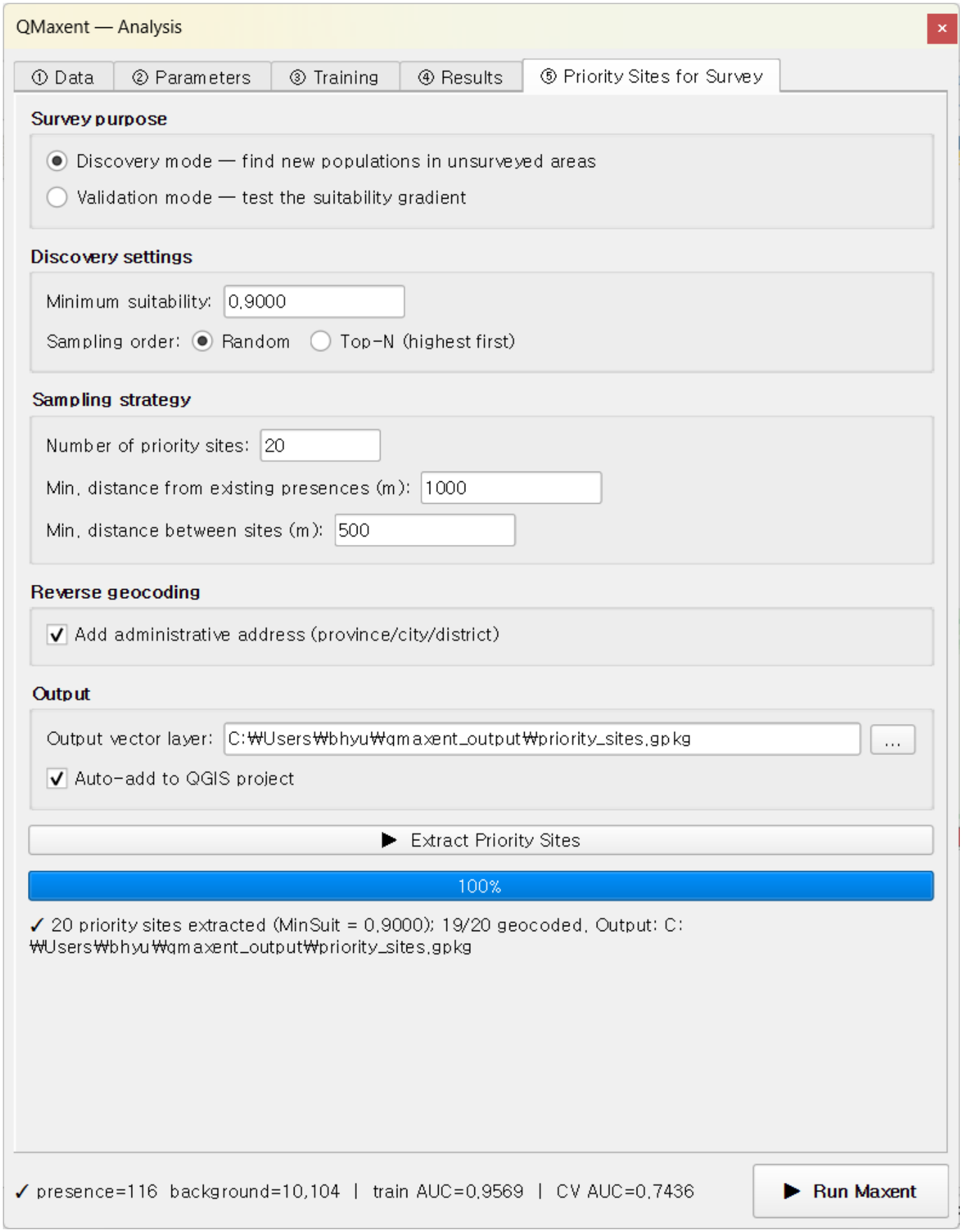
16.9 8. Survey site selection

A natural next step is to use the trained model to plan follow-up surveys. ⑤ **Survey Site Selection** offers two modes: **High-suitability search** (looking for *new* populations) and **Gradient-stratified sampling** (field checks across the suitability gradient). See ⑤ Survey Site Selection for all settings.

8.1 High-suitability search

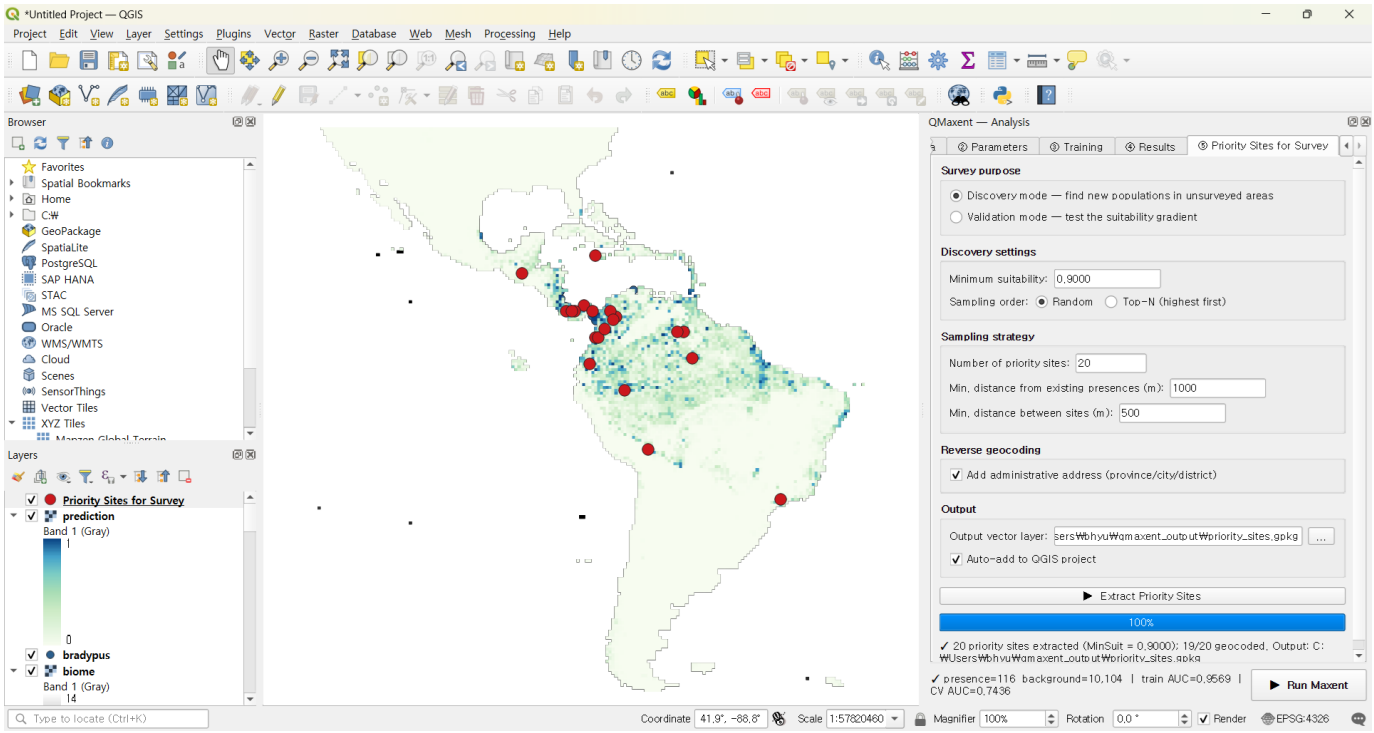
High-suitability search samples sites from a high-suitability band. Open the tab, keep **High-suitability search** as the mode, leave the automatic **Minimum suitability** (0.9 × the raster maximum) and **Sampling order: Random**, keep 20 sites and the 1,000 m / 500 m spacing defaults, and click ▶ **Select Survey Sites**:

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20 sites (red dots) appear on the suitability map as the layer **Survey sites** (`survey_sites.gpkg`), with their attribute table populated by OpenStreetMap reverse geocoding. With the default **Address language: Local names**, the addresses are in the local language of each country (Spanish or Portuguese across most of the Bradypus range):

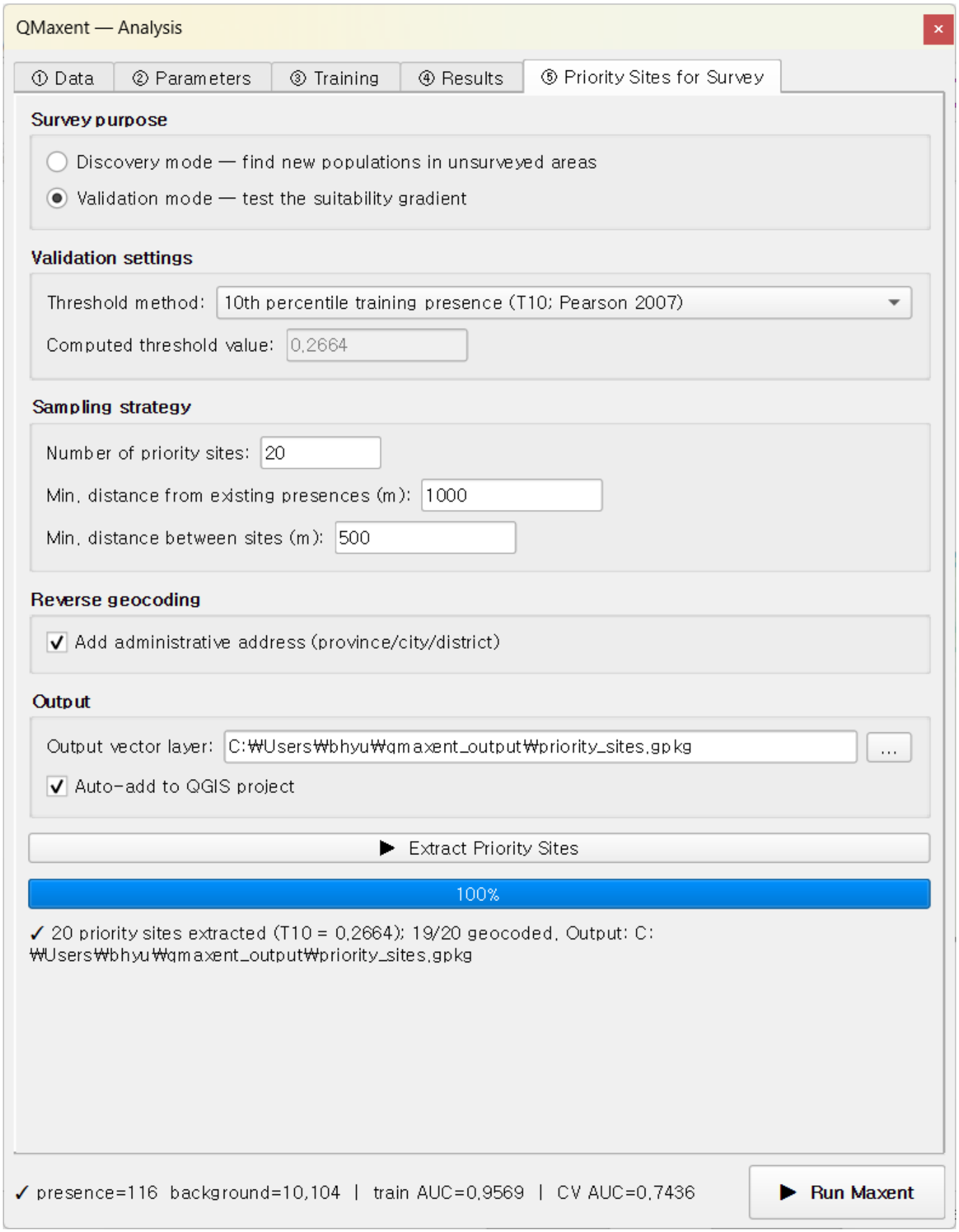


Each site is at least 1 km from any known occurrence and at least 500 m from any other site, so a single field trip can plausibly cover several at once. High-suitability search is the right mode when the question is *"where could the species be that we have not yet looked?"*

8.2 Gradient-stratified sampling

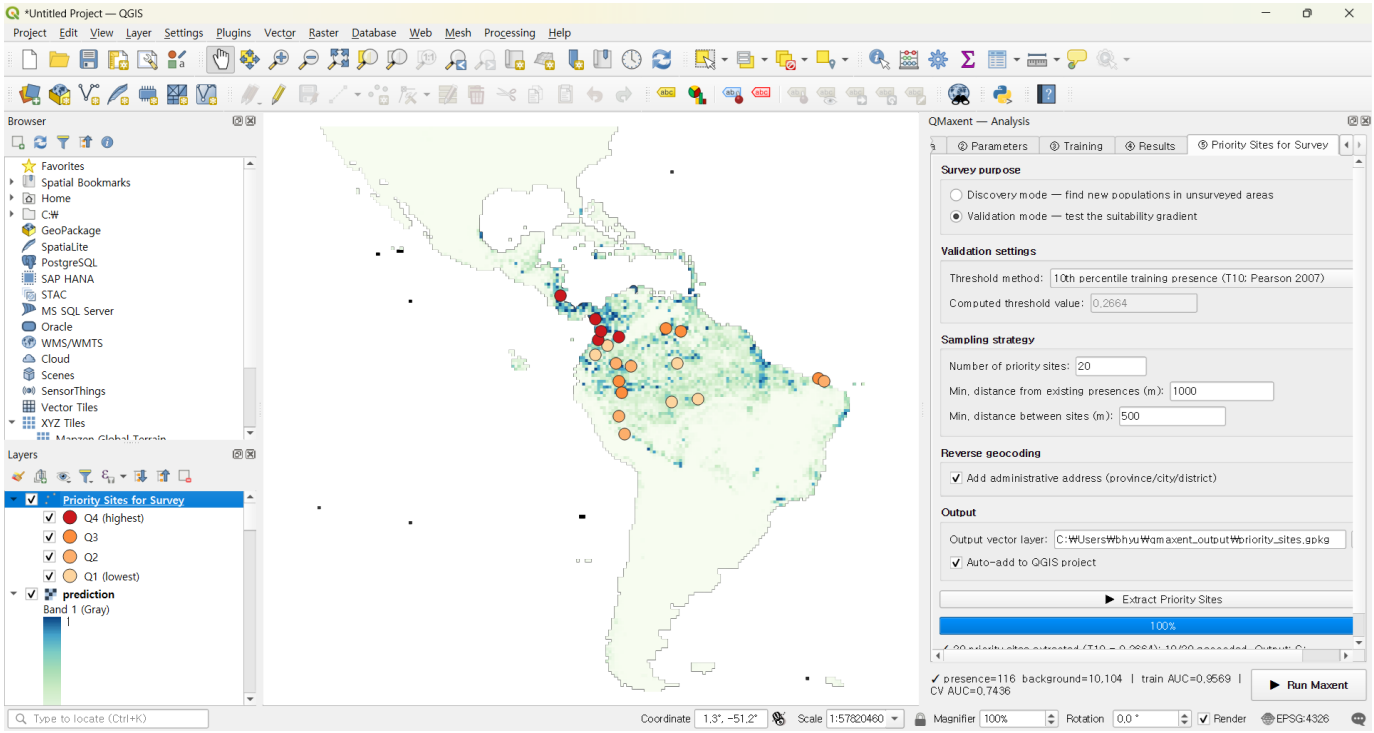
Switching the **Survey purpose** to **Gradient-stratified sampling** draws a different kind of sample. A threshold (T10 by default; MTP, MaxSSS or a custom value are also offered) only defines the sampling frame. The cells above it are split into four suitability quartiles, and the same number of sites is drawn from each (Rhoden et al. 2017). No binary map is produced. A field trip can then test whether the species is found more often where the model predicts higher suitability, rather than visiting only the high-suitability core:

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The resulting sites span low- through high-suitability cells in roughly equal numbers, each tagged with its quartile in the attribute table:



Gradient-stratified sampling is the right mode for checking the model in the field rather than for discovery: it is the field-side complement of the cross-validation AUC we computed in § 4.

16.10 Next steps

- **The same workflow with messy rasters**: Ariolimax example starts from rasters that do not share a CRS or resolution — exercising the Check + Harmonize tools.
- **Compare your workflow to a published study**: Pitta nympha example reproduces a published Java MaxEnt analysis in QMaxent and discusses where the two pipelines agree and differ.
- **Deeper theory**: Methodological background explains why each default we accepted in this tour is the right choice.

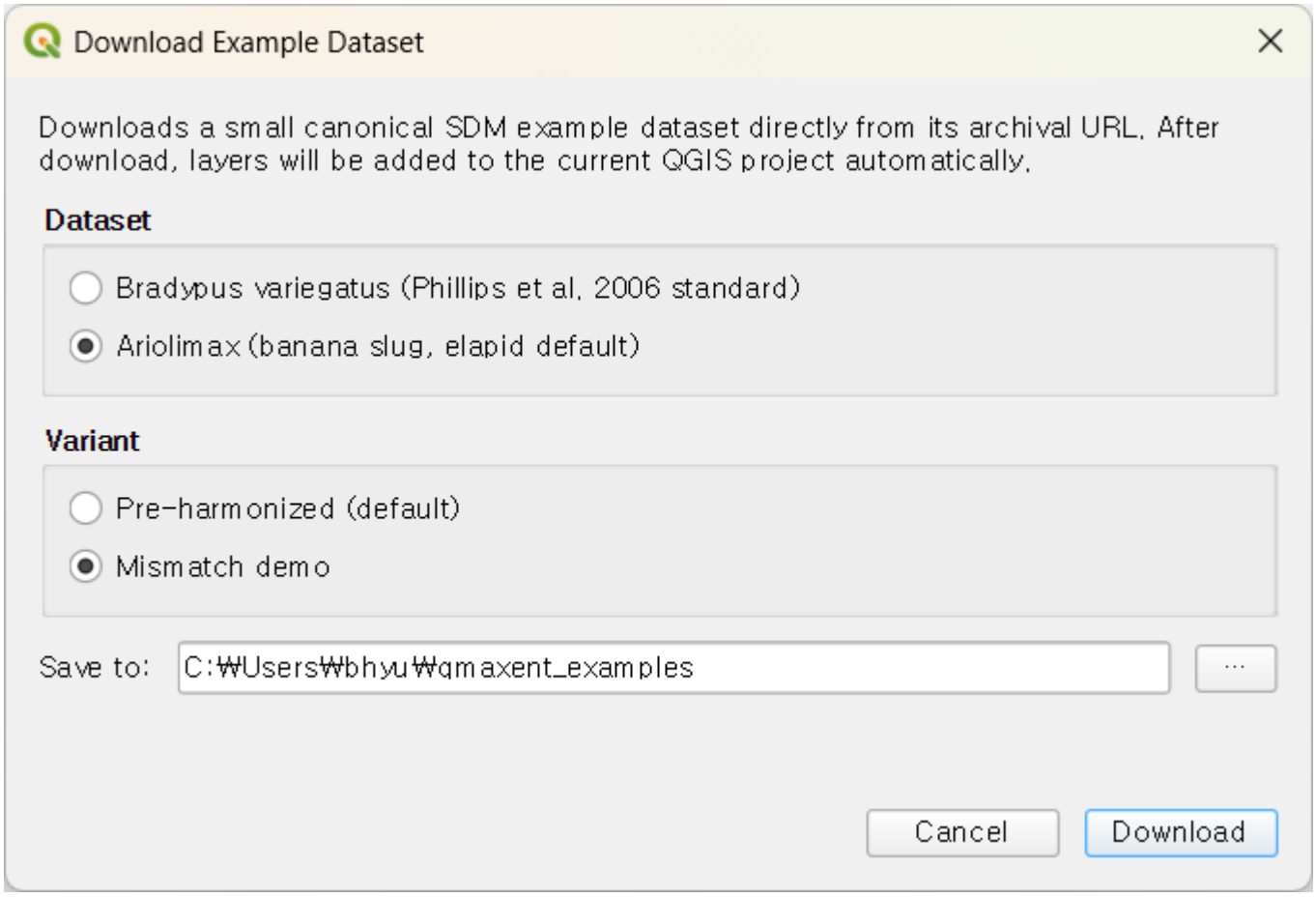
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17. Ariolimax

The Pacific banana slug *Ariolimax columbianus* is the second worked example. Its purpose is different from Bradypus's feature tour — this dataset is deliberately *messy*: the supplied environmental rasters do **not** share a common CRS, extent, or resolution. The example walks through QMaxent's **Check Raster Consistency** preflight and **Harmonize to Folder...** workflow, showing the silent-failure mode you would otherwise hit and the one-click fix that makes the data Maxent-ready.

17.1 1. Dataset

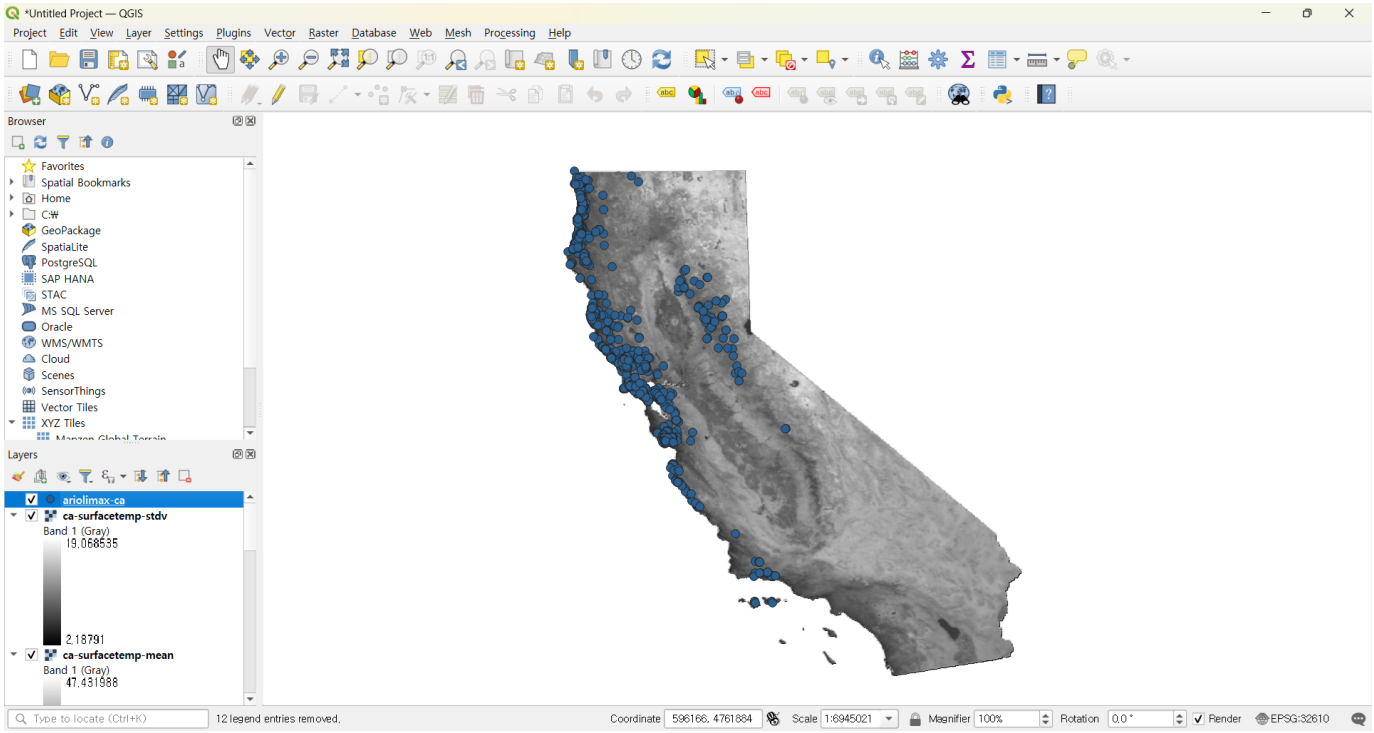
The Ariolimax dataset is the default that ships with elapid. It comes in two variants chosen via the **Download Example Dataset** dialog:



- **Pre-harmonized (default)** — the same six rasters already reprojected and resampled onto a common grid. Use this if you want to jump straight to model fitting.
- **Mismatch demo** — the *original* tiles with their original CRS, extent, and resolution intact. Use this if you want to exercise the Check + Harmonize tooling.

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This walkthrough uses the **Mismatch demo** variant. After clicking **Download**, the layers appear on the QGIS canvas spanning California's coastal range:

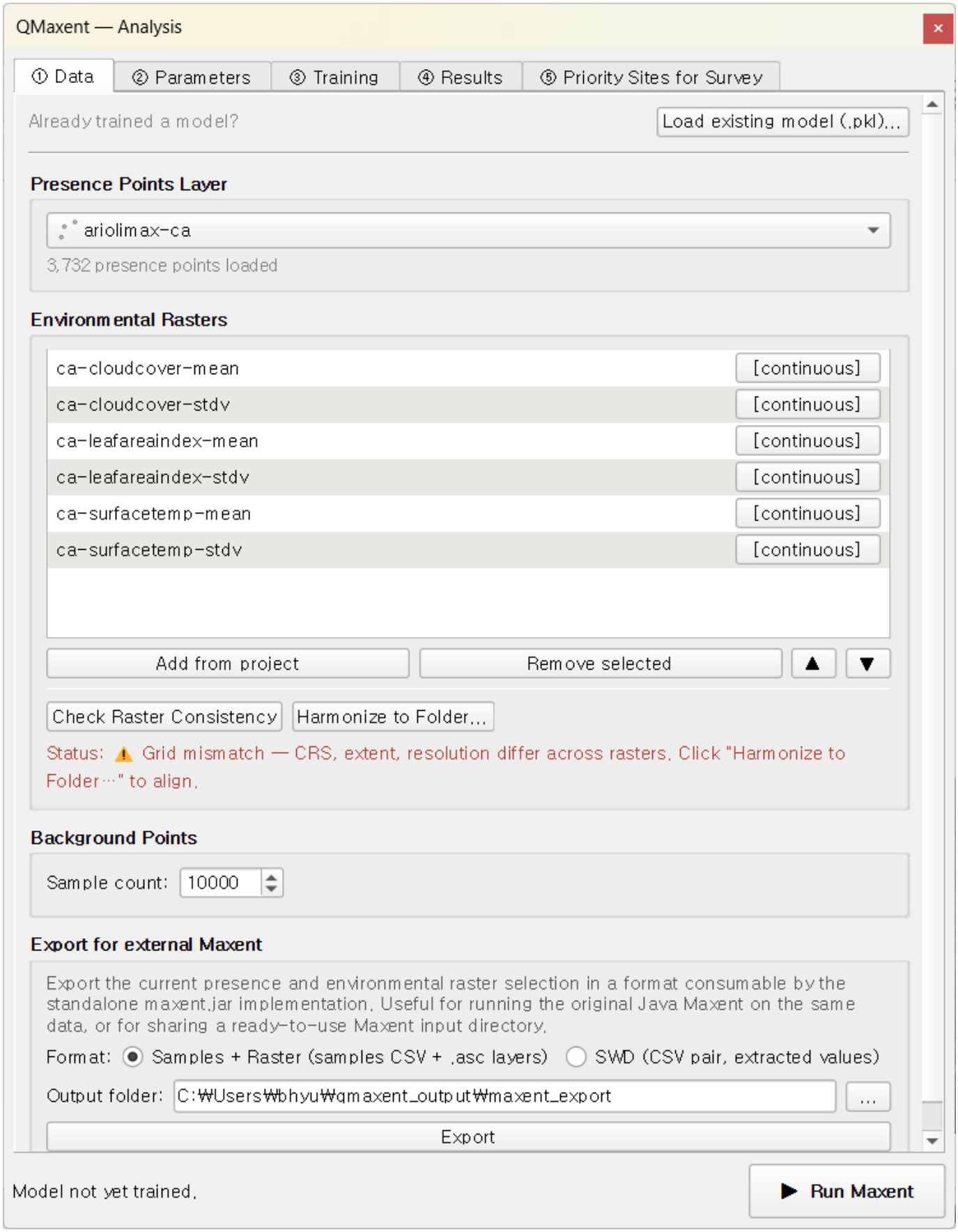


Visually the data already looks unified. The raster tiles, however, were authored by different remote-sensing pipelines and inherit different projections and resolutions — exactly the situation that breaks Maxent silently.

17.2 2. The mismatch problem

Open **Plugins → QMaxent → QMaxent Analysis**. On ① **Data**, choose the `ariolimax-ca` presence layer (3,732 points), then **Add from project** to register every loaded raster (six in total — `ca-cloudcover-mean`, `ca-cloudcover-stdv`, `ca-leafareaindex-mean`, `ca-leafareaindex-stdv`, `ca-surfacetemp-mean`, `ca-surfacetemp-stdv`). Click **Check Raster Consistency**:

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The status line turns amber and reports:

⚠ Grid mismatch — CRS, extent, resolution differ across rasters. Click "Harmonize to Folder..." to align.

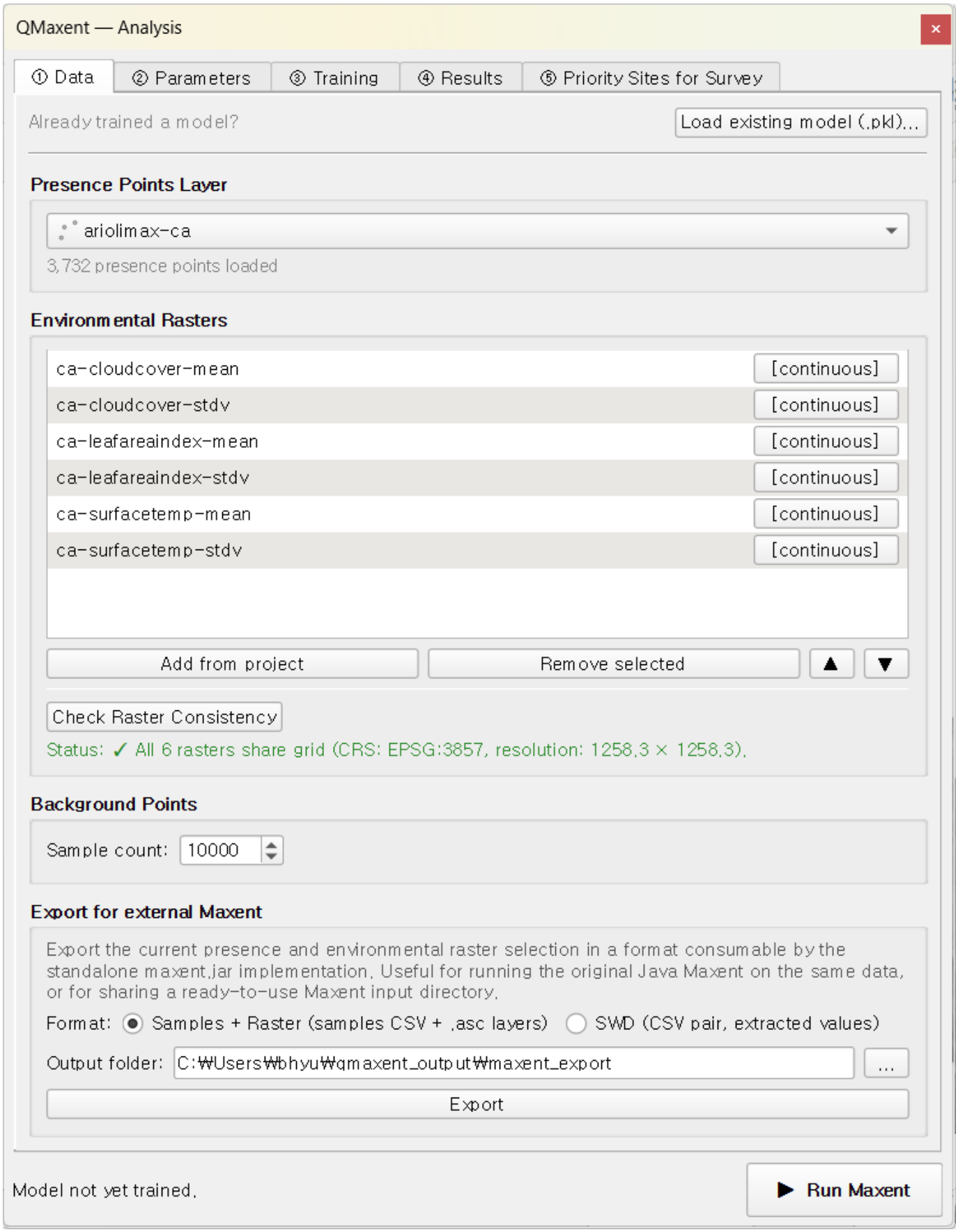
Crucially, **the Run Maxent button is not blocked** — Maxent itself would still produce numbers. Those numbers, however, would be silently wrong: covariates would be sampled from the cells *nominally* underneath each presence point but actually belonging to misaligned rasters. This is the single most common silent-failure mode in operational SDM and the entire reason this preflight exists.

17.3 3. Running Harmonize to Folder…

A new button appears next to **Check Raster Consistency** as soon as a mismatch is detected: **Harmonize to Folder...**. Click it and choose an output directory. QMaxent picks the **highest-resolution** raster as the reference grid, reprojects every other raster to that grid via `gdalwarp` under the hood (nearest-neighbour for categoricals, bilinear for continuous), and writes new GeoTIFFs into the chosen folder. The new files are auto-loaded into the project and the old ones are removed from the QMaxent raster list.

The Data tab refreshes to show the harmonized stack:

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The status line is now green:

✓ All 6 rasters share grid (CRS: EPSG:3857, resolution: 1258.3 × 1258.3).

Harmonized rasters get a numeric prefix (`00_`, `01_`, …) that locks their order. This survives a `.qgz` save+reload cycle — the model variable order is part of the model's identity, and the prefix makes that order visible at the file-system level too.

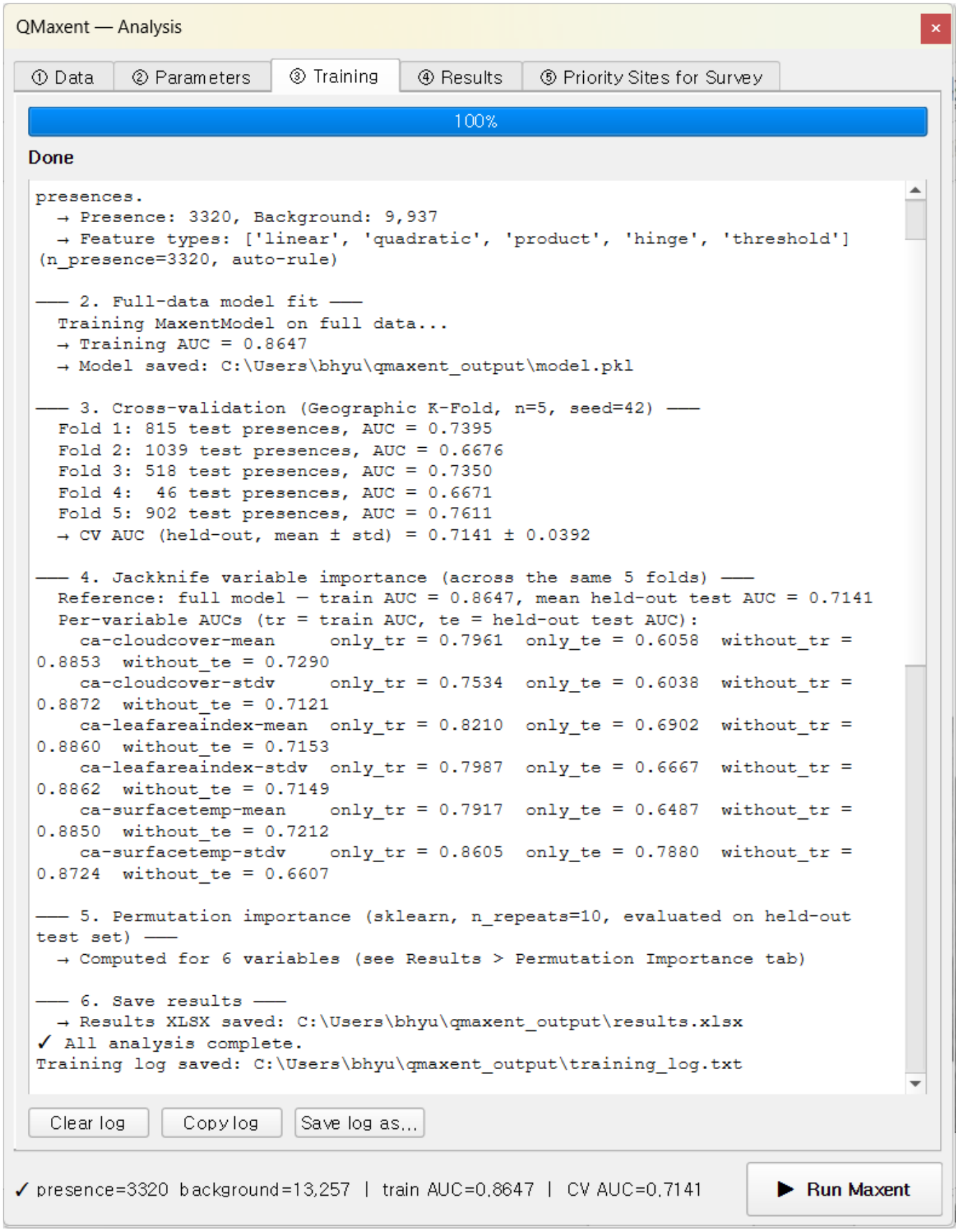
17.4 4. Running the model

With the stack harmonized, the rest of the workflow is identical to Bradypus. Accept the defaults on ② **Parameters**, click ▶ **Run Maxent**, and let the training complete:

Numbers from an earlier version

The values in this section were recorded with QMaxent v0.1.x, which used LQPHT under Auto and drew a new background sample on every run. v0.2.0 uses LQPH for these 3,000+ presences and a seeded background, so your values will differ slightly.

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The status bar at the bottom summarises:
`presence=3320 background=13,257 | train AUC=0.8647 | CV AUC=0.7141`. Reading the log:

- **Full-data model** — `Training AUC = 0.8647`.
- **Cross-validation** — Geographic K-Fold n=5, seed=42:

Fold	Test presences	AUC
1	815	0.7395
2	1,039	0.6676
3	518	0.7350
4	46	0.6671
5	902	0.7611

Pooled mean ± std = **0.7141 ± 0.0392**.

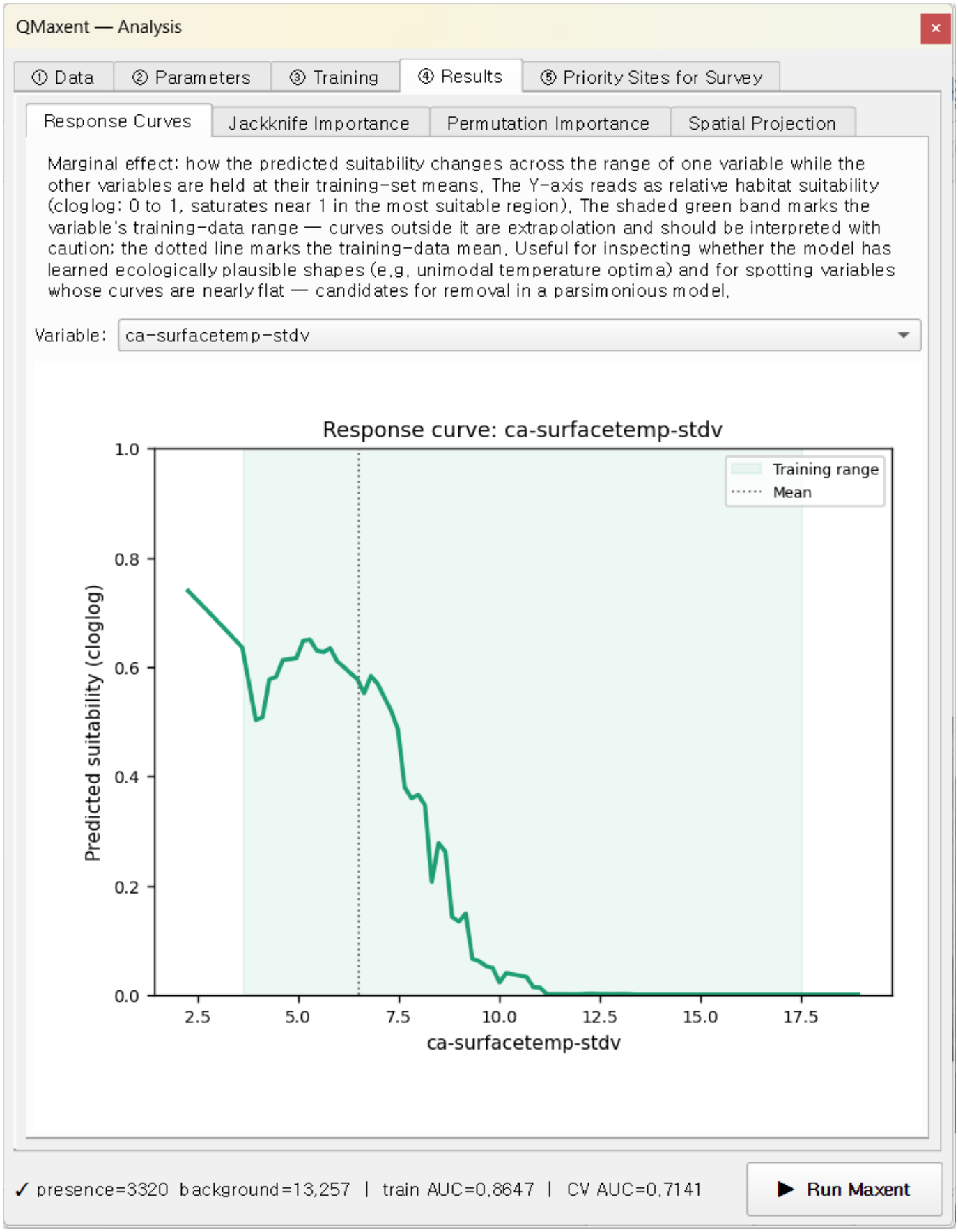
The much tighter ± 0.04 standard deviation (compared to Bradypus's ± 0.075) reflects Ariolimax's larger and more uniformly distributed presence sample — when each spatial fold contains hundreds rather than a handful of presences, the per-fold AUC stabilises.

17.5 5. Variable behaviour

Response curve

`ca-surfacetemp-stdv` (variability of land-surface temperature) carries the strongest stand-alone signal — biologically sensible for an organism whose activity windows depend on cool, moist microclimates:

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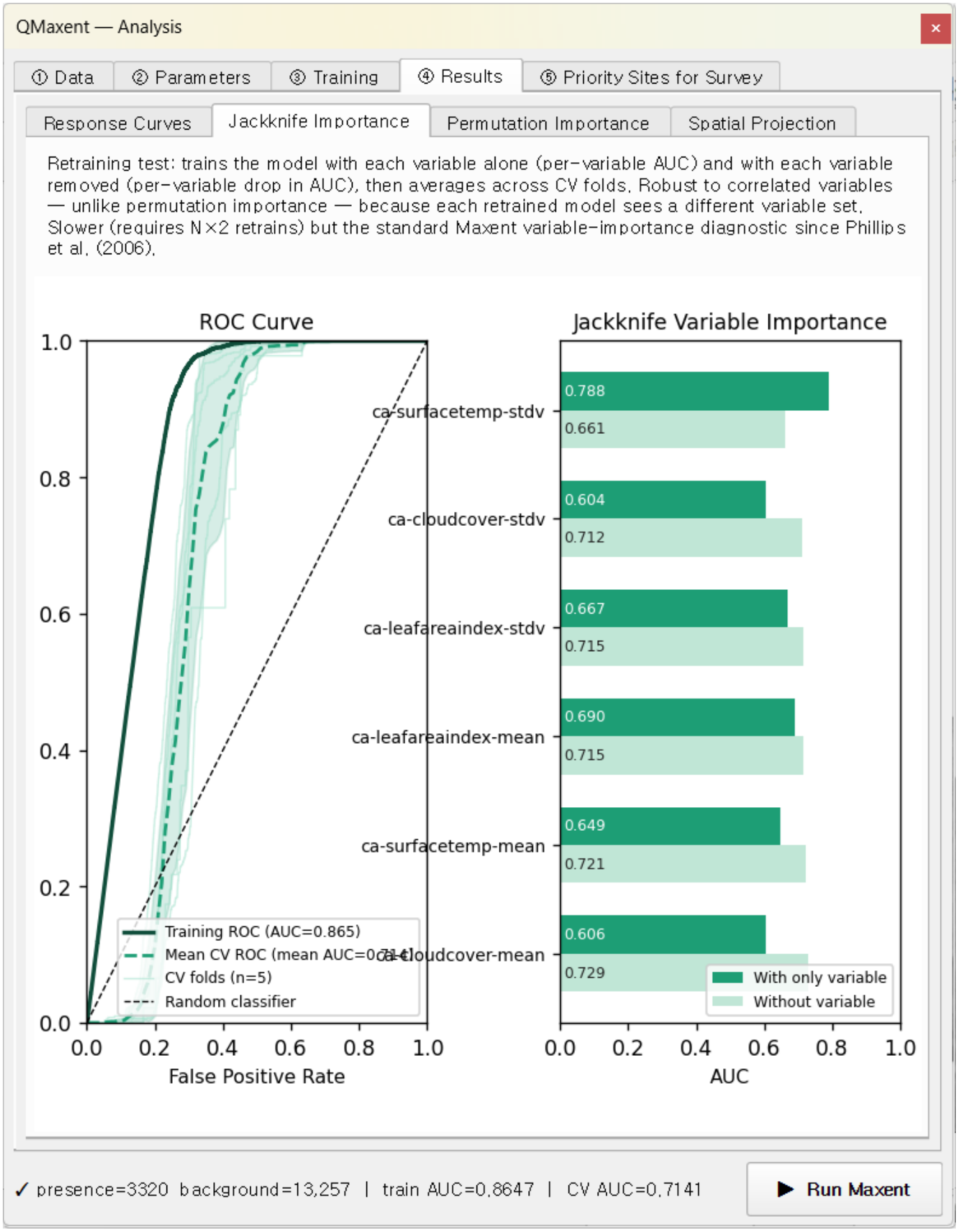
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The curve shows suitability rising as temperature variability falls below ~ 2 K and dropping to near zero above ~ 8 K — the classic preference of a moisture-dependent species for thermally stable maritime climates.

Jackknife importance

The Jackknife panel ranks every variable by both stand-alone power and removal cost:

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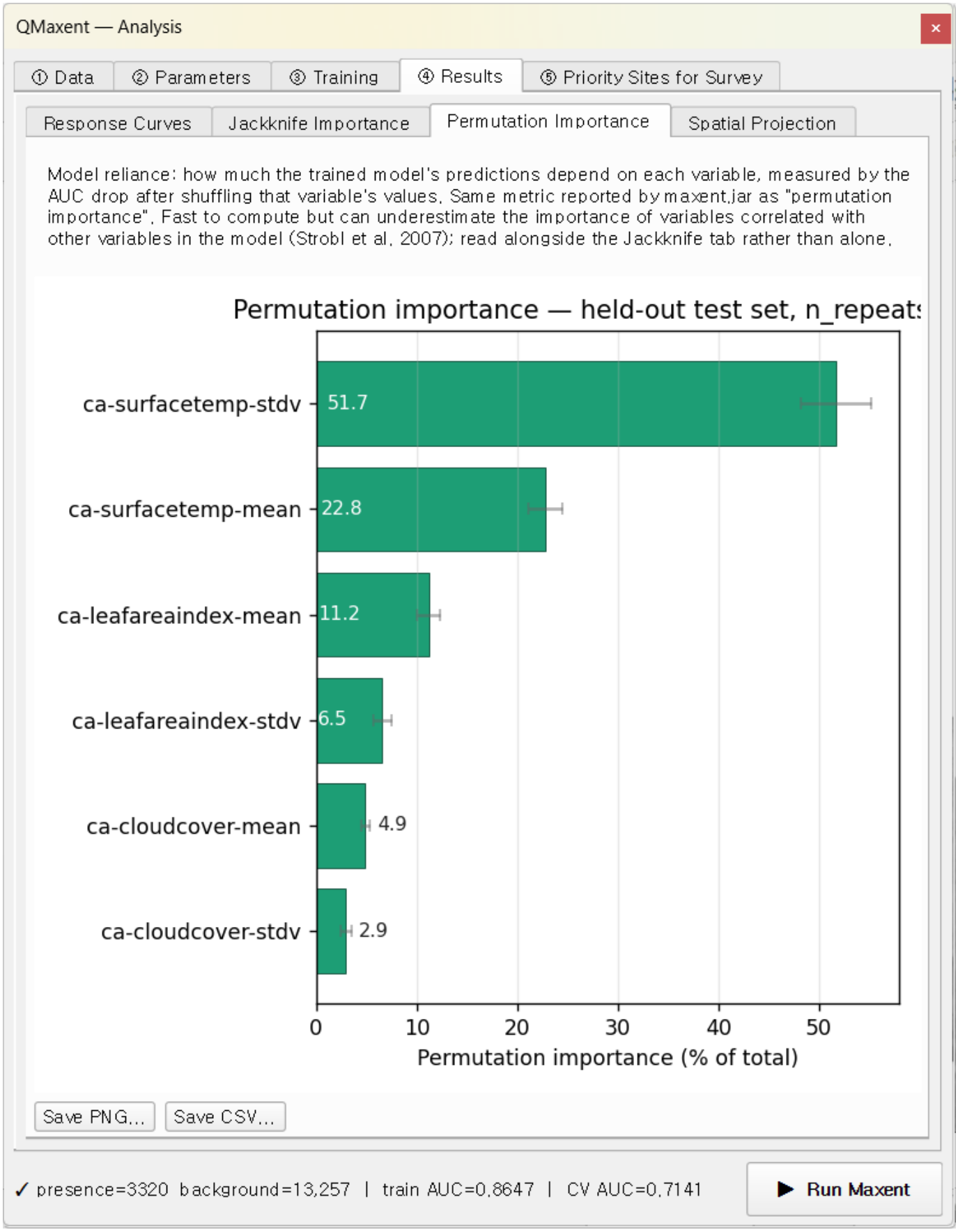
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`ca-surfacetemp-stdv` and `ca-leafareaindex-mean` lead, with the cloud-cover variables weakest. The "without" bars cluster tightly above 0.85 — same correlation-pattern argument as Bradypus.

Permutation importance

The permutation view distributes the total importance across all variables and is directly comparable to maxent.jar's per-variable percentage table:

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17.6 6. Comparing models with and without harmonization

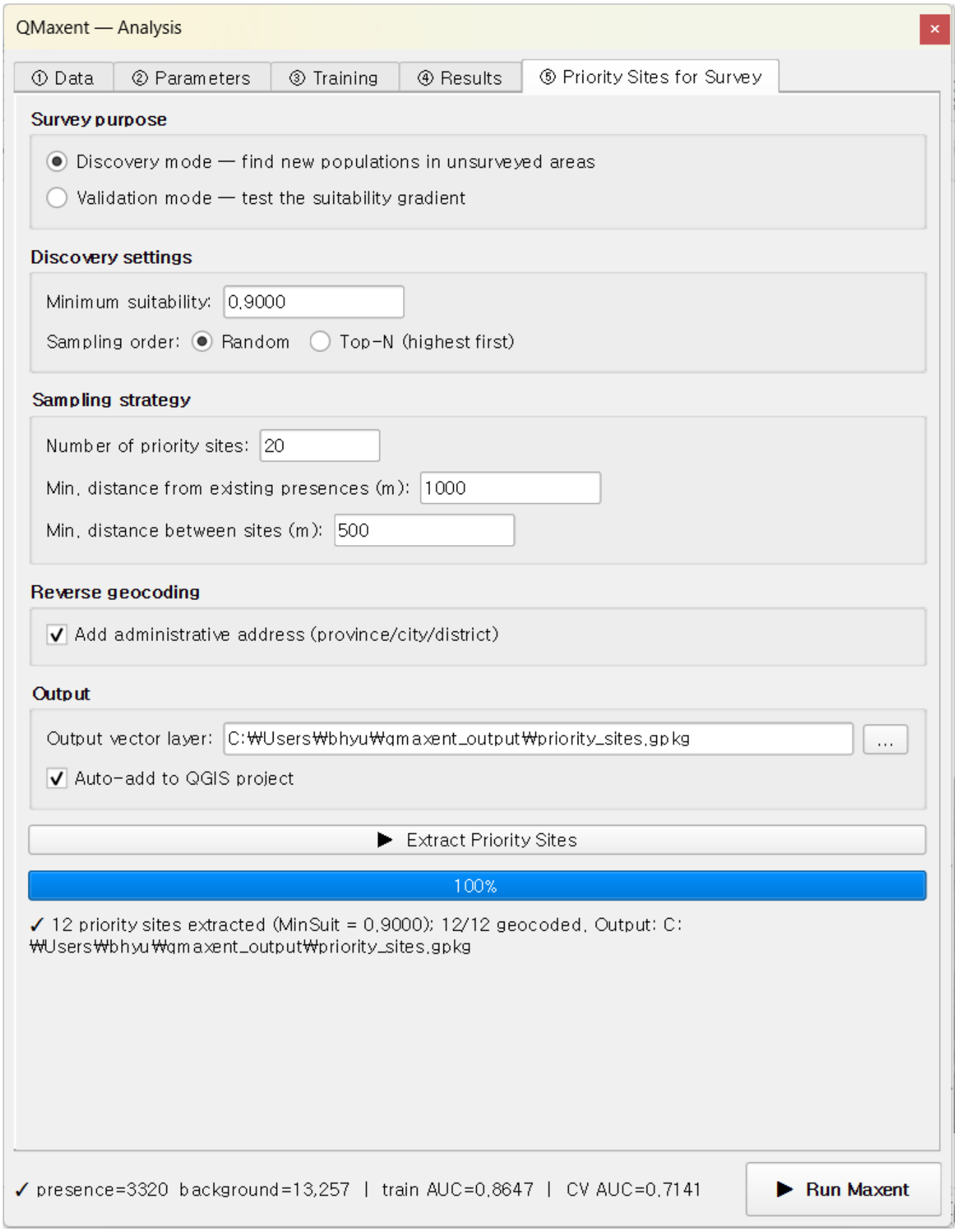
We strongly recommend running the model once on the *unharmonized* stack as a teaching exercise. With Maxent's permissive raster handling you will get a finished model and a finished AUC, but the AUC will typically be 0.05–0.10 higher than the harmonized run — *not because the model is better*, but because covariate misalignment introduces spurious patterns that the model fits to. The cross-validation gap (training vs. CV AUC) widens correspondingly.

Always run Check Raster Consistency before drawing conclusions.

17.7 7. Survey site selection

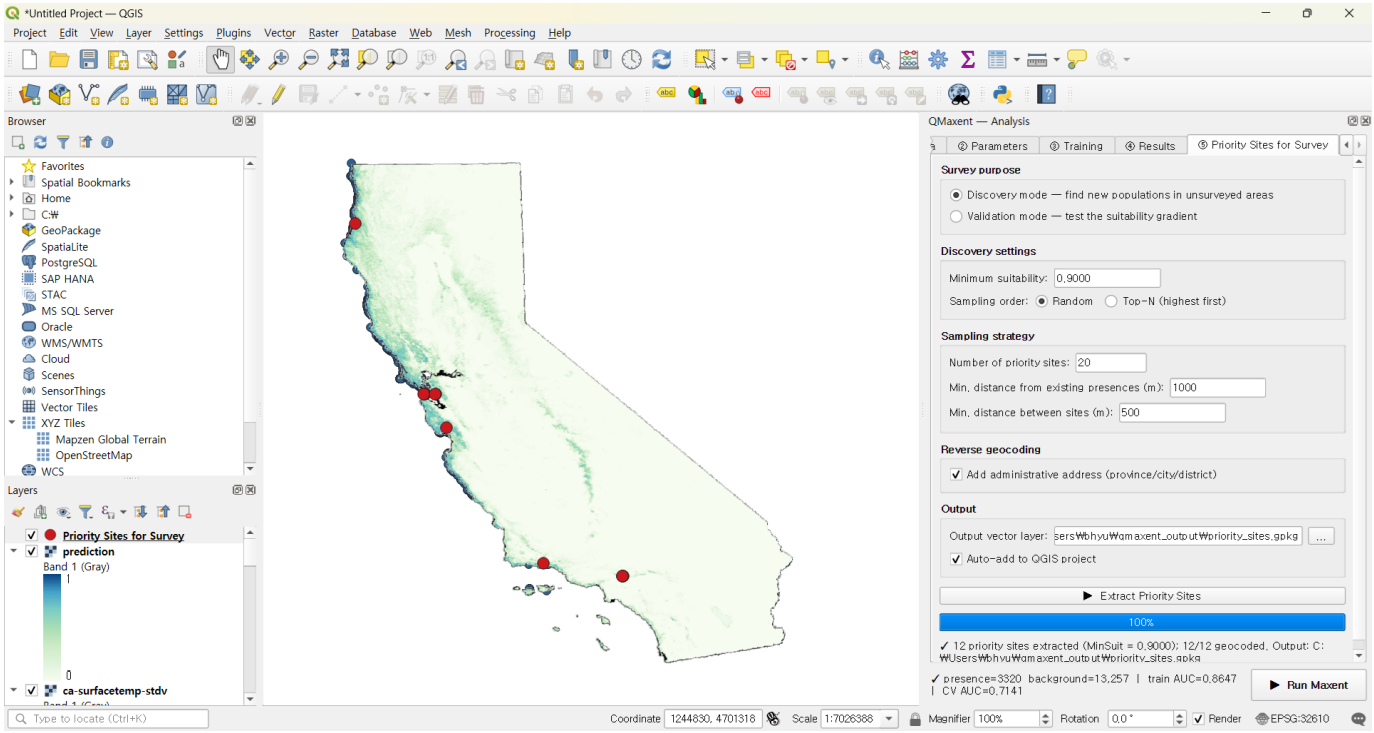
After projection, switch to ⑤ **Survey Site Selection**, choose **High-suitability search** mode, and click ▶ **Select Survey Sites**. The defaults work well for Ariolimax's smaller study area:

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The sites land on the coastal mountain ranges that the suitability map highlighted, and survey teams can take the resulting GeoPackage straight to the field:



17.8 What this example demonstrates

1. **The two example variants** (Pre-harmonized vs Mismatch demo) for didactic use of the same dataset.
2. **The silent-failure mode of Maxent** when rasters disagree.
3. **QMaxent's preflight + harmonize tooling** that turns a project-killing mistake into a one-click fix.
4. **Sample-size effects on spatial CV variability** — 3,732 presences produce a tighter ± std than Bradypus's 116.

Carry the same habit into your own work: every time you assemble a new raster stack, run Check Raster Consistency *before* training. If it fails, harmonize first, train second.

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18. Pitta nympha

The fairy pitta *Pitta nympha* is a long-distance migratory passerine that breeds in dense, multi-strata broadleaved forest. The dataset used here comes from a published field study, Lee et al. (2025, *Global Ecology and Conservation* 64:e03939), which surveyed nest sites on Geoje Island (Geoje-si, South Korea), selected Maxent settings with ENMeval and fitted the final model with linear, quadratic and hinge features (LQH) and a regularization multiplier of 4. Re-running the same data in QMaxent serves two purposes:

1. **Worked example for real, small-n field data** (47 nest locations) with a mix of continuous topographic and forest-structure variables and two categorical variables.
2. **Cross-implementation comparison** with maxent.jar on identical inputs, at the default regularization (β = 1) and at the settings of Lee et al. (β = 4). See § 7 below.

18.1 1. Dataset

The ten covariates are those of Lee et al. (2025, Table 1):

Layer	Type	Description
`pitta_nympha_occurrence`	Vector point	**47** nest locations on Geoje Island
`TWI`	Continuous raster	Topographic wetness index
`TIN`	Continuous raster	Elevation
`ASPECT`	Continuous raster	Aspect
`SLOPE`	Continuous raster	Slope (°)
`SMI`	Continuous raster	Soil moisture index
`DBH`	Continuous raster	Mean trunk diameter at breast height
`HEIGHT`	Continuous raster	Mean tree height
`AGE`	**Categorical** raster	Forest age class
`CANOPY_COVER`	Continuous raster	Canopy closure (%)
`SPECIES`	**Categorical** raster	Dominant tree species (class code)

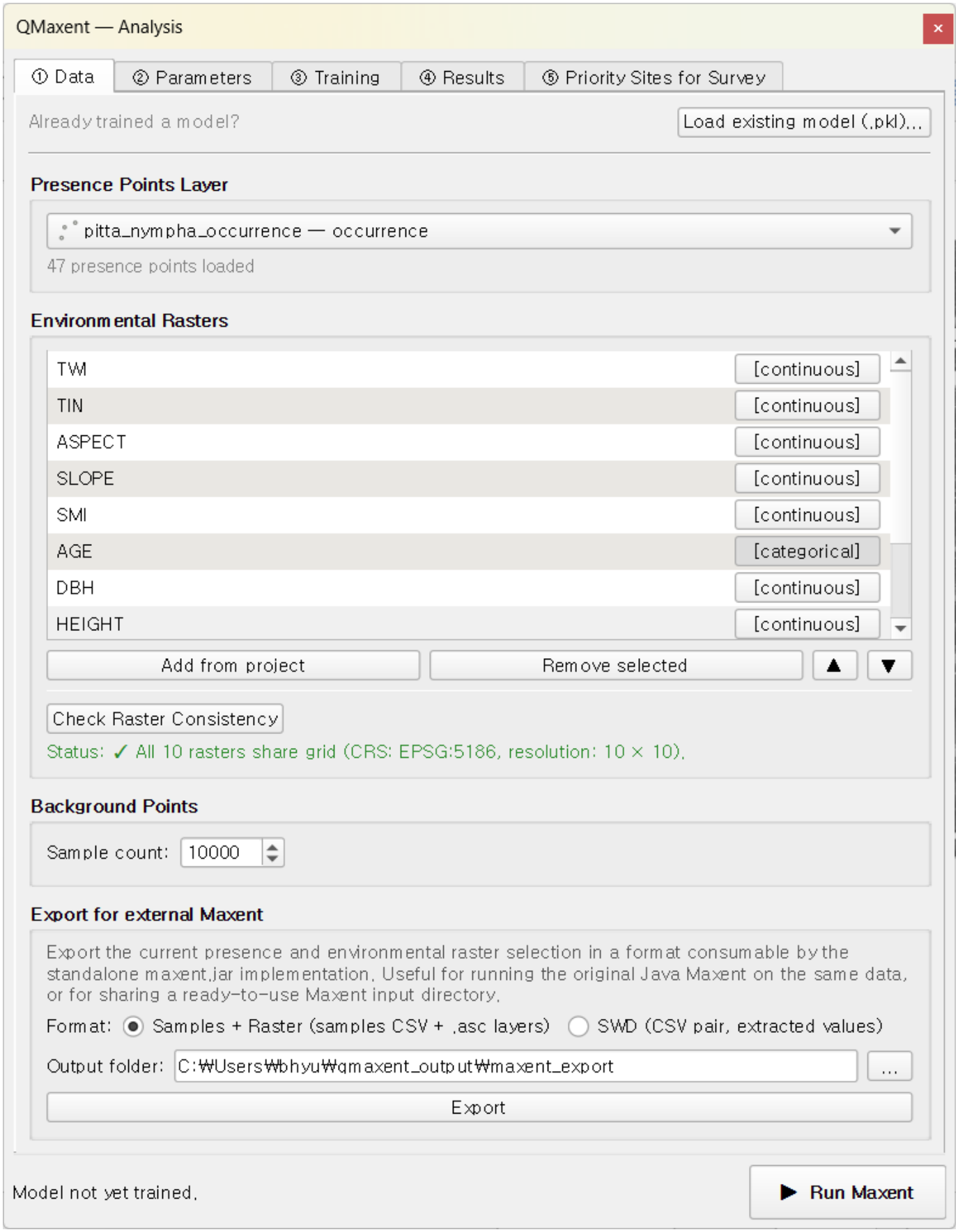
All ten rasters share a common grid (EPSG:5186 KGD2002 / Central Belt, 10 m × 10 m). This is real survey data: the dataset is not bundled with the plugin's Example Dataset Downloader; the corresponding author of the manuscript holds the canonical copy (contact bhyu@knps.or.kr).

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18.2 2. Loading data

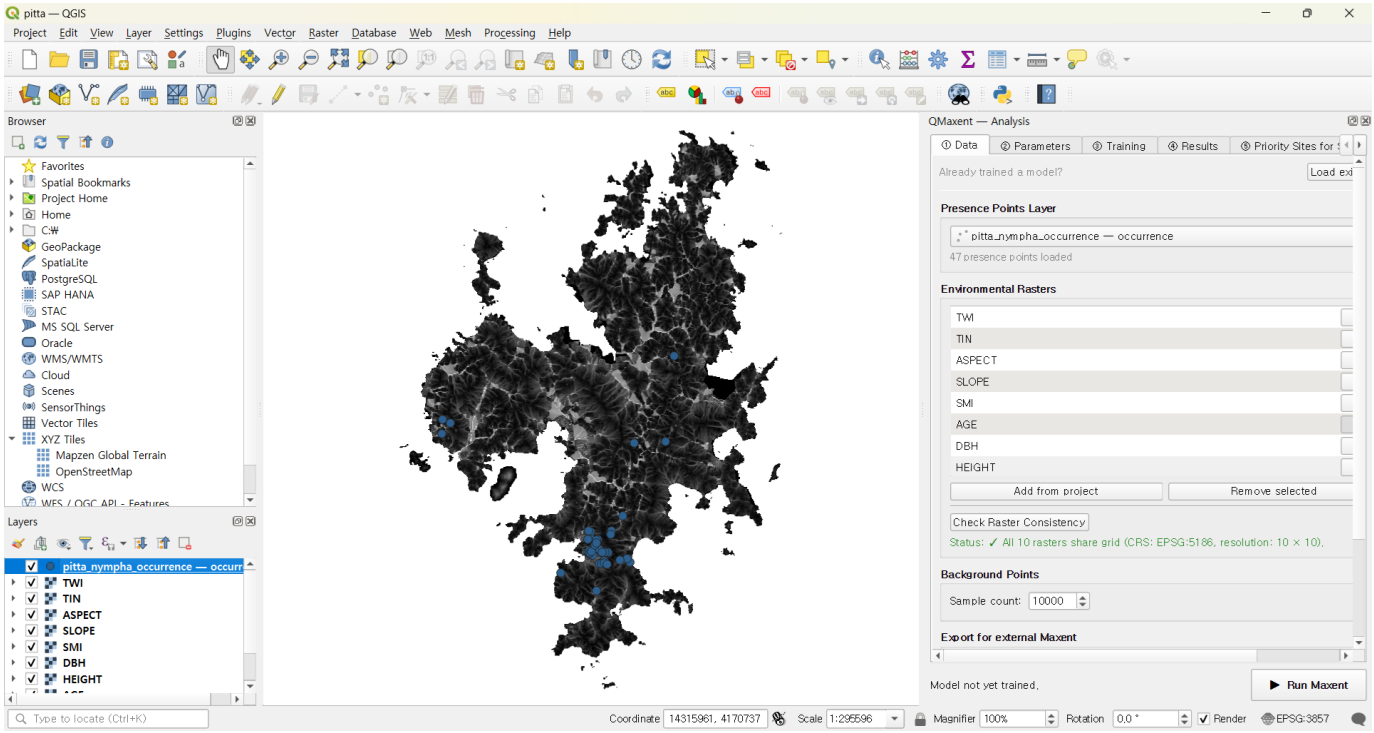
On ① **Data**, pick `pitta_nympha_occurrence` from the Presence Points Layer drop-down (47 points), add all ten rasters from the project, **mark `AGE` and `SPECIES` as `[categorical]`**, and click **Check Raster Consistency**:

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The status line reads `✓ All 10 rasters share grid (CRS: EPSG:5186, resolution: 10 × 10)`. The presence points cluster on Geoje Island's central forested ridges:

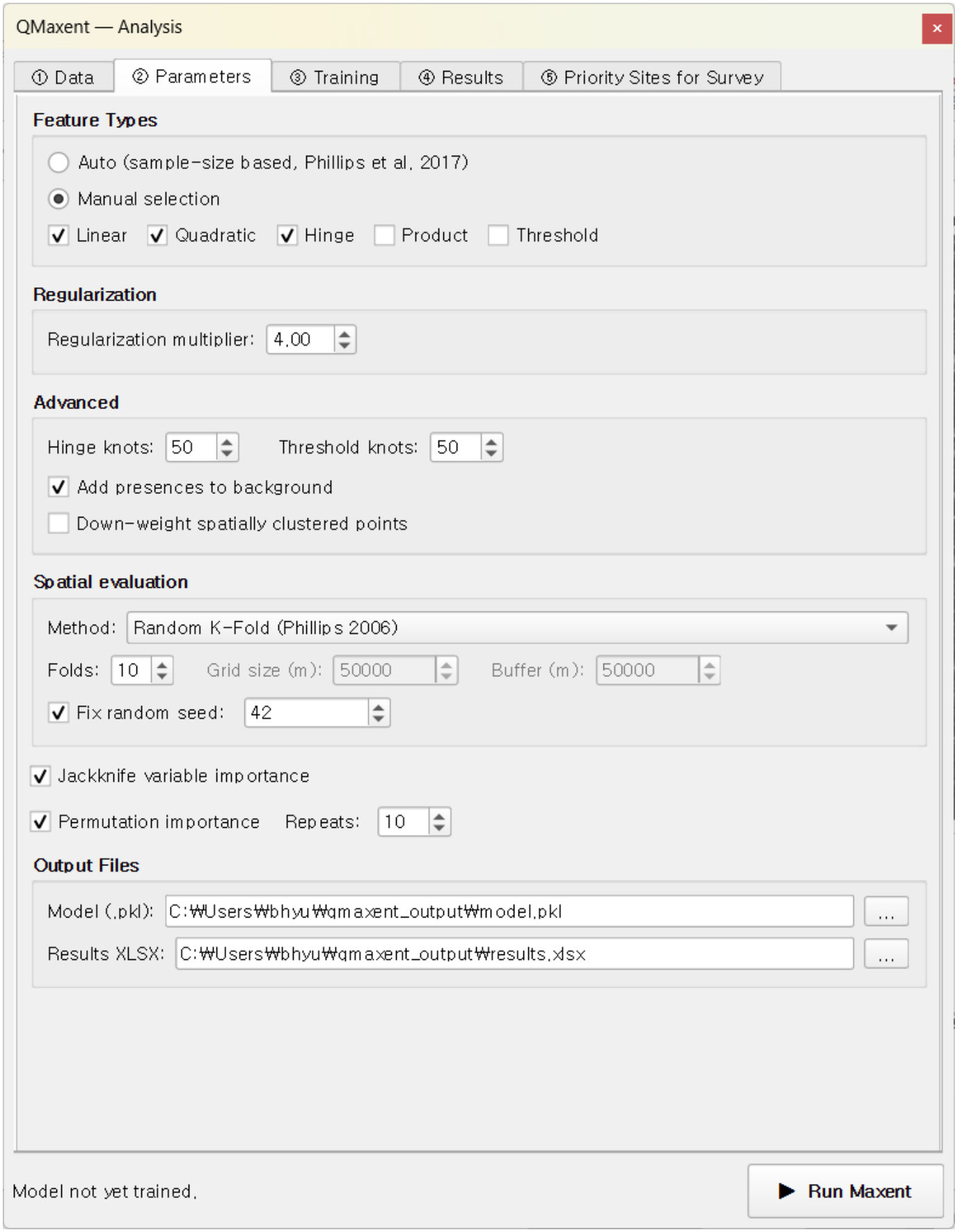


Keep the raster order unchanged between runs: background points are drawn from the first raster in the list, so the order is part of what makes a run reproducible.

18.3 3. Lee-matched parameters

On ② **Parameters**, switch to **Manual selection** for Feature Types and tick **Linear**, **Quadratic**, **Hinge** (untick Product and Threshold). Set **Regularization multiplier = 4.00**. For Spatial evaluation, choose **Random K-Fold (Phillips 2006)** with **Folds = 10**, and leave **Fix random seed** on with the value 42. Leave Jackknife and Permutation importance enabled (10 repeats, **Evaluate on: Training data (maxent.jar convention)**):

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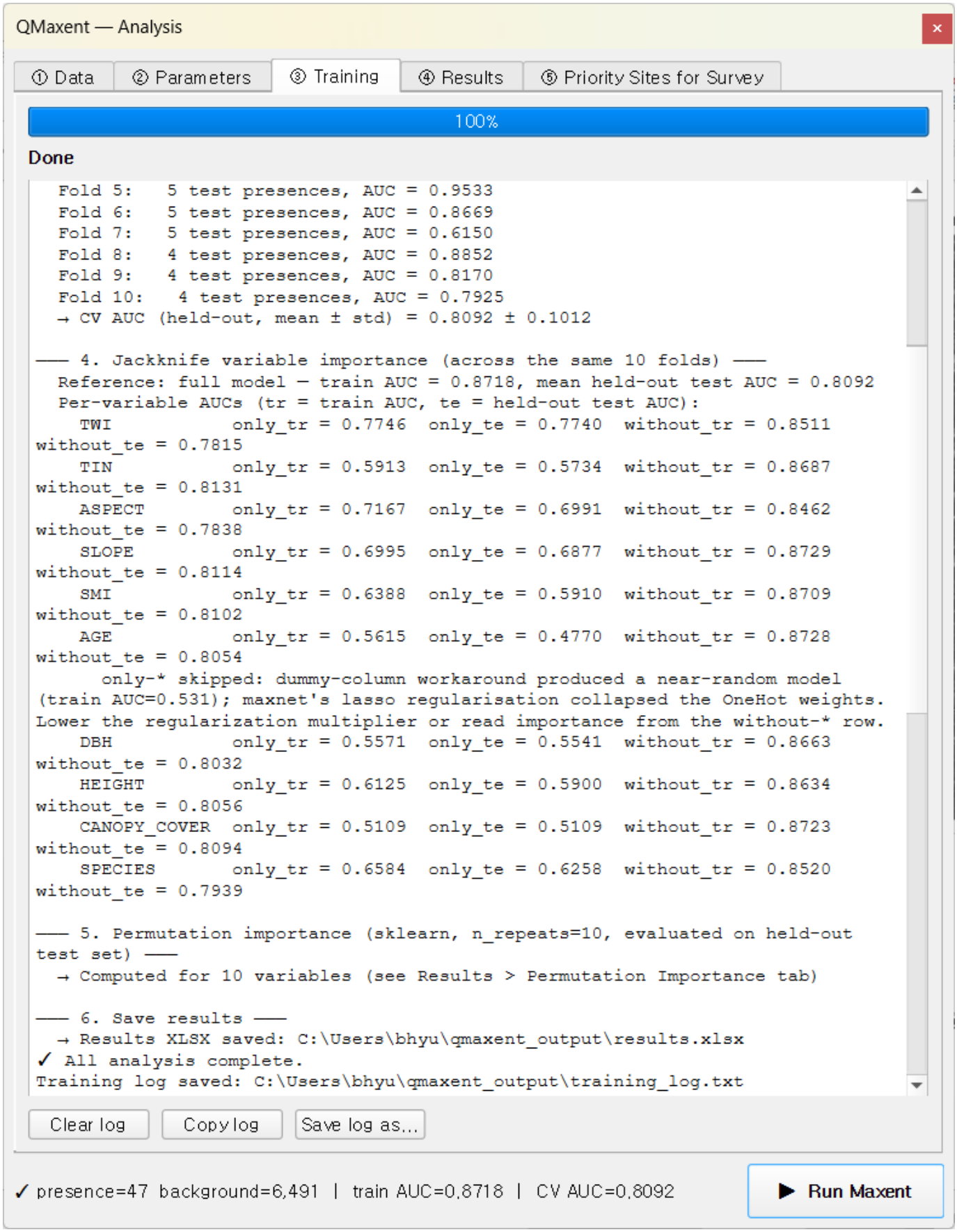
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This is the configuration that ENMeval selected in Lee et al. (2025). Note that **Auto (maxnet default rule by sample size)** would also give LQH for 47 presences (15 to 79 presences); the manual selection makes the choice explicit. Lee et al. evaluated the final model with ten random 75/25 splits of the presences; random 10-fold CV is the closest option in QMaxent.

18.4 4. Training

Click ▶ **Run Maxent**. The Training tab finishes in ~ 20 seconds:

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With the fixed seed, QMaxent v0.2.0 uses 47 presences and 9,384 background points. With v0.2.0:

- **Full-data model**: training AUC = 0.8645.
- **Cross-validation**: Random K-Fold n=10, seed=42, CV AUC = **0.812 ± 0.110** (mean ± SD across folds). The per-fold AUCs spread widely, reflecting the small sample (4–5 test presences per fold) typical of field-survey datasets.

The table below shows how the CV AUC depends on the regularization multiplier and on the partition (QMaxent v0.2.0, mean ± SD across folds):

β	Random 10-fold	Geographic 5-fold
1	0.805 ± 0.108	0.713 ± 0.138
4	0.812 ± 0.110	0.739 ± 0.134

Geographic folds give lower AUCs than random folds, as expected when the test presences are spatially separated from the training presences (Roberts et al. 2017). The stronger regularization of β = 4 helps slightly under both partitions.

Jackknife with a categorical variable

The log can surface a diagnostic that does not appear in the Bradypus / Ariolimax runs:

`only-*` skipped: dummy-column workaround produced a near-random model (train AUC = 0.531); maxnet's lasso regularisation collapsed the OneHot weights. Lower the regularization multiplier or read importance from the `without-*` row.

In the *only-this-variable* jackknife pass a categorical variable such as `AGE` must be one-hot encoded, but at β = 4 the L1 lasso penalty can collapse every OneHot weight back to near-zero, and Maxent has nothing left to score with. QMaxent detects this collapse, skips the affected `only-*` row, and tells you in plain English. The `without-*` row remains informative and is the right place to read the variable's incremental contribution.

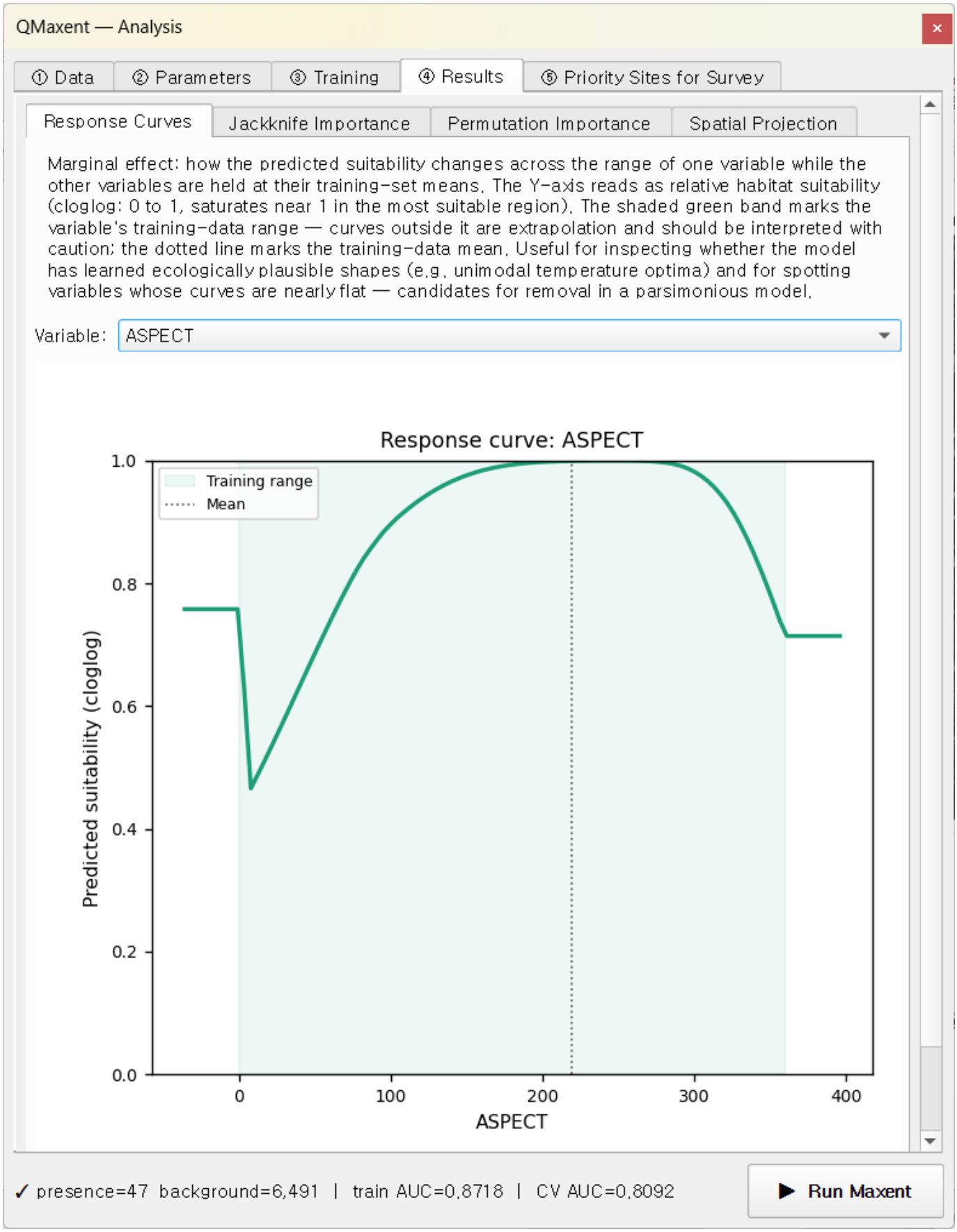
This is exactly the over-regularisation failure mode Merow et al. 2013 describe. A β = 1 run recovers a meaningful `only-*` AUC.

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18.5 5. Variable behaviour

Response curve — `ASPECT`

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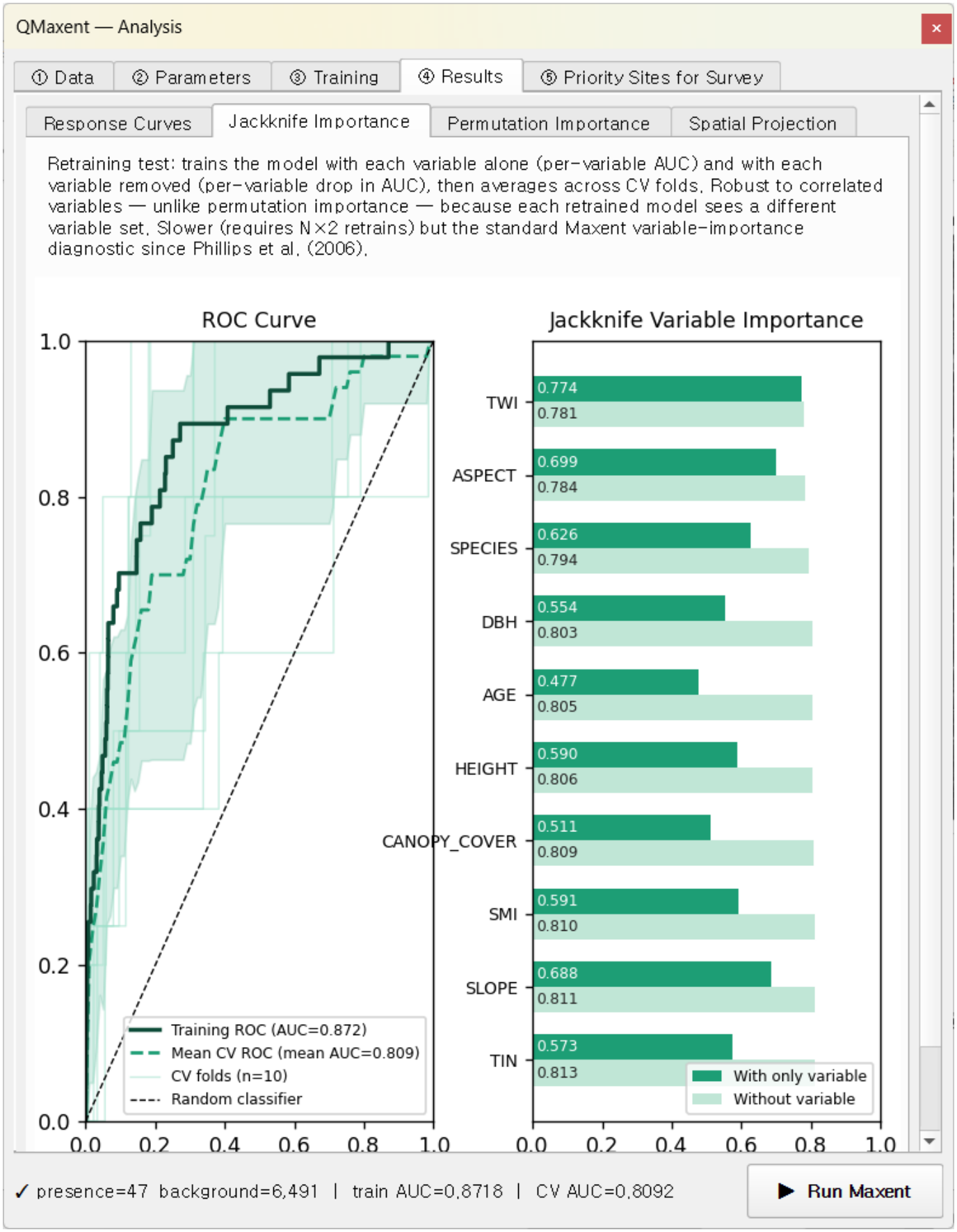


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The model assigns highest suitability to north-facing aspects (roughly 270°–360°), consistent with the fairy pitta's known preference for shaded, cool, humid microclimates on ridge shoulders.

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Jackknife importance



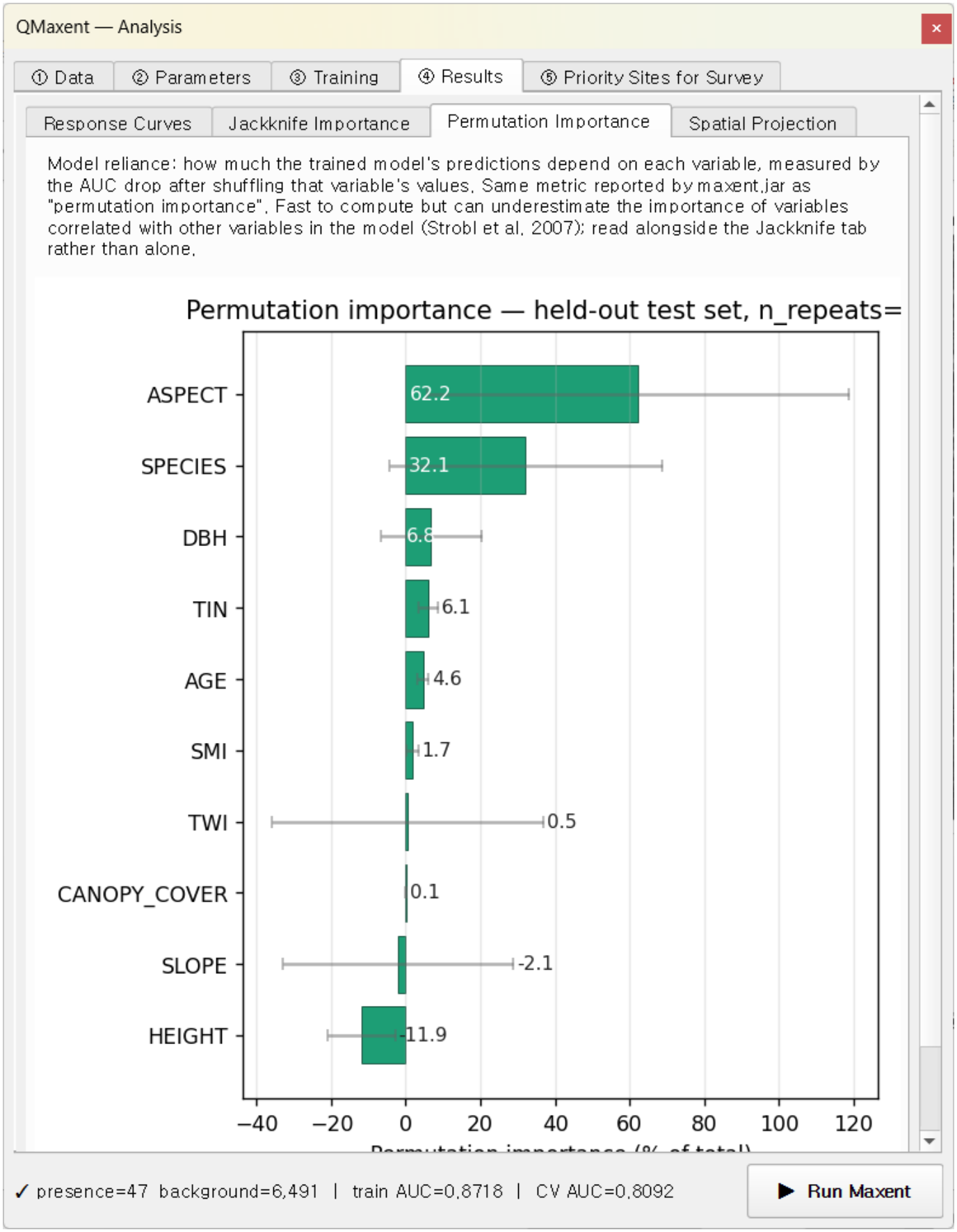
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`ASPECT`, `TWI`, and `SPECIES` carry the strongest *without-row* signals: removing them costs the most. A skipped `only-*` bar means the collapse described in § 4 above occurred; read that variable's `without-*` bar instead.

Permutation importance

The permutation pass shuffles each variable in the fitted model, so it is not affected by the lasso collapse that can disable a categorical variable's `only-*` jackknife row, and all ten variables get a comparable percentage:

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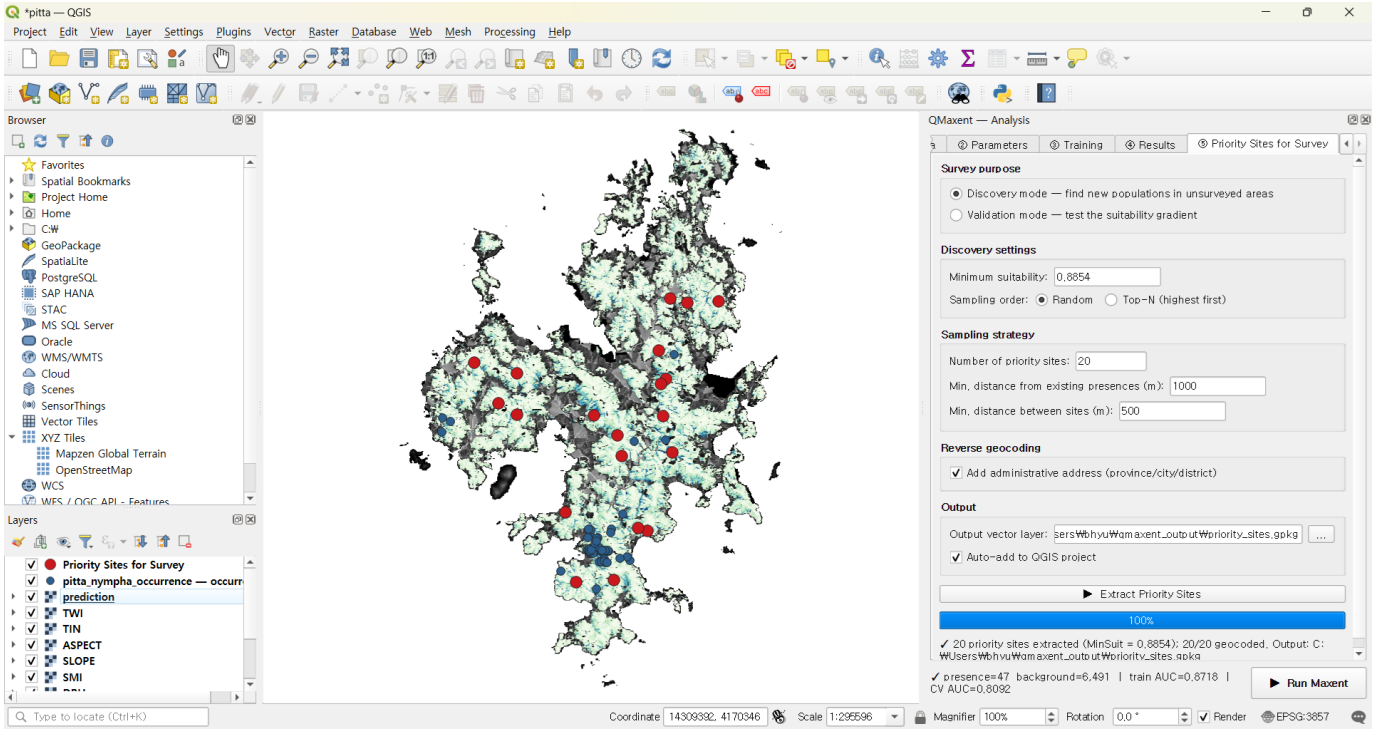


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With **Evaluate on: Training data (maxent.jar convention)** the ranking is computed the same way as in maxent.jar and in Lee et al. (2025). `TWI`, `ASPECT` and `SPECIES` are the top three variables in QMaxent, in maxent.jar and in Lee et al. (2025, Table 4); see § 7.

18.6 6. Survey site selection

After projection, the ⑤ **Survey Site Selection** tab in **High-suitability search** mode produces field-trip candidates on Geoje. With **Minimum suitability** at its automatic value of 0.900 (0.9 × the raster maximum), **Sampling order: Random**, 20 sites, at least 1,000 m from known nests and at least 500 m apart, QMaxent returns a tractable list of twenty new search locations:



Reverse geocoding fills the `country`, `province`, `city_county` and `district` fields from OpenStreetMap. With **Address language: Local names** (the default) the names are in Korean; choose **English** for English names. A level that OpenStreetMap does not hold for a location is left blank. This is a one-step path from model to field-trip planning.

18.7 7. Comparison with maxent.jar

QMaxent v0.2.0 gave maxent.jar 3.4.4 and QMaxent **identical inputs**: the same 47 presences and the same 9,384 background points (sampled by QMaxent with seed 42 and exported in SWD format), with LQH features. Permutation importance was evaluated on the training data in both programs.

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Full-data fits on identical inputs

Setting	Training AUC, maxent.jar	Training AUC, QMaxent	Prediction correlation (Pearson r)	Permutation-importance rank agreement (Spearman ρ)
β = 1 (default)	0.8795	0.8738	0.957	0.648
β = 4 (Lee et al. settings)	0.8520	0.8645	0.894	0.867 (p = 0.001)

The two programs agree closely but not to the third decimal. elapid's scikit-learn backend applies one regularization strength to all features, while maxent.jar regularizes each feature class separately, so the fitted coefficients differ somewhat. Rank agreement of the variable importances is moderate at β = 1 and strong at β = 4.

Lee et al. (2025) protocol

Following the final-model protocol of Lee et al. (LQH, β = 4, ten random 75/25 splits of the presences, identical splits in both programs), the QMaxent permutation-importance ranking agrees with Lee et al. (2025, Table 4) at Spearman ρ = 0.830. The top three variables are the same in maxent.jar, QMaxent and Lee et al.: `TWI`, `ASPECT` and `SPECIES`. The bias file of Lee et al. was not reproduced.

18.8 8. What this example demonstrates

1. **Real-data workflow with small-n field surveys** (47 presences, 10 covariates, real-world spatial scale).
2. **Categorical variable handling** (`AGE`, `SPECIES`) plus the OneHot-collapse diagnostic when over-regularised.
3. **The Lee-matched (β = 4) configuration** and its comparison with maxent.jar on identical inputs (§ 7).
4. **A different, narrower kind of survey site selection**: a targeted search around known sub-populations rather than continental discovery.

The comparison values are recorded in `tests/fixtures/pitta_golden_values.json` in the QMaxent repository.

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19. Methodological background

This chapter collects the *why* behind the *how* — the academic reasoning that justifies QMaxent's defaults. It is intended for users who want to understand or cite the model choices, not for first-time users (who should read the User guide chapters first).

19.1 Maxent and the principle of maximum entropy

Maxent fits the **maximum-entropy distribution** consistent with the empirical moments observed at the presence locations (Phillips, Anderson & Schapire 2006). Intuitively: among all distributions over the study area that produce the same average value of each environmental variable as the presence sample, the maximum- entropy distribution is the *least informative* — the one that adds no assumptions beyond what the data actually says.

The estimator is mathematically equivalent to a regularized inhomogeneous Poisson point-process model fit to the presence-background contrast (Fithian & Hastie 2013) — and that equivalence is what elapid's implementation exploits to delegate the optimisation to scikit-learn's logistic-regression routines (Anderson 2023; Pedregosa et al. 2011).

19.2 The maxnet default rule for feature selection

The Maxent feature classes (Linear, Quadratic, Product, Hinge, Threshold) trade off **flexibility** against **over-fit risk**. maxent.jar and its R counterpart maxnet both choose default feature classes from the number of presence points. QMaxent's **Auto (maxnet default rule by sample size)** mode applies the maxnet default (Phillips et al. 2017; `maxnet.formula(classes = "default")`):

Presences	Feature classes
fewer than 10	L
10 to 14	L, Q
15 to 79	L, Q, H
80 or more	L, Q, P, H

Unlike the maxent.jar rule, which QMaxent used up to v0.1.x, the maxnet default never adds threshold features. This matters for QMaxent's engine: elapid fits the model with scikit-learn, which applies one regularization strength to all features, whereas maxent.jar regularizes each feature class separately. Under a single penalty the many step-shaped threshold features overfit. On the Bradypus benchmark (random 10-fold CV), the test AUC was 0.873 with LQPHT and 0.890 with LQPH, against 0.895 for maxent.jar with LQPHT. Threshold features are still available under **Manual selection**.

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The rule is a default, not a guarantee of the best model. The most suitable feature classes and regularization depend on the data (Merow, Smith & Silander 2013; Radosavljevic & Anderson 2014). For a publication, compare alternatives with **Manual selection** and spatial cross-validation, or with a model-selection tool such as ENMeval (Muscarella et al. 2014).

19.3 Why spatial cross-validation matters

Random K-Fold CV gives over-optimistic AUCs on spatial data because random folds are likely to contain points geographically *adjacent to* training points. Spatial autocorrelation makes the held-out points trivially predictable.

Roberts et al. 2017 demonstrate this empirically across many SDM benchmarks: random-K-Fold AUCs systematically over-estimate the performance you would actually see on a new region by 0.05–0.15. They recommend spatially-blocked CV as the default for any spatial model.

QMaxent therefore defaults to **Geographic K-Fold (k = 5, fixed seed)** — the recommended Roberts et al. design. *Random* K-Fold is offered only to allow exact reproduction of older studies.

19.4 Pooled vs per-fold AUC

Two summary statistics across folds are commonly reported:

- **Per-fold AUC then averaged** — compute AUC inside each fold, average across folds, report mean ± SD.
- **Pooled AUC** — concatenate all held-out predictions across folds and compute a single AUC.

QMaxent reports the **per-fold mean ± SD** as the headline number because it carries the across-fold variance, which is itself an informative diagnostic — high variance signals that the model's geographic transferability is unstable. The pooled-AUC alternative is included in Table 3 of the XLSX as a secondary metric.

19.5 Jackknife variable importance

The classical Maxent jackknife (Phillips, Anderson & Schapire 2006) fits, for each variable, two extra models: one with that variable alone (*with-only*) and one with that variable removed (*without*). The drop in test AUC from full to *without* identifies variables whose unique contribution is hard to recover from the others.

QMaxent reports four AUCs per variable (only-train, only-test, without-train, without-test) plus the test-side drop column, sorted by descending drop so the most uniquely-informative variables are at the top. This is the format used in the published Maxent literature since Phillips et al. 2006.

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19.6 Permutation importance and its evaluation data

Permutation importance shuffles one variable at a time and measures the drop in AUC. The result depends on the data the AUC is computed on, so QMaxent lets you choose it (**Evaluate on** on the ② Parameters tab):

- **Training data (maxent.jar convention)** (default): the full model on all training presences and background points. This is how maxent.jar computes permutation importance, so rankings can be compared directly with maxent.jar studies.
- **Held-out CV test folds (all folds pooled)**: each fold model on its own test fold, with the results of all folds pooled. This measures how much predictive performance at new locations depends on each variable. It needs K-Fold or Checkerboard cross-validation.

Up to v0.1.x QMaxent used the test data of the first CV fold only, which made rankings unstable and not comparable with maxent.jar.

19.7 Threshold methods (MTP / T10 / MaxSSS)

When you need a binary suitable/unsuitable map (e.g. for reserve-design applications), you must choose a threshold of the cloglog output. The SDM literature standardises on three options (Liu, Newell & White 2013):

- **MTP — Minimum Training Presence** — the lowest cloglog value at any training presence. Most permissive; favours sensitivity.
- **T10 — 10th percentile training presence** — drops the bottom 10% of training presences as outliers. The classical Java-MaxEnt default.
- **MaxSSS — Max Sensitivity + Specificity** — the threshold that maximises the sum of sensitivity and specificity on the training ROC. Best calibration in most published comparisons.

QMaxent reports all three thresholds in Table 3 of the XLSX export. Choose **MaxSSS** unless you are reproducing a Java-MaxEnt study (T10) or have a specific reason to prefer permissive (MTP) or strict classification.

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19.8 Output transforms compared

Three output transforms are available on **Spatial Projection**:

Transform	Range	Interpretation
cloglog	0–1	Probability of presence given a typical sample, Phillips et al. 2017 recommended default
logistic	0–1	Older, Phillips & Dudík 2008 parameterisation; still used in some baselines
raw	unbounded	Unnormalised exponential output; advanced post-processing

QMaxent defaults to **cloglog** because it is the form Phillips et al. 2017 explicitly recommend as the modern default and because its "probability of presence" interpretation is the most directly communicable to non-modelling stakeholders.

19.9 Treatment of categorical variables

Java MaxEnt encodes categorical rasters via the raster's attribute table; elapid encodes them via **one-hot expansion** internally (Anderson 2023). The two approaches are equivalent for training but diverge at projection time when the raster contains class codes the model has not seen during training:

- **Java MaxEnt** silently maps unseen codes to a random class, producing arbitrary suitability values at affected cells.
- **QMaxent** detects unseen codes via the unified preflight dialog and **auto-masks them to NoData** rather than extrapolating. If an unseen class still reaches the model, the projection does not fail (since v0.2.0): those cells get no categorical contribution, as in maxent.jar.

This is the recommended practice (Elith, Kearney & Phillips 2010) — extrapolation beyond the training domain is fundamentally undefined and should be flagged, not silently filled in.

19.10 Bias correction

When occurrences are spatially clustered (e.g. road-bias, citizen- science clusters), the model fits the *bias surface* as well as the species' habitat preference. Two complementary corrections exist:

- **Sample-weight down-weighting** (Phillips et al. 2009) — applies fractional weights to clustered presences. QMaxent implements this as the *Down-weight spatially clustered points* option.
- **External KDE bias raster** — a raster representing the survey effort, used as an explicit covariate. Not implemented in QMaxent v0.1.x; the closest analog is the down-weight option, which performs comparably for many published studies.

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For a more comprehensive treatment, Boria et al. 2014 recommend **spatial thinning** of the presence layer before modeling as a complementary step — implementable in QGIS via the *NNJoin* plugin or the standalone *spThin* tool.

19.11 Why these choices

The defaults assembled in QMaxent are the union of (a) what the benchmark literature shows works well across many species and study areas, and (b) what minimises the silent-failure modes documented in Roberts et al. 2017, Araújo et al. 2019, and Elith, Kearney & Phillips 2010. Diverging from them is sometimes the right move for a specific study; the Pitta nympha worked example shows what that looks like for the LQH/RM=4 fixed-hyperparameter setting of Lee et al. 2025.

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20. References

QMaxent's defaults, evaluation procedures, and survey-planning workflows are grounded in the species distribution modeling (SDM) literature. This chapter groups the works the plugin draws on by topic, with a one-line annotation on how each reference is used inside QMaxent. The plugin source code carries inline citations to these works in the relevant modules (`workers/maxent_worker.py`, `bridge/elapid_bridge.py`, `bridge/priority_sites.py`, `core/venv_manager.py`).

20.1 Maxent — core methodology

Phillips, S. J., Anderson, R. P., & Schapire, R. E. (2006). Maximum entropy modeling of species geographic distributions. *Ecological Modelling*, 190(3-4), 231–259. *The original Maxent paper. QMaxent's feature-class taxonomy (LQPHT), jackknife importance, and ROC-based evaluation all come from this formulation.*

Phillips, S. J., & Dudík, M. (2008). Modeling of species distributions with Maxent: new extensions and a comprehensive evaluation. *Ecography*, 31(2), 161–175. *The benchmark paper behind the maxent.jar sample-size feature-class rule (used by QMaxent's Auto mode up to v0.1.x) and the regularization-multiplier default of 1.0.*

Phillips, S. J., Anderson, R. P., Dudík, M., Schapire, R. E., & Blair, M. E. (2017). Opening the black box: an open-source release of Maxent. *Ecography*, 40(7), 887–893. *The release of the open-source Maxent and the maxnet R package, and the formal recommendation to adopt the cloglog output transform, adopted as QMaxent's default. Since v0.2.0, QMaxent's "Auto (maxnet default rule by sample size)" mode follows the maxnet default feature classes (`maxnet.formula(classes = "default")`).*

Fithian, W., & Hastie, T. (2013). Finite-sample equivalence in statistical models for presence-only data. *The Annals of Applied Statistics*, 7(4), 1917–1939. *The paper that establishes the formal equivalence between Maxent and infinitely-weighted logistic regression — the foundation for elapid's implementation.*

20.2 elapid — software backend

Anderson, C. B. (2023). elapid: Species distribution modeling tools for Python. *Journal of Open Source Software*, 8(84), 5435. *The peer-reviewed release of the elapid library that QMaxent uses for its `MaxentModel`, `GeographicKFold`, and feature transformers.*

Pedregosa, F., et al. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830. *The numerical core that elapid's regularized-likelihood maximisation delegates to.*

Harris, C. R., et al. (2020). Array programming with NumPy. *Nature*, 585, 357–362. *The array library underlying elapid, scikit-learn, and the QMaxent raster-extraction layer.*

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20.3 Plugin architecture

Wu, Q. (2026). GeoAI: A Python package for integrating artificial intelligence with geospatial data analysis and visualization. *Journal of Open Source Software*, 11(118), 9605. https://doi.org/10.21105/joss.09605 *The QGIS plugin from which QMaxent's dependency-installation workflow in* `core/venv_manager.py` *was adapted — in particular the* `_get_qgis_python()` *pattern that resolves the real Python interpreter on Windows (where* `sys.executable` *points to* `qgis-bin.exe`*) and the subprocess pipe handling used during* `pip install`*.*

20.4 Cross-validation and model evaluation

Roberts, D. R., et al. (2017). Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*, 40(8), 913–929. *The paper that established* ***spatial cross-validation as the default*** *for SDMs — directly shaping QMaxent's Geographic K-Fold default and the warning text in The Parameters tab.*

Hijmans, R. J. (2012). Cross-validation of species distribution models: removing spatial sorting bias and calibration with a null model. *Ecology*, 93(3), 679–688. *Provides the formal argument for buffered leave-one-out CV when sample sizes are small; QMaxent's *Buffered LOO* option implements this.*

Valavi, R., Elith, J., Lahoz-Monfort, J. J., & Guillera-Arroita, G. (2019). blockCV: An R package for generating spatially or environmentally separated folds for k-fold cross-validation of species distribution models. *Methods in Ecology and Evolution*, 10(2), 225–232. *The block-CV recipes implemented by QMaxent's Checkerboard option.*

Muscarella, R., et al. (2014). ENMeval: An R package for conducting spatially independent evaluations and estimating optimal model complexity for Maxent ecological niche models. *Methods in Ecology and Evolution*, 5(11), 1198–1205. *The hyperparameter-tuning framework whose checkerboard partitioning is implemented as a QMaxent CV option.*

Kass, J. M., et al. (2018). Wallace: A flexible platform for reproducible modeling of species niches and distributions built for community expansion. *Methods in Ecology and Evolution*, 9(4), 1151–1156. *The interactive ENM platform whose hyperparameter-tuning workflow QMaxent intentionally complements rather than duplicates.*

20.5 Sample bias correction

Phillips, S. J., et al. (2009). Sample selection bias and presence-only distribution models: implications for background and pseudo-absence data. *Ecological Applications*, 19(1), 181–197. *The paper underlying QMaxent's *Down-weight spatially clustered points* option in the Parameters → Advanced section.*

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Boria, R. A., Olson, L. E., Goodman, S. M., & Anderson, R. P. (2014). Spatial filtering to reduce sampling bias can improve the performance of ecological niche models. *Ecological Modelling*, 275, 73–77. *Cited in Data tab as the reason to spatially-thin presence layers before modeling.*

Elith, J., Kearney, M., & Phillips, S. (2010). The art of modelling range-shifting species. *Methods in Ecology and Evolution*, 1(4), 330–342. *The MESS-style extrapolation analysis whose logic underlies QMaxent's unified projection-preflight dialog.*

20.6 SDM best-practice reviews

Elith, J., et al. (2011). A statistical explanation of MaxEnt for ecologists. *Diversity and Distributions*, 17(1), 43–57. *The standard pedagogical reference for understanding what Maxent does and does not assume.*

Merow, C., Smith, M. J., & Silander Jr, J. A. (2013). A practical guide to MaxEnt for modeling species' distributions: what it does, and why inputs and settings matter. *Ecography*, 36(10), 1058–1069. *The companion practical guide to Elith et al. 2011.*

Araújo, M. B., et al. (2019). Standards for distribution models in biodiversity assessments. *Science Advances*, 5(1), eaat4858. *The IPBES-aligned reporting standard QMaxent's XLSX export was designed to satisfy. Cited throughout this manual.*

Radosavljevic, A., & Anderson, R. P. (2014). Making better Maxent models of species distributions: complexity, overfitting and evaluation. *Journal of Biogeography*, 41(4), 629–643. *The empirical paper documenting how feature-class tuning without spatial CV inflates AUC, and that the best settings depend on the data; underlies the advice in the Parameters tab to treat the Auto rule as a default and compare alternatives.*

20.7 Survey design

Williams, J. N., et al. (2009). Using species distribution models to predict new occurrences for rare plants. *Diversity and Distributions*, 15(4), 565–576. *The original survey-design paper behind the *High-suitability search* mode of the Survey Site Selection tab.*

Rhoden, C. M., Peterman, W. E., & Taylor, C. A. (2017). Maxent- directed field surveys identify new populations of narrowly endemic habitat specialists. *PeerJ*, 5, e3632. *The model-validation companion to Williams et al. 2009; QMaxent's *Gradient-stratified sampling* mode (equal numbers of sites from four suitability quartiles) reproduces this design.*

Stevens, D. L. Jr., & Olsen, A. R. (2004). Spatially balanced sampling of natural resources. *Journal of the American Statistical Association*, 99(465), 262–278. *The spatially-balanced-sampling theory underlying QMaxent's between-candidate spacing constraint.*

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Robinson, N. M., et al. (2018). Refining survey effort: how detection probability shapes the design of biodiversity surveys. *Methods in Ecology and Evolution*, 9(3), 575–587. *Cited in Survey Site Selection for typical survey-budget heuristics.*

20.8 Worked example reference

Lee, S.-J., et al. (2025). Breeding habitat prediction and nest-site characteristics of the fairy pitta (*Pitta nympha*) in Geoje-si, South Korea — Insights from a species distribution model. *Global Ecology and Conservation*, 64, e03939. *The published Java-MaxEnt analysis reproduced in the Pitta nympha worked example.*

20.9 How QMaxent cites itself

If you publish a study using QMaxent, please cite:

Yu, B.-H. (2026). QMaxent: Maxent species distribution modeling in QGIS. *Software*. https://github.com/osgeokr/qmaxent

The exact CITATION.cff is the canonical source on the repository.

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21. FAQ & troubleshooting

Common questions, error messages, and workarounds. If you do not see your issue here, please open an issue on GitHub.

21.1 Installation issues

"QMaxent environment not ready" stays red after install

Check the **Plugins → QMaxent → Manage Dependencies** dialog. If any of the required packages (`elapid`, `numpy`, `scikit-learn`, `pandas`, `openpyxl`) is reported missing, click **Install missing**.

If the dialog itself errors out, your QGIS Python may be missing `pip`. On Windows, run **OSGeo4W Shell** → `python -m ensurepip --upgrade`; on macOS/Linux, install QGIS via the OSGeo4W or Conda channel that ships with `pip` enabled.

Windows: "ensurepip ... returned non-zero exit status 1"

This error comes from QMaxent 0.1.x with QGIS 3.44.15 or later on Windows, where creating a Python virtual environment fails. QMaxent 0.2.0 installs the packages with the QGIS Python's own pip into the QMaxent package folder instead, without a virtual environment. Upgrade the plugin and click **Install / Update Dependencies** again. See Dependencies.

"Could not find a version that satisfies the requirement elapid"

This usually means the QGIS Python is older than 3.9. QMaxent requires **Python ≥ 3.9** because that is the floor elapid sets. Upgrade QGIS to the **LTR ≥ 3.34** release; the bundled Python is 3.9+.

Plugin Manager shows QMaxent but "Install" is greyed out

You probably already have QMaxent installed in a *user* plugin directory and the Plugin Manager is showing the *system* package list. Check **Plugins → Manage and Install Plugins → Installed** — uninstall the older copy first, then re-install the latest.

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21.2 Modeling errors

"ValueError: All presence points fell on NoData cells"

The presence layer's CRS or extent does not overlap the raster stack. Check on ① **Data**:

1. Click on a presence point in QGIS to read its coordinates.
2. Click on a raster cell in roughly the same area to confirm the raster has a value there.
3. If they disagree, the layer is misprojected — re-export the presence layer with the rasters' CRS using **Vector → Save Layer As...**.

"Convergence not reached after 500 iterations"

The model could not find a likelihood maximum within the iteration cap. Three common causes:

- **Too few presences** for the chosen feature classes. Switch to **Auto** mode on ② **Parameters** and the maxnet rule will simplify the feature set automatically.
- **Highly collinear variables**. Use **Jackknife Importance** on the results tab to identify near-zero contributors and drop them.
- **Categorical variable with very rare classes**. Re-bin the rare classes into an `Other` category before modeling.

"These rasters cannot be aligned because of their coordinate reference system"

Check Raster Consistency found a raster with no CRS, or with a local (engineering) CRS without a datum, which PROJ cannot reproject. This is typical of a `.prj` file that only names a projection, such as `NAD83(NSRS2007) / Michigan South`. The message names the file and, when possible, the matching EPSG code. In the **Layers** panel, right-click the raster, choose **Layer CRS > Set Layer CRS...**, pick the correct CRS, and run the check again; Check and **Harmonize to Folder...** then use that CRS. See ① Data tab.

"QMaxent uses one raster band per environmental variable"

A raster in the list has several bands. Split it into single-band files, for example with **Processing > GDAL > Raster conversion > Rearrange bands** (one band per output), and add the single-band files instead.

"Grid mismatch — CRS, extent, resolution differ across rasters"

This is the **Check Raster Consistency** failure mode. Click **Harmonize to Folder...** and let QMaxent reproject every raster to the highest-resolution one. The Ariolimax worked example walks through this.

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21.3 Performance

Training is very slow (>10 minutes for 100 presences)

Two common causes:

- **Very large raster stack** (e.g. 30-m resolution over a continental extent). Background-point extraction is the bottleneck. Either reduce the raster resolution to a value appropriate for the species' home-range size, or restrict the analysis to a relevant ROI by clipping the rasters first.
- **Too many background points**. The default 10,000 is appropriate for most studies; a value of 50,000+ rarely helps and slows the run proportionally.

Spatial projection is fast but uses lots of RAM

Projection loads the full raster stack into memory cell-by-cell. For a continental 1-km stack of 10 variables, expect ~2 GB peak. If RAM is tight, downsample the rasters before projection — projection at the training resolution is methodologically defensible (Elith, Kearney & Phillips 2010).

21.4 Output and projection

"Why are some cells in my projection NoData when the rasters cover them?"

The unified preflight dialog auto-masks two cell categories to NoData (a categorical class missing from the training data no longer makes the projection fail; see ④ Results tab):

1. Categorical class codes that did not appear during training.
2. Cells where any continuous variable lies outside its training range — *only when the relevant safety mask is enabled*.

This is intentional (Elith, Kearney & Phillips 2010) — extrapolation beyond the training domain is undefined and should be flagged, not silently filled in.

Can I project onto a different region than I trained on?

Yes — but the preflight dialog will flag every continuous variable whose new-region range exceeds its training range, and will auto-mask unseen categorical codes. Read the dialog before clicking **Yes**; treat any reported extrapolation as a finding to discuss in the paper.

The output cloglog values look uniformly low

Probably normal — cloglog values (Phillips et al. 2017) are calibrated as "probability of presence given a typical sample of the species at this location", and that is genuinely low (<0.5) over most of any species'

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range. The peaks (0.7+) are what matter; the absolute scale is not directly comparable across species or studies.

21.5 How to ask a good question

When opening an issue on GitHub, please include:

1. The **QMaxent version** (Plugins → Manage and Install Plugins → Installed, or `metadata.txt`).
2. The **QGIS version** (Help → About).
3. The **exact error message** (copy from the Training tab log or the QGIS Python console).
4. A **minimum reproducer** — ideally a small dataset that triggers the problem, or screenshots of the ① **Data** and ② **Parameters** tabs so we can see the configuration.

The fastest issues to resolve are the ones that ship with everything a maintainer needs to recreate the failure on their own machine.

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